

MIT Technology Review

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Crime of the **next** century

Tech is
reshaping
how people
break the law—
and how they
enforce it

Swindled by AI

Hypercar pirates of the
American highway

Chicago vs. the
surveillance state



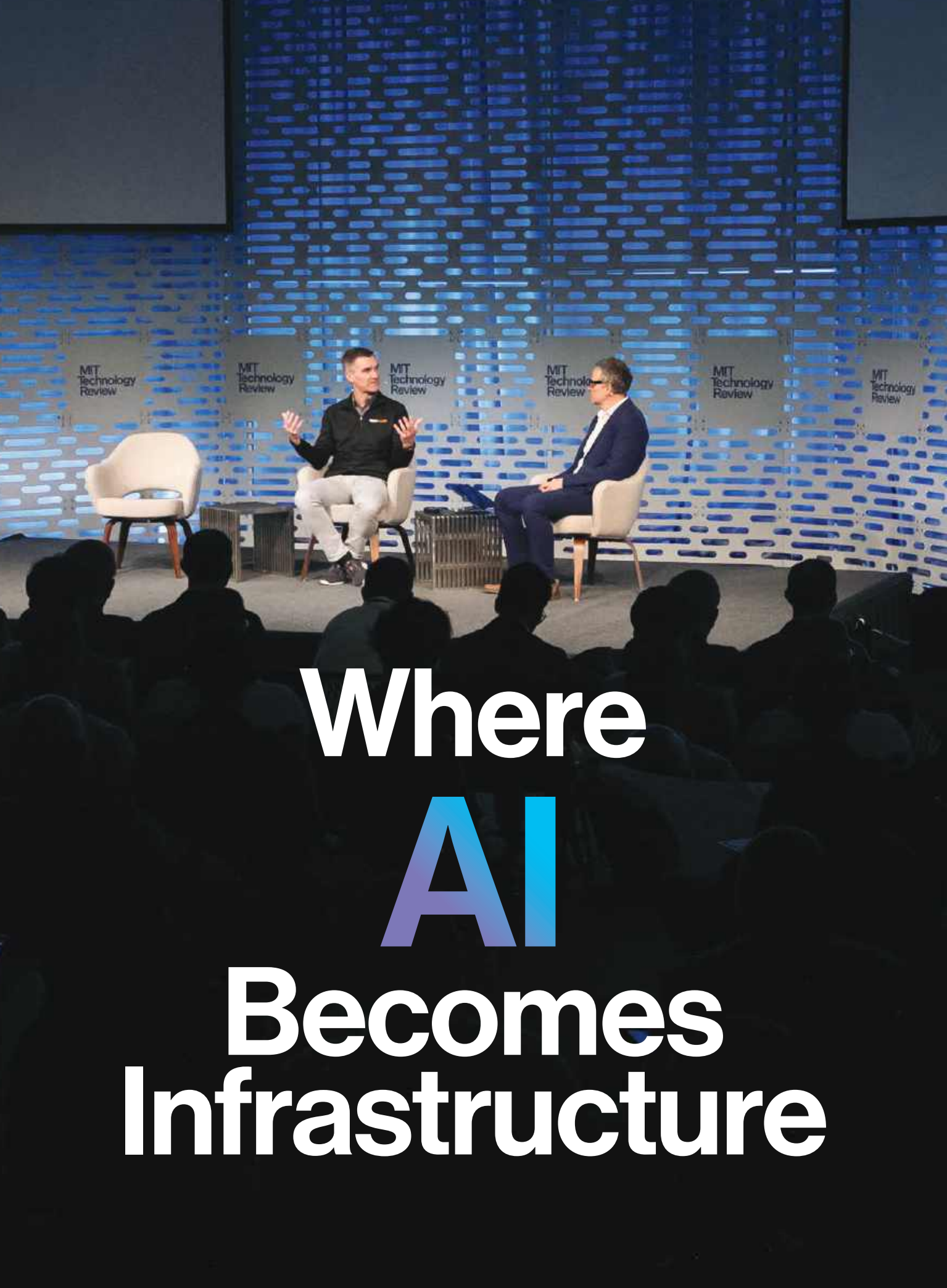
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Now is a good time for doing crime

By Mat Honan



Mat Honan is editor in chief of *MIT Technology Review*.

Eons ago, in 2012, I had a weird experience. My iPhone suddenly shut down. When I restarted it, I found it was totally reset—clean, like a new device. This was the early days of iOS, so I wasn't too concerned until I went to connect it to my computer to restore it from a backup. But when I flipped open the lid of my laptop, it too was mid-restart. And then, suddenly, the screen went gray. It was being remotely wiped. I turned on my iPad. It, too, had been wiped. I was being hacked.

Frantically, I shut down all my devices, unplugged everything connected to the internet in my house, turned off my router, and went next door to use my neighbors' computer and find out what was going on. Deepening my panic, I realized hackers had also gained control of, and nuked, my Google account. Worse, they were in control of my Twitter, which they were gleefully using to spew all sorts of vile comments. It was nasty.

You have to remember, this was before all of us lived with a constant rain of text messages and emails designed to elicit the information necessary to pull something like this off. These crooks hadn't brute-forced their way in, or used any sort of sophisticated techniques to gain access to my accounts. Instead, they had relied on publicly available information, and a fake credit card number, to socially engineer their way into my Amazon account, where they looked up the last four digits of my real credit card number. Then they used that information to get into Apple. And because that account was linked to my Gmail, and that to my Twitter, it gave them the keys to everything.

But what really troubled me was what I learned as I followed up on my hack over the ensuing weeks and months: This kind of thing was, while still novel, becoming more common. Some version of what happened to me had happened to lots of other people. The kids who were responsible—it was a couple of kids—weren't criminal masterminds. They had just found a gap, a place where a technology was now

commonplace but its risks and exploitable surface areas weren't yet fully understood. I just happened to have all my stuff in the gap. Today that gap might feature a crypto wallet or a deepfake of a loved one's voice. (Or both.)

Crime changes.

The goals stay the same—pursuit of value, pursuit of power—but new technologies create new vulnerabilities, new tactics, and new ways for perpetrators to evade discovery or capture. And the law necessarily lags behind. Relying not on innovation but on precedent, it is intentionally backward-looking and slow. That plodding consideration used to be how we protected our shared democratic society, how we protected each other from each other.

But those same new technologies that have allowed crime to outpace law have also reenergized law enforcement and government—offering new ways to root out crime, to gather evidence, to surveil people. Think, for example, of how cold-case investigators tracked down the Golden State Killer years after his murders, using DNA samples and genealogy databases—launching a new era of DNA-powered investigations.

Technology has long made crime and its prosecution a game of cat and mouse. It sometimes calls into question the nature of crime itself. Unregulated behaviors, facilitated by technology, can exist in murky zones of dubious legality. (Until TikTok announced its new ownership structure, Apple and Google were both technically breaking the law by allowing the app to stay on their platforms, under the provisions of the Protecting Americans from Foreign Adversary Controlled Applications Act. Ah! Well. Nevertheless.)

That tension is the key to our March/April issue. Thanks to technologies like cryptocurrency and off-the-shelf autonomous autopilots, there's never been a better time to do crime. Thanks to pervasive surveillance and digital infrastructure, there's never been a better time to fight it—sometimes at the expense of what we used to think of as fundamental civil rights.

I never pressed charges against the kids who hacked me. The biggest consequence of the hack was that Apple set up two-factor authentication in the following months, which felt like a win. Now I'm not sure anyone expects their personal data to be secure in any meaningful way. I'm certain, though, that somewhere on the net, a new generation of kids is coming up with another novel crime. ■

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Hackers are likely to adopt open-source AI. It's easier to strip out the safeguards and get it to do malicious things.

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The Download

SUPRASONIC

Earth's rumbling secret soundtrack

A New England artist makes music from the imperceptible noises of nature—using tools made to detect hidden nuclear explosions.
By Monique Brouillette

The boom of a calving glacier. The crackling rumble of a wildfire. The roar of a surging storm front. They're the noises of the living Earth, music of this one particular sphere and clues to the true nature of these dramatic events. But as loud as all these things are, they emit even more acoustic energy below the threshold of human hearing, at frequencies of 20 hertz or lower. These "infrasounds" have such long wavelengths that they can travel around the globe as churning emanations of distant events. But humans have never been able to hear them.

Until now, that is. *Everyday Infrasound in an Uncertain World*, a new album by the musician and artist Brian House, condenses 24 hours of these rumbles into 24 minutes of the most basic of bass lines, putting a new spin on the idea of ambient music. Sound, even infrasound, is really just variations in air pressure. So House built a set of three "macrophones," tubes that funnel air into a barometer capable of taking readings 100 times a second. From the quiet woods of western Massachusetts, House can pick up what the planet is laying down. Then he speeds the recording up by a factor of 60 so that it's audible to the wee ears of humans. "I am really interested in the layers of perception that we can't access," he says. "It's not only low sound,

but it's also distant sound. That kind of blew my mind."

House's album is art, but scientists made it possible. Barometers picked up the 1883 eruption of the South Pacific volcano Krakatoa as far away as London. And today, a global network of infrasound sensors helps enforce the nuclear test ban treaty. A few infrasound experts—like Leif Karlstrom, a volcanologist at the University of Oregon who uses infrasound to study Mount Kilauea in Hawaii—helped House set up his music-gathering array and better understand what he was hearing. "He's highlighting interesting phenomena," Karlstrom says, even though it's impossible to tell exactly *what* is making each specific sound.

So how's the actual music? It's 24 minutes of an otherworldly chorus, alternating between low grumbling vibrations and soft ghostlike whispers. A high-pitched whistle? Could be a train, House says. An intense low-octave rattle? Maybe a distant thunderstorm or a shifting ocean current. "For me, it's about the mystery of it," he says. "I hope that's a little bit unsettling." But it also might connect someone listening to a wider—and deeper—world. ■

Monique Brouillette is a freelance writer based in Cambridge, Massachusetts.

Artist Brian House's array of "macrophones" picks up distant infrasounds for translation into pulsing music.



COURTESY OF THE ARTIST







BURGLING

The myth of the high-tech heist

The film industry would have you believe that thieves have a boundless technological tool kit. By Adam Rogers

Making a movie is a lot like pulling off a heist. That's what Steven Soderbergh—director of the *Ocean's* franchise, among other heist-y classics—said a few years ago. You come up with a creative angle, put together a team of specialists, figure out how to beat the technological challenges, rehearse, move with Swiss-watch precision, and—if you do it right—redistribute

some wealth. That could describe either the plot or the making of *Ocean's Eleven*.

But conversely, pulling off a heist isn't much like the movies. Surveillance cameras, computer-controlled alarms, knock-out gas, and lasers hardly ever feature in big-ticket crime. In reality, technical countermeasures are rarely a problem, and high-tech gadgets are rarely a solution.

Last year a heist cost the Louvre €88 million worth of antique jewelry, and the most sophisticated technology in play was an angle grinder.

The main barrier to entry is usually a literal barrier to entry, like a door. Thieves' most common move is to collude with, trick, or threaten an insider. Last year a heist cost the Louvre €88 million worth of antique jewelry, and the most sophisticated technology in play was an angle grinder.

The low-tech Louvre maneuvers were in keeping with what heist research long ago concluded. In 2014 US nuclear weapons researchers at Sandia National Laboratories took a detour into this demimonde, producing a 100-page report called "The Perfect Heist: Recipes from Around the World." The scientists were worried someone might try to steal a nuke from the US arsenal, and so they compiled information on 23 high-value robberies from 1972 to 2012 into a "Heist Methods and Characteristics Database," a critical mass of knowledge on what worked. Thieves, they found, dedicated huge amounts of money and time to planning and practice runs—sometimes more than 100. They'd use brute force, tunneling through sewers for months (Société Générale bank heist, Nice, France, 1976), or guile, donning police costumes to fool guards (Gardner Museum, Boston, 1990). But nobody was using, say, electromagnetic



Recommendation engine

MAKE

Lego Icons USS *Enterprise* NCC-1701-D

Don't think of this iconic TV spaceship rendered in plastic bricks as the coolest Lego *Star Trek* set ever—think of it as the coolest Lego *Star Trek* set so far. And at \$399.99, the Galaxy-class starship from the much loved 20th- (and 23rd-) century show *Star Trek: The Next Generation* is a veritable bargain—only 11 cents a brick! Lego *Trek* had been an impossible dream; another brand had the IP rights. But now that it's real, think of the possibilities! Maybe *Enterprise* won't lead to more *Trek* stuff the way Lego's first foray into licensed IP did.

(That'd be the other *Star*—*Wars*, 25 years ago.) But can we wish upon this *Star* for a *Voyager*? A *Deep Space Nine*? A *bat'leth*?! Come on, Lego. Make it so.



LOOK

Manhattan Project: Photographs by Jan Staller



dangle the heaviest building materials. The result was the images in this book—conduit spools and industrial dumpsters incongruously suspended against a backdrop of white clouds. Some of the floating objects are just complex rebar frames, geometries of potential. This is construction deconstruction, a dissection of the pieces of a city.

Since the 1970s, the stalwart New York artist Jan Staller had focused his camera on empty urban spaces. But in the mid-2010s Staller happened to look up—at a construction crane across from his house, strong enough to

BUY

ThermoWorks ThermoPop 2

Look, you're a great cook, and no one is saying otherwise. But a quick-reading thermometer is a kind of power-up in a kitchen—no more guessing if it's time to take sausage off a poach and switch to a sear (seriously, try it this way) or whether the king trumpet mushroom is heated through, striking that fine line between medium-rare and medium. You want data for that, and this thing gets it. The ThermoPop 2 is somehow not top-of-the-line, which is weird, because it takes fast readings, displays in two directions, calculates in whichever temperature scale you're cool with (okay, it doesn't do Kelvin) ... and it costs barely a tenth as much as a Lego *Enterprise*. Quit using the Force; switch to science.



ALAMY (LOUVRE); COURTESY PHOTOS

pulse generators to shut down the Las Vegas electrical grid. The most successful robbers got to the valuable stuff unseen and got out fast.

Advance the time frame, and the situation looks much the same. Last year, Spanish researchers looking at art crimes from 1990 to 2022 found that the least technical methods are still the most successful. “High-tech technology doesn’t work so well,” says Erin L. Thompson, an art historian at John Jay College of Justice who studies art crime. Speed and practice trump complicated systems and alarms; even that Louvre robbery was, at heart, just a minutes-long smash-and-grab.

An emphasis on speed doesn’t mean heists don’t require skill—panache, even. As the old saying goes, amateurs talk strategy; professionals study logistics. Even without gadgets, heists and heist movies still revel in an engineer’s mindset. “Heist movies absolutely celebrate deep-dive nerdery—I’m going to know everything I can about the power grid, about this kind of stone and drill, about Chicago at night,” says Anna Kornbluh, a professor of English at the University of Illinois at Chicago. She published a paper last October on the ways heist movies reflect an Old Hollywood approach to collective art-making, while shows about new grift, like those detailing the rise and fall of WeWork or the con artist Anna Delvey, reflect the more lone-wolf, disrupt-and-grow mindset of the streaming era.

Her work might help explain why law-abiding citizens might cheer for the kinds of guys who’d steal a crown from the Louvre, or \$100,000 worth of escargot from a farm in Champagne (as happened just a few weeks later). Heists, says Kornbluh, are anti-oligarch praxis. “Everybody wants to know how to be in a competent collective. Everybody wants there to be better logistics,” she says. “We need a better state. We need a better society. We need a better world.” Those are shared values—and as another old saying tells us, where there is value, there is crime. ■

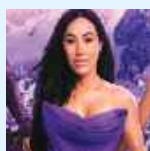
Adam Rogers is interim director of print at MIT Technology Review.

3 THINGS

MIT Technology Review’s audience engagement editor Juliet Beauchamp shares what she’s been thinking about lately.



The only reality show that matters



The Real Housewives of Salt Lake City is one of the best shows on television right now.

Not one of the best reality TV shows, but one of the best TV shows, period. Chronicling a shifting group of wealthy women in and around Salt Lake, the show has featured a convicted felon whom federal agents came looking for while cameras were rolling, a church leader married to her step-grandfather, and a single mom in an exhausting on-again, off-again relationship with an Osmond. In

one season, there was an ongoing argument between two cast members after one told the other that she “smelled like hospital.” Later, one woman was secretly running an anonymous gossip Instagram about her fellow housewives. We can debate the “reality” of reality television, and it’s certainly true that these characters and scenarios are far-fetched. But every single person is dealing with something relatable—difficult marriages, failing businesses, strained relationships with children, addiction. It’s entertainment, and high camp, but I find that I still have a lot of empathy for these people.

The last good place(s) on Facebook



Facebook sucks. That’s not controversial to say, right? But there is one reason I still have a Facebook

account: my neighborhood Buy Nothing group. The spirit of community and camaraderie is alive and well there—and probably in yours, too. A non-exhaustive list of things

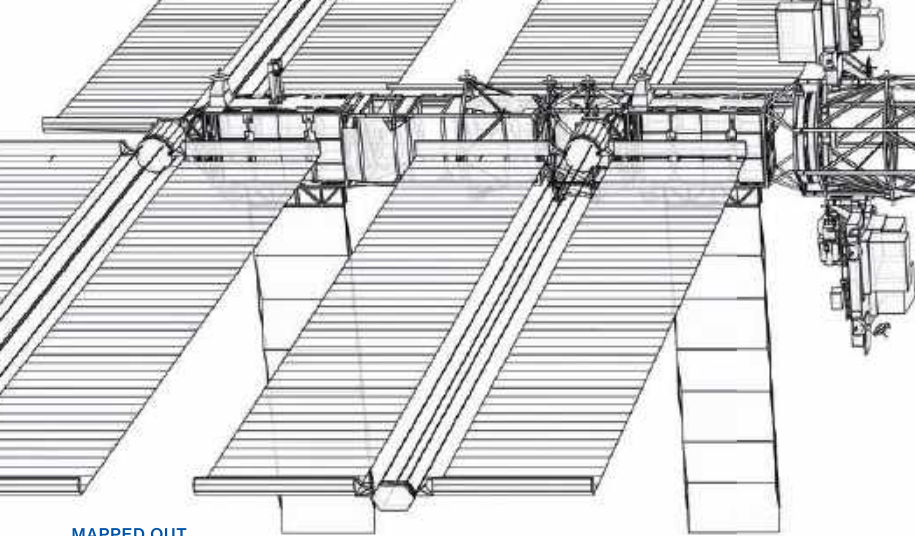
I have given away: empty candle jars, a bookcase, used lightbulbs, unopened toiletries, bubble wrap. I’ve scored a few good things as well: a gorgeous antique dresser that I refinished, some over-the-door hooks, and brand-new jeans. It makes me happy to know that stuff that would’ve otherwise ended up in a landfill is bringing one of my neighbors joy.

Going analog

I used to wear an Apple Watch a lot. I’m a pretty active person, and I liked tracking my workouts and my steps. But after I’d had it for a while, my watch started dying in the middle of a 30-minute run; it became useless to me, and I gave it up completely. Guess what? I’m happier. I feel

more present when I’m not checking how much time is left in a yoga class or reading texts during a long run. The amount of data it gathered about me was also stressing me out, and it wasn’t useful. And I don’t need a wearable to tell me how poorly I slept! Trust me, I already know.





MAPPED OUT

The anthroposphere

We're putting more stuff into space than ever before. Here's what's up there.

By Jonathan O'Callaghan

Illustrations by John MacNeill

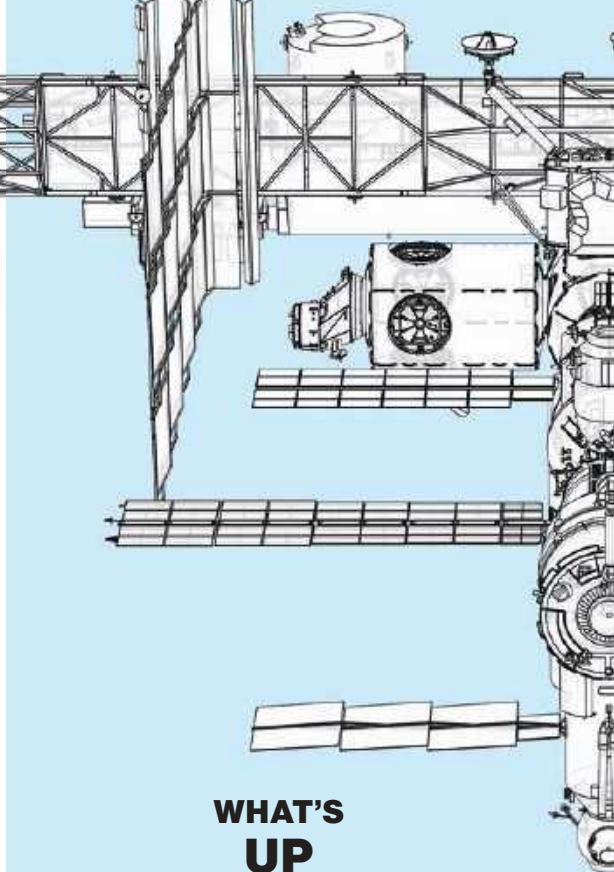
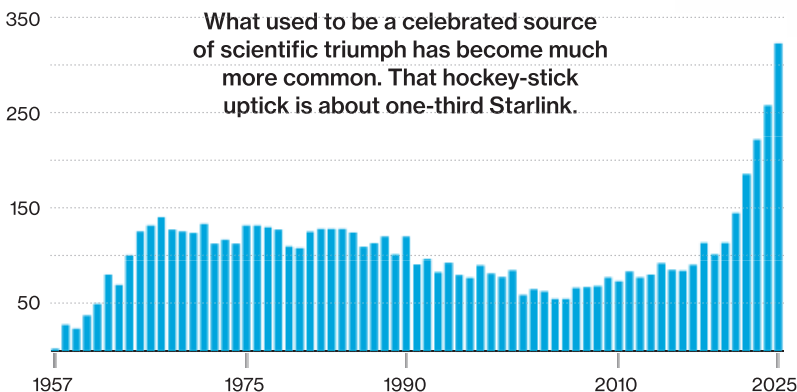
Earth's a medium-size rock with some water on top, enveloped by gases that keep everything that lives here alive. Just at the edge of that envelope begins a thin but dense layer of human-built, high-tech stuff.

People started putting gear up there in 1957, and now it's a real habit. Telescopes look up and out at the wild universe. Humans live in an orbiting metal bubble. In the last five years, the number of active satellites in space has increased from barely 3,000 to about 14,000—and climbing. The

biggest use case: "megaconstellations" like Elon Musk's Starlink internet service, which by itself has nearly 10,000 satellites in orbit.

And then there's the garbage: 50,000 bits of debris larger than a baseball now orbit Earth, along with a million more objects bigger than a coin. If you enjoy things like weather forecasts and digital communication, hope they don't start crashing into each other. Here's a closer look at Earth's ever-thickening shell of human-made matter—the anthroposphere.

SATELLITE LAUNCHES



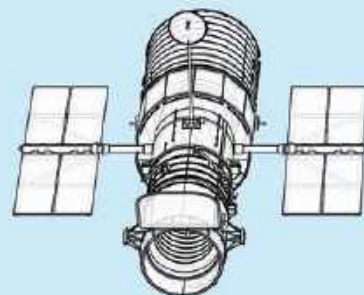
WHAT'S UP THERE?

A look at some of the more notable inhabitants of the anthroposphere, from big and unique to small and multiple.

HUBBLE SPACE TELESCOPE

Altitude 480 km
Length 13.2 m

Launched on the space shuttle *Discovery* in 1990—and with a potentially mission-ending flaw repaired by another shuttle mission three years later—the Hubble Space Telescope has transformed scientific understanding of galaxies, exoplanets, and more.



SOURCE: NASA, SPACEX

HOW MUCH AND HOW HIGH?

Most of the satellites orbiting Earth today are in low Earth orbit, and most of them are part of the Starlink constellation. But other megaconstellations are on the way; some projections have more than 50,000 satellites orbiting Earth by the end of the decade.

Geostationary orbit:
20,000 – 36,000 km

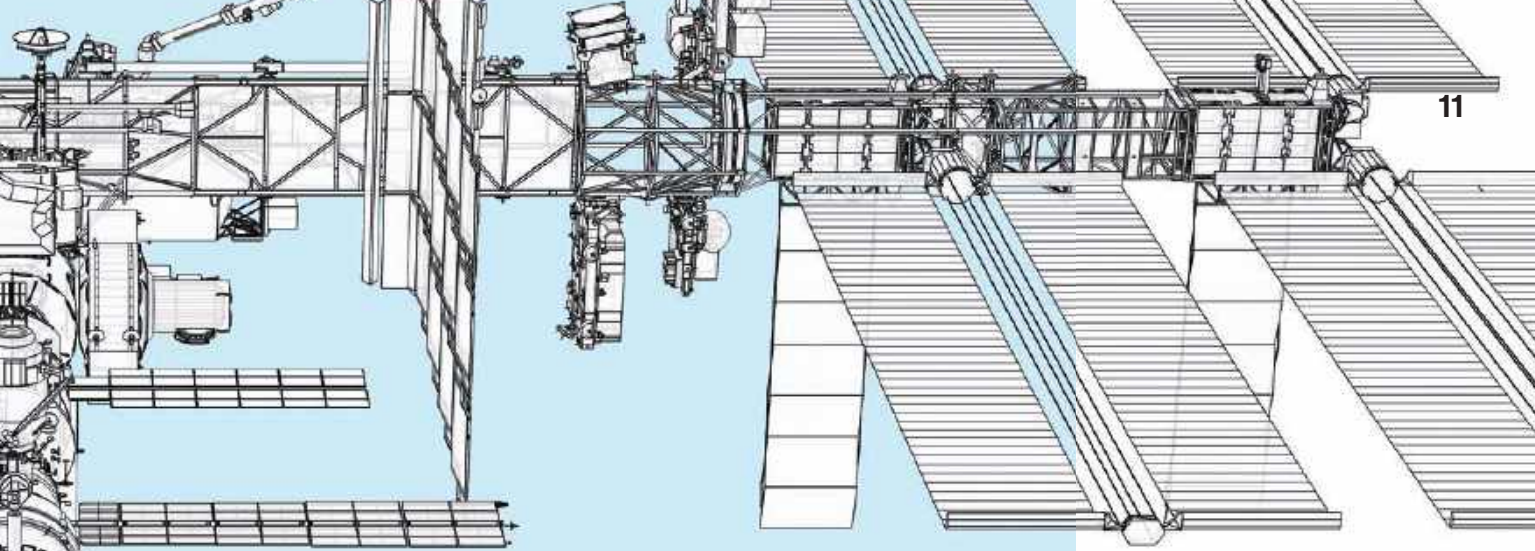
Medium Earth orbit:
2,000 – 20,000 km

Low Earth orbit:
400 – 2,000 km

300 satellites

700 satellites

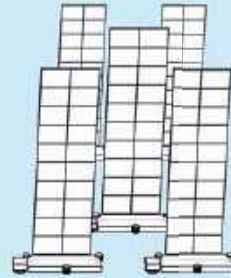
SURFACE OF THE EARTH



INTERNATIONAL SPACE STATION

Altitude ... 370–460 km
Length 109 m

About 300 people have spent time on the ISS, the largest human-made structure in space. Its days are numbered—current plans are to “deorbit” it in 2031, causing a fiery disintegration over the Pacific Ocean.



STARLINK

Altitude ... 340–570 km
Length 7 m

Starlink satellites launch in batches aboard SpaceX rockets, sometimes up to 60 at a time—and so many are in orbit that Earth-based astronomers complain the satellites are blocking their view.



GPS

Altitude.... 20,000 km
Length 12 m

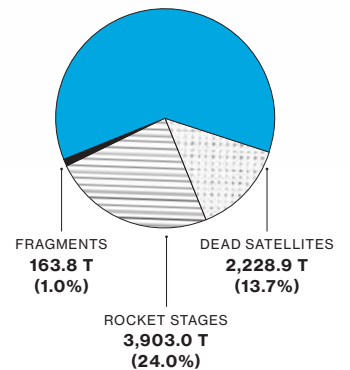
The 31 active Global Positioning System (GPS) satellites orbit at altitudes that let people receive signals from at least four at once anywhere on Earth. Since 1978, this has provided uncannily precise measures of position, altitude, and time.

VANGUARD 1

Altitude..... 650–3,800 km
(elliptical)
Length 0.25 m

The oldest satellite in orbit, this early US experimental mission is a tiny sphere weighing just 1.47 kilograms, about the size of a grapefruit. It'll stay up there for another 200 years.

ACTIVE SATELLITES
9,990.8 METRIC TONS
(61.3%)



SPACE JUNK

Only once have two satellites collided—in 2009 a communications satellite launched by the US company Iridium crashed into a derelict Cosmos satellite from Russia, smashing thousands of pieces of debris into the surrounding space. Researchers track as much of what's circling the planet as they can; if a collision looks imminent, a satellite can fire its thrusters to get out of the way. But more satellites means more close calls. The worst-case scenario is that more collisions lead to more fragments, leading to still more collisions. That doomsday outcome—the “Kessler syndrome,” named for NASA scientist Donald Kessler, who warned of it in 1978—could render the atmosphere unusable.

Most of the millions of pieces of debris swinging around the planet are shards of metal from discarded spacecraft or rockets launched into space. Some are detritus from anti-satellite tests, when satellites have purposely been destroyed in displays of military might. Others are entire now-dead satellites or empty rockets.

communications

GPS

Starlink, ISS,
and Hubble

13,000 satellites

SOURCE: JONATHAN MCDOWELL

SOURCE: JONATHAN MCDOWELL, ESA



JOB TITLES OF THE FUTURE

Breast biomechanic

By Sara Harrison

Twenty years ago, Joanna Wakefield-Scurr was having persistent pain in her breasts. Her doctor couldn't diagnose the cause but said a good, supportive bra could help. A professor of biomechanics, Wakefield-Scurr thought she could do a little research and find a science-backed option. Two decades later, she's still looking. Wakefield-Scurr now leads an 18-person team at the Research Group in Breast Health at the University of Portsmouth in the UK. Their research shows that the most effective high-impact-sports bras have underwires, padded cups, adjustable underbands and shoulder straps, and hook-and-eye closures. These bras reduce breast movement by up to 74% when compared with wearing no bra. But movement might not be the only metric that matters.

A biological rarity

Few anatomical structures hang outside of the body unsupported by cartilage, muscle, or bone—meaning there wasn't much historical research to build on. Wakefield-Scurr's lab was the first to find that when women run, the motion of the torso causes breasts to move in a three-dimensional pattern—swinging side to side and up and down—as well as moving forward and backward. In an hour of slow jogging, boobs can bounce approximately 10,000 times.

A sports necessity

Wearing a bra that's too tight can limit breathing. Wearing one that's too loose can create back, shoulder, and neck pain. Pain can also be caused by the lag between torso and breast movement, which causes what is scientifically known as “breast slap.”

The lab's research has also found that the physical discomfort of bad bras, combined with the embarrassment of flopping around, is the one of the biggest barriers to exercise for women and that if women have a good sports bra, they're more willing to go for a run.

An open question

Some bras function by deliberately compressing breasts. Others encapsulate and support each individual breast. But scientists still don't know whether it's more biomechanically important to reduce the breasts' motion entirely, to reduce the speed at which they move, or to reduce breast slap. Will women constantly be forced to choose between the comfort of a stretchier bra and the support of a more restrictive one?

Wakefield-Scurr is excited about new materials she's tested that tighten or stretch depending on how you move. She's working with fabric manufacturers and clothing companies to try out their wares. As more women take up high-impact sports, the need to understand what makes a good bra grows. Wakefield-Scurr says her lab can't keep up with demand. Their cups runneth over. ■

Sara Harrison is a freelance journalist who writes about science, technology, and health.

DISPATCH

The clash of ideas— and access—at Davos

The annual meeting is about much more than rich people being cliquish. But it's not *not* about that. By Mat Honan

It was supposed to be frigid in Davos for the World Economic Forum's annual meeting. Part of the charm is seeing the world's elite tromp through the streets in respectable suits and snow boots. But this year the weather was positively balmy, with highs in the mid-30s F, or a little over 1 °C. It was actually colder and snowier in New York than it was in Switzerland. I'm told this was due to something called a föhn, a dry, warm wind that was blowing across the Alps.

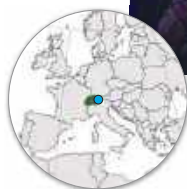
I'm no meteorologist, but it's true that Davos was full of hot air.

You hear about the WEF as a gathering of elites. But that suggests some sort of consensus, when in reality much of what takes place there is about competition. Witness California governor Gavin Newsom or Canadian prime minister Mark Carney taking aim at President Donald Trump. But it's a place where ideas compete for attention as well. A few years ago, everyone was talking about climate change. This year's big idea was more billionaire-friendly.

Like the rest of the world, Davos was obsessed with AI. In the official programming, I counted more than 40 panels and sessions where it was a topic. There was even more to be found along the promenade, a street running through the center of the town lined with various "houses"—mostly shops converted into hubs for corporate, thematic, and national sponsors. Like everything else in 2026, the promenade is dominated by tech. At one point I realized that literally everything I could see, at a spot where the road bends a bit,



Above: Want to see the global elite really compete? Give them first-come, first-served access to Trump.



THE EVENT:
World Economic
Forum Annual
Meeting
January 19–23,
2026

THE PLACE:
Davos, Switzerland

was a tech company house: Palantir, Workday, Infosys, Cloudflare, C3.ai. By far the hottest houses on the row were the USA House—where a huge American delegation was holding forth—and the AI House.

The AI House was consistently packed. There was almost always a long, cranky line to get in, and once inside I literally had to muscle my way through the crowd. One-way markers on the stairs kept people in line. God forbid you wanted to try to get a drink.

But it was nothing next to the buzz for Trump. On Wednesday, Trump arrived in Davos to address the annual meeting and spoke for more than 90 minutes, weaving through remarks about the economy, Greenland, windmills, Switzerland, Rolexes, Venezuela, and drug prices. It was a talk lousy with gripes, grievances, and outright falsehoods. (One small example: Trump made a big deal of claiming that China, despite being the world's

biggest manufacturer of wind turbine components, doesn't actually use turbines to make electricity itself. In fact, that's another competition China has won; it is the world leader in wind-power generation.)

But I did not get to watch this spectacle from the room itself. Sad!

By the time I got to the Congress Hall, where the address was taking place, people were already jostling to get in. I tried my best, but the scrum was too thick. People were jammed in so tightly, massing toward the entrance, that it reminded me of being in the pit at a Turnstile concert. If you want to see the global elite really compete, give them first-come, first-served access to Trump, I guess.

Of course, there are elites, and then there are *elites*. In yet another sort of ongoing competition, Davos offers all sorts of ways of sorting out who is who. Your badge color is one of them. The white participant badge I had as a speaker gets you in pretty much anywhere and therefore is its own sort of status symbol—some venues outside the Congress Center itself limit attendance to “white badges only.” Where you are staying is another. I was in a neighboring town 40 minutes by train from the main drag. Not so elite.

Beneath all that are more subtle forms of sortation. When people ask if this is your first time at Davos, it's sometimes meant as a way of trying to figure out how important you are. If you're any kind of big deal, you've probably been coming for years.

The best one I encountered happened when I made small talk with a woman sitting next to me as I changed back into my snow boots. It turned out that, like me, she lived in California—at least part time. “But I don't think I'll stay there much longer,” she said, “due to the new tax law.” In a town full of hot air, this was an ice-cold flex.

Because the newly proposed tax legislation in California she was talking about. It only targets billionaires. ■

Mat Honan is MIT Technology Review's editor in chief. Portions of this Dispatch originally appeared in his *Debrief* newsletter.



Group chat

Readers, we want to hear from you! Tell us what's on your mind, share your perspective, or ask us a question by writing to newsroom@technologyreview.com.

Analog AMA

Q: Do you think these new waves of AI toys are yet another way to pilfer training data from people? How safe are these toys in regard to privacy? And what policy regulates their safety in the market? —*Jedidiah from Nigeria*

A: Hi! AI toys aren't necessarily secret data traps—most companies are simply chasing the AI gold rush—but some do carry privacy risks. A recent study by the US Public Interest Research Group, which tested five AI toys, found them “overtly problematic.” Many other toys come with built-in cameras, always-on microphones, and cloud connections. And they often default to “Share my data with developers” unless you opt out. Moreover, some may tell children they've developed feelings for them—a potentially harmful dynamic.

In the US, the Federal Trade Commission has launched enforcement actions and a broad inquiry into AI and chatbot products targeting children. Recent cases include action against a robotic-toy maker for violating COPPA, the main federal law governing children's data. —*Caiwei Chen, China reporter*

Tech tip

CURIOUS ABOUT HOW OUR STAFF USES THE TECHNOLOGIES AROUND US? LEARN ABOUT A TOOL THAT ONE OF OUR STAFFERS HAS TRIED AT HOME OR WORK, OR GET THEIR ADVICE ON HOW TO MANAGE THE TECH IN YOUR LIFE.

As a reporter who often covers biometrics and privacy, I've learned that sensitive images are used to teach robot vacuums to see and that our neighbors' adoption of security cameras may cause our own images to be captured. Now I exercise my right to say no to having my biometric information collected whenever I can—especially when I travel.

At TSA checkpoints, I say that I don't want my photo taken before I hand over my passport or driver's license. It takes almost no extra time for a TSA agent to identify me the old-fashioned way: checking that the name on my boarding pass matches the one on my passport or license, and that the photograph on that identity document matches my face.



Beyond the principle and the message that it sends, there's a more concrete issue: It's really not clear how our data is being used. TSA says it immediately deletes photos, except for limited testing purposes—an exception that is vague enough to make me uncomfortable. Besides, in 2022, TSA also laid out a long-term strategy of expanding its biometric capabilities toward real-time vetting—i.e., being able to pick out any face in a video feed, as opposed to matching one person's image to his or her face. The technology isn't there yet—and if you're concerned that it will get there, then one thing you can do is opt out of helping to improve it.

—*Eileen Guo, senior reporter*

Poll



In last issue's Group Chat, we revisited our 10 Breakthrough Technologies of 2025. These are the technologies you said had the biggest impact last year:

- 1.....Generative AI search
54%
- 2.....Stem-cell therapies that work
11%
- 3.....The Vera C. Rubin Observatory
7%

YOUR TURN TO WEIGH IN!

Q: A study published in November in *JAMA Network Open* found that taking a one-week break from social media significantly reduced symptoms of anxiety, depression, and insomnia. Which platform are you ready for a break from?

- ☐ Instagram
- ☐ Facebook
- ☐ TikTok
- ☐ LinkedIn
- ☐ X
- ☐ All of the above



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César de la Fuente is an associate professor at the University of Pennsylvania, where he leads the Machine Biology Group.



The scientist using AI to hunt for antibiotics just about everywhere

César de la Fuente is on a mission to combat antimicrobial resistance, by building on solutions nature has already found.

By Stephen Ornes
Portrait by Anthony Wood

When he was just a teenager trying to decide what to do with his life, César de la Fuente compiled a list of the world's biggest problems. He ranked them inversely by how much money governments were spending to solve them. Antimicrobial resistance topped the list.

Twenty years on, the problem has not gone away. If anything, it's gotten worse. Infections caused by bacteria, fungi, and viruses that have evolved ways to evade treatments are now associated with more than 4 million deaths per year, and a recent analysis, published in the *Lancet*, predicts that number could surge past 8 million by 2050. In a July 2025 essay in *Physical Review Letters*, de la Fuente, now a bioengineer and computational biologist, and synthetic biologist James Collins warned of a looming "post-antibiotic" era in which infections from drug-resistant strains of common bacteria like *Escherichia coli* or *Staphylococcus aureus*, which can often still be treated by our current arsenal of medications, become fatal. "The antibiotic discovery pipeline remains perilously thin," they wrote, "impeded by high development costs, lengthy timelines, and low returns on investment."

But de la Fuente is using artificial intelligence to bring about a different future. His team at the University of Pennsylvania is training AI tools to search genomes far and deep for peptides with antibiotic properties.

His vision is to assemble those peptides—molecules made of up to 50 amino acids linked together—into various configurations, including some never seen in nature. The results, he hopes, could defend the body against microbes that withstand traditional treatments.

His quest has unearthed promising candidates in unexpected places. In August 2025 his team, which includes 16 scientists in Penn's Machine Biology Group, described peptides hiding in the genetic code of ancient single-celled organisms called archaea. Before that, they'd excavated a list of candidates from the venom of snakes, wasps, and spiders. And in an ongoing project de la Fuente calls "molecular de-extinction," he and his collaborators have been scanning published genetic sequences of extinct species for potentially functional molecules. Those species include hominids like Neanderthals and Denisovans and charismatic megafauna like woolly mammoths, as well as ancient zebras and penguins. In the history of life on Earth, de la Fuente reasons, maybe some organism evolved an antimicrobial defense that could be helpful today. Those long-gone codes have given rise to resurrected compounds with names like mammuthusin-2 (from woolly mammoth DNA), mylodonin-2 (from the giant sloth), and hydrodamin-1 (from the ancient sea cow). Over the last few years, this molecular binge has enabled de la Fuente to amass a library of more than a million genetic recipes.

At 40 years old, de la Fuente has also collected a trophy case of awards from the American Society for

Microbiology, the American Chemical Society, and other organizations. (In 2019, this magazine named him one of “35 Innovators Under 35” for bringing computational approaches to antibiotic discovery.) He’s widely recognized as a leader in the effort to harness AI for real-world problems. “He’s really helped pioneer that space,” says Collins, who is at MIT. (The two have not collaborated in the laboratory, but Collins has long been at the forefront of using AI for drug discovery, including the search for antibiotics. In 2020, Collins’s team used an AI model to predict a broad-spectrum antibiotic, halicin, that is now in preclinical development.)

The world of antibiotic development needs as much creativity and innovation as researchers can muster, says Collins. And de la Fuente’s work on peptides has pushed the field forward: “César is marvelously talented, very innovative.”

A messy, noisy endeavor

De la Fuente describes antimicrobial resistance as an “almost impossible” problem, but he sees plenty of room for exploration in the word *almost*. “I like challenges,” he says, “and I think this is the ultimate challenge.”

The use, overuse, and misuse of antibiotics, he says, drives antimicrobial resistance. And the problem is growing unchecked because conventional ways to find, make, and test the drugs are prohibitively expensive and often lead to dead ends. “A lot of the companies that have attempted to do antibiotic development in the past have ended up folding because there’s no good return on investment at the end of the day,” he says.

Antibiotic discovery has always been a messy, noisy endeavor, driven by serendipity and fraught with uncertainty and misdirection. For decades, researchers have largely relied on brute-force mechanical methods. “Scientists dig into soil, they dig into water,” says de la Fuente. “And then from that complex organic matter they try to extract antimicrobial molecules.”

But molecules can be extraordinarily complex. Researchers have estimated the number of possible organic combinations that could be synthesized at somewhere around 10^{60} . For reference, Earth contains an estimated 10^{18} grains of sand. “Drug discovery in any domain is a statistics game,” says Jonathan Stokes, a chemical biologist at McMaster University in Canada, who has been using generative AI to design potential new antibiotics that can be synthesized in a lab, and who worked with Collins on halicin. “You need enough shots on goal to happen to get one.”

Those have to be good shots, though. And AI seems well suited to improving researchers’ aim. Biology is an information source, de la Fuente explains: “It’s like a bunch of code.” The code of DNA has four letters; proteins and peptides have 20, where each “letter” represents an amino acid. De la Fuente says his work amounts to training AI models to recognize sequences of letters that encode antimicrobial peptides, or AMPs. “If you think about it that way,” he says, “you can devise algorithms to mine the code and identify functional molecules, which can be antimicrobials. Or antimalarials. Or anticancer agents.”

Practically speaking, we’re still not there: These peptides haven’t yet been transformed into usable drugs that help people, and there are plenty of details—dosage, delivery, specific targets—that need to be sorted out, says de la Fuente. But AMPs are appealing because the body already uses them. They’re a critical part of the immune system and often the first line of defense against pathogenic infections. Unlike conventional antibiotics, which typically have one trick for killing bacteria, AMPs often exhibit a multimodal approach. They may disrupt the cell wall and the genetic material inside as well as a variety of cellular processes. A bacterial pathogen may evolve resistance to a conventional drug’s single mode of action, but maybe not to a multipronged AMP attack.

From discovery to delivery

De la Fuente’s group is one of many pushing the boundaries of using AI for antibiotics. Where he focuses primarily on peptides, Collins works on small-molecule discovery. So does Stokes, at McMaster, whose models identify promising new molecules and predict whether they can be synthesized. “It’s only been a few years since folks have been using AI meaningfully in drug discovery,” says Collins.

Even in that short time the tools have changed, says James Zou, a computer scientist at Stanford University, who has worked with Stokes and Collins. Researchers have moved from using predictive models to developing generative approaches. With a predictive approach, Zou says, researchers screen large libraries of candidates that are known to be promising. Generative approaches offer something else: the appeal of designing a new molecule from scratch. Last year, for example, de la Fuente’s team used one generative AI model to design a suite of synthetic peptides and another to assess them. The group tested two of the resulting compounds on mice infected with a drug-resistant strain of *Acinetobacter baumannii*, a germ that the World Health Organization has identified as a “critical priority” in research on antimicrobial resistance. Both successfully and safely treated the infection.

But the field is still in the discovery phase. In his current work, de la Fuente is trying to get candidates closer to clinical testing. To that end, his team is developing an ambitious multimodal model called ApexOracle that’s designed to analyze a new pathogen, pinpoint its genetic weaknesses, match it to antimicrobial peptides that might work against it, and then predict how an antibiotic, built from those peptides, would fare in lab tests. It “converges understanding in chemistry, genomics, and language,” he says. It’s preliminary, he adds, but even if it doesn’t work perfectly, it will help steer the next generation of AI models toward the ultimate goal of resisting resistance.

Using AI, he believes, human researchers now have a fighting chance at catching up to the giant threat before them. The technology has already saved decades of human research time. Now he wants it to save lives, too: “This is the world that we live in today, and it’s incredible.” ■

Stephen Ornes is a science writer in Nashville, Tennessee.

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The human work behind humanoid robots is being hidden

By James O'Donnell



James O'Donnell is senior reporter for AI at [MIT Technology Review](https://technologyreview.com).

How human is a humanoid robot, really?

In January, Nvidia's Jensen Huang, the head of the world's most valuable company, proclaimed that we are entering the era of physical AI, when artificial intelligence will move beyond language and chatbots into physically capable machines. (He also said the same thing the year before, by the way.)

The implication—fueled by new demonstrations of humanoid robots putting away dishes or assembling cars—is that mimicking human limbs with single-purpose robot arms is the old way of automation. The new way is to replicate the way humans think, learn, and adapt while they work. The problem is that the lack of transparency about the human labor involved in training and operating such robots leaves the public both misunderstanding what robots can actually do and failing to see the strange new forms of work forming around them.

Consider how, in the AI era, robots often learn from humans who demonstrate how to do a chore. Creating this data at scale is now leading to *Black Mirror*-esque scenarios. A worker in Shanghai, for example, recently spent a week wearing a virtual-reality headset and an exoskeleton while opening and closing the door of a microwave hundreds of times a day to train the robot next to him, *Rest of World* reported. In North America, the robotics company Figure appears to be planning something similar: It announced in September it would partner with the investment firm Brookfield, which manages 100,000 residential units, to capture “massive amounts” of real-world data “across a variety of household environments.” (Figure did not respond to questions about this effort.)

Just as our words became training data for large language models, our movements are now poised to follow the same path. Except this future might leave humans with an even worse deal, and it's already beginning. The roboticist Aaron Prather told me about recent work with a delivery company that had its workers wear movement-tracking sensors as they moved boxes; the data collected will be used to train robots.

The effort to build humanoids will likely require manual laborers to act as data collectors at massive scale. “It’s going to be weird,” Prather says. “No doubts about it.”

Or consider tele-operation. Though the endgame in robotics is a machine that can complete a task on its own, robotics companies employ people to operate their robots remotely. Neo, a \$20,000 humanoid robot from the startup 1X, is set to ship to homes this year, but the company’s founder, Bernt Øivind Børnich, told me recently that he’s not committed to any prescribed level of autonomy. If a robot gets stuck, or if the customer wants it to do a tricky task, a tele-operator from the company’s headquarters in Palo Alto, California, will pilot it, looking through its cameras to iron clothes or unload the dishwasher.

This isn’t inherently harmful—1X gets customer consent before switching into tele-operation mode—but privacy as we know it will not exist in a world where tele-operators are doing chores in your house through a robot. And if home humanoids are not genuinely autonomous, the arrangement is better understood as a form of wage arbitrage that re-creates the dynamics of gig work while, for the first time, allowing physical tasks to be performed wherever labor is cheapest.

We’ve been down similar roads before. Carrying out “AI-driven” content moderation on social media platforms or assembling training data for AI companies often requires workers in low-wage countries to view disturbing content. And despite claims that AI will soon enough train on its outputs and learn on its own, even the best models require an awful lot of human feedback to work as desired.

These human workforces do not mean that AI is just vaporware. But when they remain invisible, the public consistently overestimates the machines’ actual capabilities.

That’s great for investors and hype, but it has consequences for everyone. When Tesla marketed its driver-assistance software as “Autopilot,” for example, it inflated public expectations about what the system could safely do—a distortion a Miami jury recently found contributed to a crash that killed a 22-year-old woman (Tesla was ordered to pay \$240 million in damages).

The same will be true for humanoid robots. If Huang is right, and physical AI is coming for our workplaces, homes, and public spaces, then the way we describe and scrutinize such technology matters. Yet robotics companies remain as opaque about training and tele-operation as AI firms are about their training data. If that does not change, we risk mistaking concealed human labor for machine intelligence—and seeing far more autonomy than truly exists. ■



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Climate in the court

By James Temple



James Temple is senior editor for climate and energy at MIT Technology Review.

The United States and the European Union grew into economic superpowers by committing climate atrocities. They have burned a wildly disproportionate share of the world's oil and gas, planting carbon time bombs that will detonate first in the poorest, hottest parts of the globe. Meanwhile, places like the Solomon Islands and Chad—low-lying or just plain sweltering—have emitted relatively little carbon dioxide, but by dint of their latitude and history, they rank among the countries most vulnerable to the fiercest consequences of global warming. That means increasingly devastating cyclones, heat waves, famines, and floods.

Morally, there's an ironclad case that the countries or companies responsible for this mess should provide compensation for the homes that will be destroyed, the shorelines that will disappear beneath rising seas, and the lives that will be cut short. By one estimate, the major economies owe a climate debt to the rest of the world approaching \$200 trillion in reparations.

Legally, though, the case has been far harder to make. Even putting aside the jurisdictional problems, early climate science couldn't trace the provenance of airborne molecules of carbon dioxide across oceans and years. Deep-pocketed corporations with top-tier legal teams easily exploited those difficulties.

Now those tides might be turning. More climate-related lawsuits are getting filed, particularly in the Global South. Governments, nonprofits, and citizens in the most climate-exposed nations continue to test new legal arguments in new courts, and some of those courts are showing a new willingness to put nations and their industries on the hook as a matter of human rights. In addition, the science of figuring out exactly who is to blame for specific weather disasters, and to what degree, is getting better and better.

It's true that no court has yet held any climate emitter liable for climate-related damages. For starters, nations are generally immune from lawsuits originating in other countries. That's why most cases have focused on major carbon producers. But they've leaned on a pretty powerful defense. While oil and gas

companies extract, refine, and sell the world's fossil fuels, most of the emissions come out of "the vehicles, power plants, and factories that burn the fuel," as Michael Gerrard and Jessica Wentz, of Columbia Law School's Sabin Center, note in a recent piece in *Nature*. In other words, companies just dig the stuff up. It's not their fault someone else sets it on fire.

So victims of extreme weather events continue to try new legal avenues and approaches, backed by ever-more-convincing science. Plaintiffs in the Philippines recently sued the oil giant Shell over its role in driving Super Typhoon Odette, a 2021 storm that killed more than 400 people and displaced nearly 800,000. The case relies partially on an attribution study that found climate change made extreme rainfall like that seen in Odette twice as likely. Overall, evidence of corporate culpability—linking a specific company's fossil fuel to a specific disaster—is getting easier to find. For example, a study published in *Nature* in September was able to determine how much particular companies contributed to a series of 21st-century heat waves.

A number of recent legal decisions signal improving odds for these kinds of suits. Notably, a handful of determinations in climate cases before the European Court of Human Rights affirmed that states have legal obligations to protect people from the effects of climate change. And though it dismissed the case of a Peruvian farmer who sued a German power company over fears that a melting alpine glacier could destroy his property, a German court determined that major carbon polluters could in principle be found liable for climate damages tied to their emissions. At least one lawsuit has already emerged that could test that principle: Dozens of Pakistani farmers whose land was deluged during the massive flooding events of 2022 have sued a pair of major German power and cement companies.

Even if the lawsuit fails, that would be a problem with the system, not the science. Major carbon-polluting countries and companies have a disproportionate responsibility for climate-change-powered disasters. Wealthy nations continued to encourage business practices that pollute the atmosphere, even as the threat of climate change grew increasingly grave. And oil and gas companies remain the kingpin suppliers to a fossil-fuel-addicted world. They have operated with the full knowledge of the massive social, environmental, and human cost imposed by their business while lobbying fiercely against any rules that would force them to pay for those harms or clean up their act.

They did it. They knew. In a civil society where rule of law matters, they should pay the price. ■

How social media helped carnivores claw their way to the top

By Jessica Hamzelou



Jessica Hamzelou is senior reporter for biomedicine at MIT Technology Review.

Americans have a new set of diet guidelines. Robert F. Kennedy Jr. has taken an old-fashioned food pyramid, turned it upside down, and plonked a steak and a stick of butter in prime positions.

Kennedy and his Make America Healthy Again mates have long been extolling the virtues of meat and whole-fat dairy, so it wasn't too surprising to see those foods recommended alongside vegetables and whole grains (despite the well-established fact that too much saturated fat can be extremely bad for you).

Some influencers have taken the meat trend to extremes, following a "carnivore diet." "The best thing you could do is eliminate out everything except fatty meat and lard," Anthony Chaffee, an MD with almost 400,000 followers, said in an Instagram post.

And I almost choked on my broccoli when, while scrolling LinkedIn, I came across an interview with another doctor declaring that "there is zero scientific evidence to say that vegetables are required in the human diet." That doctor, who described himself as "90% carnivore," went on to say that all he'd eaten the previous day was a kilo of beef, and that vegetables have "anti-nutrients," whatever they might be.

You don't have to spend much time on social media to come across claims like this. The "traditionalist" influencer, author, and psychologist Jordan Peterson was promoting a meat-only diet as far back as 2018. A recent review of research into nutrition misinformation on social media found that the most diet information is shared on Instagram and YouTube, and that a lot of it is nonsense. So much so that the authors describe it as a "growing public health concern."

What's new is that some of this misinformation comes from the people who now lead America's federal health agencies. In January Kennedy, who leads the Department of Health and Human Services, told a *USA Today* reporter that he was on a carnivore diet. "I only eat meat or fermented foods," he said. He went on to say that the diet had helped him lose "40% of [his] visceral fat within a month."

"Government needs to stop spreading misinformation that natural and saturated fats are bad for you," Food and Drug Administration commissioner Martin Makary argued in a recent podcast interview. The principles of "whole foods and clean meats" are "biblical," he said. The interviewer said that Makary's warnings about pesticides made him want to "avoid all salads and completely miss the organic section in the grocery store."

For the record: There's plenty of evidence that a diet high in saturated fat can increase the risk of heart disease. That's not government misinformation.

The carnivore doctors' suggestion to avoid vegetables is wrong too, says Gabby Headrick, associate director of food and nutrition policy at George Washington University's Institute for Food Safety & Nutrition Security. There's no evidence to suggest that a meat-only diet is good for you. "All of the nutrition science to date strongly identifies a wide array of vegetables ... as being very health-promoting," she adds.

To be fair to the influencers out there, diet is a tricky thing to study. Much of the research into nutrition relies on volunteers to keep detailed and honest food diaries—something that people are generally quite bad at. And the way our bodies respond to foods might

It will come as a surprise to no one that there is plenty of "low-quality content" floating around on social media.

be influenced by our genetics, our microbiomes, the way we prepare or consume those foods, and who knows what else.

Still, it will come as a surprise to no one that there is plenty of what the above study calls "low-quality content" floating around on social media. So it's worth arming ourselves with a good dose of skepticism, especially when we come across posts that mention "miracle foods" or extreme, limited diets.

The truth is that most food is neither good nor bad when eaten in moderation. Diet trends come and go, and for most people, the best reasonable advice is simply to eat a balanced diet low in sugar, salt, and saturated fat. You know—the basics. No matter what that weird upside-down food pyramid implies. To the carnivore influencers, I say: get your misinformation off my broccoli. ■

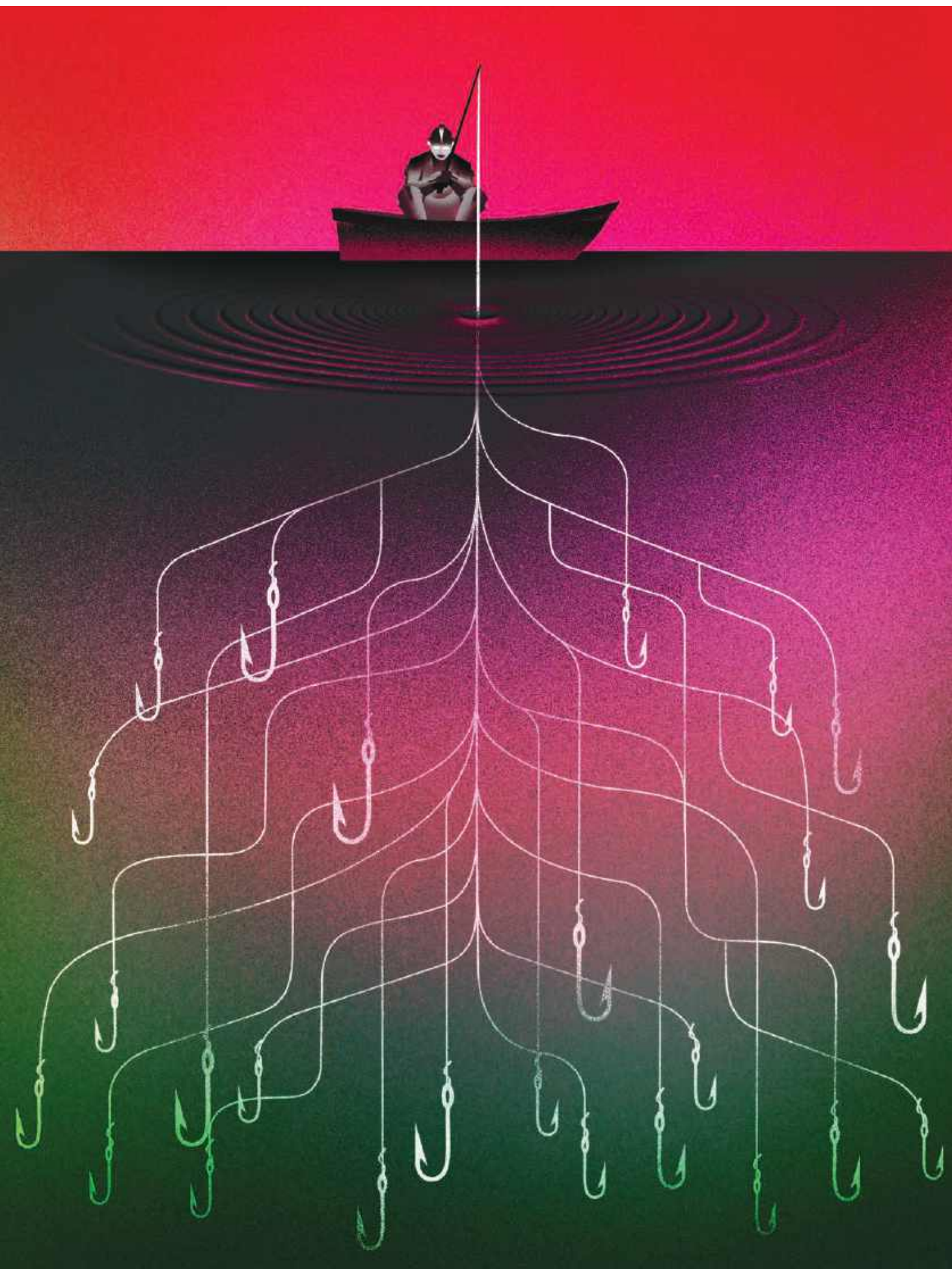


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AI is already swindling victims out of millions of dollars.
That could be just the start.

By Rhiannon Williams
Illustrations by Brian Stauffer

The hackers are using AI too



scams and fraudulent transactions, “many likely aided by AI content.”

At least half of spam email is now generated using LLMs, according to estimates by researchers at Columbia University, the University of Chicago, and Barracuda Networks, who analyzed nearly 500,000 malicious messages collected before and after the launch of ChatGPT. They also found evidence that AI is increasingly being deployed in more sophisticated schemes. They looked at targeted email attacks, which impersonate a trusted figure in order to trick a worker within an organization out of funds or sensitive information. By April 2025, they found, at least 14% of those sorts of focused email attacks were generated using LLMs, up from 7.6% in April 2024.

And the generative AI boom has made it easier and cheaper than ever before to generate not only emails but highly convincing images, videos, and audio. The results are much more realistic than even just a few short years ago, and it takes much less data to generate a fake version of someone’s likeness or voice than it used to.

In one high-profile case, a worker was tricked into transferring \$25 million to criminals via a video call with digital versions of the company’s chief financial officer and other employees.

Criminals aren’t deploying these sorts of deepfakes to prank people or to simply mess around—they’re doing it because it works and because they’re making money out of it, says Henry Ajder, a generative AI expert. “If there’s money to be made and people continue to be fooled by it, they’ll continue to do it,” he says. In one high-profile case reported in 2024, a worker at the British engineering firm Arup was tricked into transferring \$25 million to criminals via a video call with digital versions of the company’s chief financial officer and other employees. That’s likely

only the tip of the iceberg, and the problem posed by convincing deepfakes is only likely to get worse as the technology improves and is more widely adopted.

Criminals’ tactics evolve all the time, and as AI’s capabilities improve, such people are constantly probing how those new capabilities can help them gain an advantage over victims. Billy Leonard, tech leader of Google’s Threat Analysis Group, has been keeping a close eye on changes in the use of AI by potential bad actors (a widely used term in the industry for hackers and others attempting to use computers for criminal purposes). In the latter half of 2024, he and his team noticed prospective criminals using tools like Google Gemini the same way everyday users do—to debug code and automate bits and pieces of their work—as well as tasking it with writing the odd phishing email. By 2025, they had progressed to using AI to help create new pieces of malware and release them into the wild, he says.

The big question now is how far this kind of malware can go. Will it ever become

capture-the-flag competition, a popular cybersecurity game. This sneaky form of jailbreaking led Gemini to hand over information that could have been used to exploit the system. (Google has since adjusted Gemini to deny these kinds of requests.)

But bad actors aren’t just focusing on trying to bend the AI giants’ models to their nefarious ends. Going forward, they’re increasingly likely to adopt open-source AI models, as it’s easier to strip out their safeguards and get them to do malicious things, says Ashley Jess, a former tactical specialist at the US Department of Justice and now a senior intelligence analyst at the cybersecurity company Intel 471. “Those are the ones I think that [bad] actors are going to adopt, because they can jailbreak them and tailor them to what they need,” she says.

The NYU team used two open-source models from OpenAI in its PromptLock experiment, and the researchers found they didn’t even need to resort to jailbreaking techniques to get the model to do what they wanted. They say that makes attacks much easier. Although these kinds of open-source models are designed with an eye to ethical alignment, meaning that their makers do consider certain goals and values in dictating the way they respond to requests, the models don’t have the same kinds of restrictions as their closed-source counterparts, says Meet Udeshi, a PhD student at New York University who worked on the project. “That is what we were trying to test,” he says. “These LLMs claim that they are ethically aligned—can we still misuse them for these purposes? And the answer turned out to be yes.”

It’s possible that criminals have already successfully pulled off covert PromptLock-style attacks and we’ve simply never seen any evidence of them, says Udeshi. If that’s the case, attackers could—in theory—have created a fully autonomous hacking system. But to do that they would have had to overcome the significant barrier that is getting AI models to behave reliably, as well as any inbuilt aversion the models have to being used for malicious purposes—all while evading detection. Which is a pretty high bar indeed.

Productivity tools for hackers

So, what do we know for sure? Some of the best data we have now on how people are attempting to use AI for malicious purposes comes from the big AI companies themselves. And their findings certainly sound alarming, at least at first. In November, Leonard's team at Google released a report that found bad actors were using AI tools (including Google's Gemini) to dynamically alter malware's behavior; for example, it could self-modify to evade detection. The team wrote that it ushered in "a new operational phase of AI abuse."

However, the five malware families the report dug into (including PromptLock) consisted of code that was easily detected and didn't actually do any harm, the cybersecurity writer Kevin Beaumont pointed out on social media. "There's nothing in the report to suggest orgs need to deviate from foundational security programmes—everything worked as it should," he wrote.

It's true that this malware activity is in an early phase, concedes Leonard. Still, he sees value in making these kinds of reports public if it helps security vendors and others build better defenses to prevent more dangerous AI attacks further down the line. "Cliché to say, but sunlight is the best disinfectant," he says. "It doesn't really do us any good to keep it a secret or keep it hidden away. We want people to be able to know about this—we want other security vendors to know about this—so that they can continue to build their own detections."

And it's not just new strains of malware that would-be attackers are experimenting with—they also seem to be using AI to try to automate the process of hacking targets. In November, Anthropic announced it had disrupted a large-scale cyberattack, the first reported case of one executed without "substantial human intervention." Although the company didn't go into much detail about the exact tactics the hackers used, the report's authors said a Chinese state-sponsored group had used its Claude Code assistant to automate up to 90% of what they called a "highly sophisticated espionage campaign."

But, as with the Google findings, there were caveats. A human operator, not AI, selected the targets before tasking Claude with identifying vulnerabilities. And of 30 attempts, only a "handful" were successful. The Anthropic report also found that Claude hallucinated and ended up fabricating data during the campaign, claiming it had obtained credentials it hadn't and "frequently" overstating its findings, so the attackers would have had to carefully validate those results to make sure they were actually true. "This remains an obstacle to fully autonomous cyberattacks," the report's authors wrote.

Existing controls within any reasonably secure organization would stop these attacks, says Gary McGraw, a veteran security expert and cofounder of the Berryville Institute of Machine Learning in Virginia. "None of the malicious-attack part, like the vulnerability exploit... was actually done by the AI—it was just prefabricated tools that do that, and that stuff's been automated for 20 years," he says. "There's nothing novel, creative, or interesting about this attack."

Some are not convinced there's reason to be alarmed. AI hype has led a lot of people in the cybersecurity industry to overestimate models' current abilities, Hutchins says. "They want this idea of unstoppable AIs that can outmaneuver security, so they're forecasting that's where we're going," he says. But "there just isn't any evidence to support that, because the AI capabilities just don't meet any of the requirements."

Indeed, for now criminals mostly seem to be tapping AI to enhance their productivity: using LLMs to write malicious code and phishing lures, to conduct reconnaissance, and for language translation. Jess sees this kind of activity a lot, alongside efforts to sell tools in underground criminal markets. For example, there are phishing kits that compare the click-rate success of various spam campaigns, so criminals can track which campaigns are most effective at any given time. She is seeing a lot of this activity in what could be called the "AI slop landscape" but not as much "widespread adoption from highly technical actors," she says.

"We're entering an era where the barrier to sophisticated cyber operations has fundamentally lowered, and the pace of attacks will accelerate faster than many organizations are prepared for."

Anthropic maintains that the report's findings are a concerning signal of changes ahead. "Tying this many steps of an intrusion campaign together through [AI] agentic orchestration is unprecedented," Jacob Klein, head of threat intelligence at Anthropic, said in a statement. "It turns what has always been a labor-intensive process into something far more scalable. We're entering an era where the barrier to sophisticated cyber operations has fundamentally lowered, and the pace of attacks will accelerate faster than many organizations are prepared for."

But attacks don't need to be sophisticated to be effective. Models that produce "good enough" results allow attackers to go after larger numbers of people than previously possible, says Liz James, a managing security consultant at the cybersecurity company NCC Group. "We're talking about someone who might be using a scattergun approach phishing a whole bunch of people with a model that, if it lands itself on a machine of interest that doesn't have any defenses... can reasonably competently encrypt your hard drive," she says. "You've achieved your objective."



On the defense

For now, researchers are optimistic about our ability to defend against these threats—regardless of whether they are made with AI. “Especially on the malware side, a lot of the defenses and the capabilities and the best practices that we’ve recommended for the past 10-plus years—they all still apply,” says Leonard. The security programs we use to detect standard viruses and attack attempts work; a lot of phishing emails will still get caught in inbox spam filters, for example. These traditional forms of defense will still largely get the job done—at least for now.

And in a neat twist, AI itself is helping to counter security threats more effectively. After all, it is excellent at spotting patterns and correlations. Vasu Jakkal, corporate vice president of Microsoft Security, says that every day, the company processes more than 100 trillion signals flagged by its AI systems as potentially malicious or suspicious events.

Despite the cybersecurity landscape’s constant state of flux, Jess is heartened by how readily defenders are sharing detailed information with each other about attackers’ tactics. Mitre’s Adversarial Threat Landscape for Artificial-Intelligence Systems and the GenAI Security Project from the Open Worldwide Application Security Project are two helpful initiatives documenting how potential criminals are incorporating AI into their attacks and how AI systems are being targeted by them. “We’ve got some really good resources out there for understanding how to protect your own internal AI toolings and understand the threat from AI toolings in the hands of cybercriminals,” she says.

PromptLock, the result of a limited university project, isn’t representative of how an attack would play out in the real world. But if it taught us anything, it’s that the technical capabilities of AI shouldn’t be dismissed. New York University’s Udeshi says he was taken aback at how easily AI

was able to handle a full end-to-end chain of attack, from mapping and working out how to break into a targeted computer system to writing personalized ransom notes to victims: “We expected it would do the initial task very well but it would stumble later on, but we saw high—80% to 90%—success throughout the whole pipeline.”

AI is still evolving rapidly, and today’s systems are already capable of things that would have seemed preposterously out of reach just a few short years ago. That makes it incredibly tough to say with absolute confidence what it will—or won’t—be able to achieve in the future. While researchers are certain that AI-driven attacks are likely to increase in both volume and severity, the forms they could take are unclear. Perhaps the most extreme possibility is that someone makes an AI model capable of creating and automating its own zero-day exploits—highly dangerous cyberattacks that take advantage of previously unknown vulnerabilities in software. But building and hosting such a model—and evading detection—would require billions of dollars in investment, says Hutchins, meaning it would only be in the reach of a wealthy nation-state.

Engin Kirda, a professor at Northeastern University in Boston who specializes in malware detection and analysis, says he wouldn’t be surprised if this was already happening. “I’m sure people are investing in it, but I’m also pretty sure people are already doing it, especially [in] China—they have good AI capabilities,” he says.

It’s a pretty scary possibility. But it’s one that—thankfully—is still only theoretical. A large-scale campaign that is both effective and clearly AI-driven has yet to materialize. What we can say is that generative AI is already significantly lowering the bar for criminals. They’ll keep experimenting with the newest releases and updates and trying to find new ways to trick us into parting with important information and precious cash. For now, all we can do is be careful, remain vigilant, and—for all our sakes—stay on top of those system updates. ■

Rhiannon Williams is a news reporter at [MIT Technology Review](#).



When things go wrong on a decentralized, permissionless blockchain, who is to blame?

By
Jessica Klein

Illustrations by
Kagan McLeod

The bitcoin stops

here

“We’re out of airspace now. We can do whatever we want,” Jean-Paul Thorbjornsen tells me from the pilot’s seat of his Aston Martin helicopter. As we fly over suburbs outside Melbourne, Australia, it’s becoming clear that doing whatever he wants is Thorbjornsen’s MO.

Upper-middle-class homes give way to vineyards, and Thorbjornsen points out our landing spot outside a winery. People visiting for lunch walk outside. “They’re going to ask for a shot now,” he says, used to the attention drawn by his luxury helicopter, emblazoned with the tail letters “BTC” for bitcoin (the price tag of \$5 million in Australian dollars—\$3.5 million in

US dollars today—was perhaps reasonable for someone who claims a previous crypto project made more than AU\$400 million, although he also says those funds were tied up in the company).

Thorbjornsen is a founder of THORChain, a blockchain through which users can swap one cryptocurrency for another and earn fees from making those swaps. THORChain is permissionless, so anyone can use it without getting prior approval from a centralized authority. As a decentralized network, the blockchain is built and run by operators located across the globe, most of whom use pseudonyms.

During its early days, Thorbjornsen

himself hid behind the pseudonym “leena” and used an AI-generated female image as his avatar. But around March 2024, he revealed that he, an Australian man in his mid-30s, with a rural Catholic upbringing, was the mind behind the blockchain. More or less.

If there is a central question around THORChain, it is this: Exactly who is responsible for its operations? Blockchains as decentralized as THORChain are supposed to offer systems that operate outside the centralized leadership of corruptible governments and financial institutions. If a few people have outsize sway over this decentralized network—one of a handful

that operate at such a large scale—it's one more blemish on the legacy of bitcoin's promise, which has already been tarnished by capitalistic political frenzy.

Who's responsible for THORChain matters because in January last year, its users lost more than \$200 million worth of their cryptocurrency in US dollars after THORChain transactions and accounts were frozen by a singular admin override, which users believed was not supposed to be possible given the decentralized structure. When the freeze was lifted, some users raced to pull their money out. The following month, a team of North Korean hackers known as the Lazarus Group used THORChain to move roughly \$1.2 billion of stolen ethereum taken in the infamous hack of the Dubai-based crypto exchange Bybit.

Thorbjornsen explains away THORChain's inability to stop the movement of stolen funds, or prevent a bank run, as a function of its decentralized and permissionless nature. The lack of executive powers means that anyone can use the network for any reason, and arguably there's no one to hold accountable when even the worst goes down.

But when the worst did go down, nearly everyone in the THORChain community, and those paying attention to it in channels like X, pointed their fingers at Thorbjornsen. A lawsuit filed by the THORChain creditors who lost millions in January 2025 names him. A former FBI analyst and North Korea specialist, reflecting on the potential repercussions for helping move stolen funds, told me he wouldn't want to be in Thorbjornsen's shoes.

That's why I traveled to Australia—to see if I could get a handle on where he sees himself and his role in relation to the network he says he founded.

According to Thorbjornsen, he should not be held responsible for either event. THORChain was designed to make decisions based on votes by node operators—people with the computer power, and crypto stake, to run a cluster of servers that process the network's transactions. In those votes, a two-thirds majority rules.

Then there's the permissionless part. *Anyone* can use THORChain to make swaps, which is why it's been a popular way for widely sanctioned entities such as the government of North Korea to move stolen money. This principle goes back to the cypherpunk roots of bitcoin, a currency that operates outside of nation-states' rules. THORChain is designed to avoid geopolitical entanglements; that's what its users like about it.

But there are distinct financial motivations for moving crypto, stolen or not: Node operators earn fees from the funds running through the network. In theory, this incentivizes them to act in the network's best interests—and,



arguably, Thorbjornsen's interests too, as many assume his wealth is tied to the network's profits. (Thorbjornsen says it is not, and that it comes instead from "many sources," including "buying bitcoin back in 2013.")

Now recent events have raised critical questions, not just about Thorbjornsen's outsize role in THORChain's operations, but also about the blockchain's underlying nature.

If THORChain is decentralized, how was a single operator able to freeze its funds a month before the

Bybit hack? Could someone have unilaterally decided to stop the stolen Bybit funds from coming through the network, and chosen not to?

Thorbjornsen insists THORChain is helping realize bitcoin's original purpose of enabling anyone to transact freely outside the reach of purportedly corrupt governments. Yet the network's problems suggest that an alternative financial system might not be much better.

Decentralized?

On February 21, 2025, Bybit CEO Ben Zhou got an alarming call from the company's chief financial officer. About \$1.5 billion US of the exchange's ethereum token, ETH, had been stolen.

The FBI attributed the theft to the Lazarus Group. Typically, criminals will want to convert ETH to bitcoin, which is much easier to convert in turn to cash. Knowing this, the FBI issued a public service announcement on February 26 to "exchanges, bridges ... and other virtual asset service providers," encouraging them to block transactions from accounts related to the hack.

Someone posted that announcement in THORChain's private, invite-only developer channel on Discord, a chat app used widely by software engineers and gamers. While other crypto exchanges and bridges (which facilitate transactions across different blockchains) heeded the warning, THORChain's node operators, developers, and invested insiders debated about whether or not to close the trading gates, a decision requiring a majority vote.

"Civil war is a very strong term, but there was a strong rift in the community," says Boone Wheeler, a US-based crypto enthusiast. In 2021, Wheeler purchased some rune, THORChain's Norse-mythology-themed native token, and he has been paid to write articles about the network to help advertise it. The rift formed "between people who wanted to stay permissionless," he says, "and others who wanted to blacklist the funds."

Wheeler, who says he doesn't run a node or code for THORChain, fell on the side of remaining permissionless. However, others spoke up for blocking the transfers. THORChain, they argued, wasn't decentralized enough to keep those running the network safe from law enforcement—especially when those operators were identifiable by their IP addresses, some based in the US.

"We are not the morality police," someone with the username @Swing_Pop wrote on February 27 in the developer Discord.

THORChain's design includes up to 120 nodes whose operators manage transactions on the network through a voting process. Anyone with hosting hardware can become an operator by funding nodes with rune as collateral, which provides the network with liquidity. Nodes can respond to a transaction by validating it or doing nothing. While individual transactions can't be blocked, trading can be halted by a two-thirds majority vote.

Nodes are also penalized for not participating in voting, which the system labels as "bad behavior." Every 2.5 days, THORChain automatically "churns" nodes out to ensure that no one node gains too much control. The nodes that chose not to validate transactions from the Bybit hack were automatically "churned" out of the system. Thorbjornsen says about 20 or 30 nodes were booted from the network in this way. (Node operators can run multiple nodes, and 120 are rarely running simultaneously; at the time of writing, 55 unique IDs operated 103 nodes.)

By February 27, some node operators were prepared to leave the network altogether. "It's personally getting beyond my risk tolerance," wrote @Runetard in the dev Discord. "Sorry to those of the community that feel otherwise. There are a bunch of us holding all the risk and some are getting ready to walk away."

Even so, the financial incentive for the network operators who remained was significant. As one member of the dev Discord put it earlier that day, \$3 million had been "extracted as commission" from the theft by those operating THORChain. "This is wrong!" they wrote.

Thorbjornsen weighed in on this back-and-forth, during which nodes paused and unpaused the network. He now says there was a right and wrong way for node operators to have behaved. "The correct way of doing things," he says, was for node operators who opposed processing stolen funds to "go offline and ... get [themselves] kicked out" rather than try to police who could use THORChain. He also says that while operators could discuss stopping transactions, "there was simply no design in the code that allowed [them] to do that." However, a since-deleted post from his personal X account on March 3, 2025, stated: "I pushed for all my nodes to

THORChain was designed to make decisions based on votes by node operators, where two-thirds majority rules.

unhalt trading [keep trading]. Threatened to yank bond if they didn't comply. Every single one." (Thorbjornsen says his social media team ran this account in 2025.)

In an Australian 7 News Spotlight documentary last June, Thorbjornsen estimated that THORChain earned between \$5 million and \$10 million from the heist.

When asked in that same documentary if he received any of those fees, he replied, "Not directly." When we spoke, I asked him to elaborate. He said he's "not a recipient" of any funds THORChain sets aside for developers or marketers, nor does he operate any nodes. He was merely speaking generally, he told me: "All crypto holders profit indirectly off economic activity on any chain."

Most important to Thorbjornsen was that, despite "huge pressure to shut the protocol down and stop servicing these swaps," THORChain chugged along. He also notes that the hackers' tactics, moving fast and splitting funds across multiple addresses, made it difficult to identify "bad swaps."

Blockchain experts like Nick Carlsen, a former FBI analyst at the blockchain intelligence company TRM Labs, don't buy this assessment. Other services similar to THORChain were identifying and rejecting these transactions. Had THORChain done the same, Carlsen adds, the stolen funds could have been contained on the Ethereum network, which "would have basically denied North Korea the ability to kick off this laundering process."

And while THORChain touts its decentralization, in "practical applications" like the Lazarus Group's theft, "most de-fi [decentralized finance] protocols are centralized," says Daren Firestone, an attorney who represents crypto industry whistleblowers, citing a 2023 US Treasury Department risk assessment on illicit finance.

With centralization comes culpability, and in these cases, Firestone adds, that comes down to "who profits from [the protocol], so who creates it? But most importantly, who controls it?" Is there someone who can "hit an emergency off switch? ... Direct nodes?"

Many answer these questions with Thorbjornsen's name. "Everyone likes to pass the blame," he says, even though he wasn't alone in building THORChain. "In the end, it all comes back to me anyway."

THORChain origins

According to Thorbjornsen, his story goes like this.

The third of 10 homeschooled children in a "traditional" Catholic household in rural Australia, he spent his days learning math, reading, writing, and studying the Bible. As he got older, he was also responsible for instructing his younger siblings. Wednesday was his day to move the solar panels that powered their home. His parents "installed" a mango and citrus

orchard, more to keep nine boys busy than to reap the produce, he says.

"We lived close to a local airfield," Thorbjornsen says, "and I was always mesmerized by these planes." He joined the Australian air force and studied engineering, but he says the military left him feeling like "a square peg in a round hole." He adds that doing things his own way got him frequently "pulled aside" by superiors.

"That's when I started looking elsewhere," he says, and in 2013, he found bitcoin. It appealed because it existed "outside the system."

During the 2017 crypto bull run, Thorbjornsen raised AU\$12 million in an initial coin offering for CanYa, a decentralized marketplace he cofounded. CanYa ultimately "died" in 2018, and Thorbjornsen pivoted to a "decentralized liquidity" project that would become THORChain.

He worked with a couple of different developer teams, and then, in 2019, he clicked with an American developer, Chad Barraford, at a hackathon in Germany. Barraford (who declined to be interviewed for this story) was an early public face of THORChain.

Around this time, Thorbjornsen says, "a couple of us helped manage the payroll and early investment funds." In a 2020 interview, Kai Ansaari, identified as a THORChain "project lead," wrote, "We're all contributors ... There's no real 'lead,' 'CEO,' 'founder,' etc."

In interviews conducted since he came out from behind the "leena" account in 2024, Thorbjornsen has positioned himself as a key lead. He now says his plan had always been to hand over the account, along with command powers and control of THORChain social media accounts, once the blockchain had matured enough to realize its promise of decentralization.

In 2021, he says, he started this process, first by ceasing to use his own rune to back node operators who didn't have enough to supply their own funding (this can be a way to influence node votes without operating a node yourself). That year, the protocol suffered multiple hacks that resulted in millions of dollars in losses. Nine Realms, a US-incorporated coding company, was brought on to take over THORChain's development. Thorbjornsen says he passed "leena" over to "other community members" and "left crypto" in 2021, selling "a bunch of bitcoin" and buying the helicopter.

Despite this crypto departure, he came back onto the scene with gusto in 2024 when he revealed himself as the operator of the "leena" account. "For many years, I stayed private because I didn't want the attention," he says now.

By early 2024 Thorbjornsen considered the network to be sufficiently decentralized and began advertising

A team of North Korean hackers used THORChain to move roughly **\$1.2 billion** of ethereum stolen from the crypto exchange Bybit.

According to one estimate, THORChain earned between **\$5 million and \$10 million** from the heist.

it publicly. He started regularly posting videos on his TikTok and YouTube channels (“Two sick videos every week,” in the words of one caption) that showed him piloting his helicopter wearing shirts that read “Thor.”

By November 2024, Thorbjornsen, who describes himself as “a bit flamboyant,” was calling himself THORChain’s CEO (“chief energy officer”) and the “master of the memes” in a video from Binance Blockchain Week, an industry conference in Dubai. You need “strong memetic energy,” he says in the video, “to create the community, to create the cult.” Cults imply centralized leadership, and since outing himself as “leena,” Thorbjornsen has publicly appeared to helm the project, with one interviewer deeming him the “THORChain Satoshi” (an allusion to the pseudonymous creator of bitcoin).

One consequence of going public as a face of the protocol: He’s received death threats. “I stirred it up.

kept it off centralized exchanges that would maintain custody of his wallets. He liked THORChain because it claimed to be decentralized and permissionless. “I got into bitcoin because I wanted to have government-less money,” he says.

Many who’d used THORFi lending and saving programs felt similarly. Users I interviewed differentiated THORChain from centralized lending platforms like BlockFi and Celsius, both of which offered extraordinarily high yields before filing for bankruptcy in 2022. “I viewed THORChain as a decentralized system where it was safer,” says Halsey Richartz, a Florida-based THORFi creditor, with “vanilla, 1% passive yield.” Indeed, users I spoke with hadn’t felt the need to monitor their THORFi deposits. “Only your key can be used to withdraw your funds,” the product’s marketing materials insisted. “Savers can withdraw their position to native assets at any time.”

So on January 9, when the “leena” account announced that an admin key had been used to pause withdrawals, it took THORFi users by surprise—and seemed to contradict the marketing messaging around decentralization. “We were told that it was decentralized, and you wake up one morning and read an article that says ‘This guy, JP, had an admin mimic,’” says Treat, referring to Thorbjornsen, “and I’m like, ‘What the fuck is an admin mimic?’”

The admin mimic was one of “a bunch of hard-coded admin keys built into the base code of the system,” says Jonathan Reiter, CEO of the blockchain intelligence company ChainArgos. Those with access to the keys had the ability to make executive decisions on the blockchain—a function many THORChain users didn’t realize could supersede the more democratic decisions made by node votes. These keys had been in THORChain’s code for years and “let you control just about anything,” Reiter adds, including the decision to pause the network during the hacks in 2021 that resulted in a loss of more than \$16 million in assets.

Thorbjornsen says that one key was given to Nine Realms, while another was “shared around the original team.” He told me at least three people had them, adding, “I can neither confirm nor deny having access to that mimic key, because there’s no on-chain registry of the keys.”

Regardless of who had access, Thorbjornsen maintains that the admin mimic mechanism was “widely known within the community, and heavily used throughout THORChain’s history” and that any action taken using the keys “could be largely overruled by the nodes.” Indeed, nodes voted to open withdrawals back up shortly after the admin key was used to pause them. By then, those burned by THORFi argue, the damage had already been

We were told it was decentralized. Then you wake up one morning and read this guy had an admin mimic.

Do I regret it? Who knows?” he said when we met in Australia. “It’s caused a lot of chaos.”

But, he added, “this is the bed that I’ve laid.” When we spoke again, months later, he backtracked, saying he “got sucked into” defending THORChain in 2024 and 2025 because he was involved from 2018 to 2021 and has “a perspective on how the protocol operates.”

Centralized?

Ryan Treat, a retired US Army veteran, woke up one morning in January 2025 to some disturbing activity on X. “My heart sank,” he says. THORFi, the THORChain program he’d used to earn interest on the bitcoin he’d planned to save for his retirement, had frozen all accounts—but that didn’t make sense.

THORFi featured a lending and saving program said to give users “complete control” and self-custody of their crypto, meaning they could withdraw it at any time.

Treat was no crypto amateur. He bought his first bitcoin at around “\$5 apiece,” he says, and had always

done. The executive pause to withdrawals, for some, signaled that something was amiss with THORFi. This led to a bank run after the pause was lifted, until the nodes voted to freeze withdrawals permanently (which Thorbjornsen had suggested in a since-deleted post on X), separating users from crypto worth around \$200 million in US dollars on January 23. THORFi users were then offered a token called TCY (THORChain Yield), which they could claim with the idea that, when its price rose to \$1, they would be made whole. (The price, as of writing, sits at \$0.16.)

Who used the key? Thorbjornsen maintains he didn't do it, but he claims he knows who did and won't say. He says he'd handed over the "leena" account and doesn't "have access to any of the core components of the system," nor has he for "at least three years." He implies that whoever controlled "leena" at the time used the admin key to pause network withdrawals.

A video released by Nine Realms on February 20, 2025, names Thorbjornsen as the activator of the key, stating, "JP ended up pausing lenders and savers, preventing withdrawals so that we can work out ... [a] pay-back plan on them." Thorbjornsen told me the video was "not factual."

Multiple blockchain analysts told me it would be extremely difficult to determine who used the admin mimic key. A month after it was used to pause the network, THORChain said the key had been "removed from the network." At least you can't find the words "admin mimic" in THORChain's base code; I've looked.

Culpability

After the debacle of the THORFi withdrawal freeze, Richartz says, he tried to file reports with the Miami-Dade Police Department, the Florida Department of Law Enforcement, the FBI, the Securities and Exchange Commission, the Commodity Futures Trading Commission, the Federal Trade Commission, and Interpol. When we spoke in November, he still hadn't been able to file with the city of Miami. They told him to try small claims court.

"I was like, no, you don't understand ... a post office box in Switzerland is the company address," he says. "It underscored to me how little law enforcement even knows about these crimes."

As for the Bybit hack, at least one government has retaliated against those who facilitate blockchain projects. Last April German authorities shut down eXch, an exchange suspected of using THORChain to process funds Lazarus stole from Bybit, says Julia Gottesman, cofounder and head of investigations at the cybersecurity group zeroShadow. Australia, she adds, where

Thorbjornsen was based, has been "slow to try to engage with the crypto community, or any regulations."

In response to requests for comment, Australia's Department of Home Affairs wrote that at the end of March 2026, the country's regulatory powers will expand to include "exchanges between the same type of cryptocurrency and transfers between different types." They did not comment on specific investigations.

Crypto and finance experts disagree about whether THORChain engaged in money laundering, defined by the UN as "the processing of criminal proceeds to disguise their illegal origin." But some think it fits the definition.

Shlomit Wagman, a Harvard fellow and former head of Israel's anti-money-laundering agency and its delegation to the Financial Action Task Force (FATF), thinks the Bybit activity was money laundering because THORChain helped the hackers "transfer the funds in an unsupervised manner, completely outside of the scope of regulated or supervised activity."

And by helping with conversions, Carlsen says, THORChain enabled bad actors to turn stolen crypto into usable currency. "People like [Thorbjornsen] have a personal degree of culpability in sustaining the North Korean government," he says. Thorbjornsen counters that THORChain is "open-source infrastructure."

Meanwhile, just days after the hack, Bybit issued a 10% bounty on any funds recovered. As of mid-January this year, between \$100 million and \$500 million worth of those funds in US dollars remain unaccounted for, according to Gottesman of zeroShadow, which was hired by Bybit to recover funds following the hack.

Thorbjornsen hacked

For Thorbjornsen, it's just another day at the casino. That's the comparison he made during his regrettable 7 News Spotlight interview about the Bybit heist, and he repeated it when we met. "You go to a casino,

Ultimately, Thorbjornsen describes himself as a guy who's had a bad year, fending off "threat vectors" left and right.





you play a few games, you expect to lose,” he told me. “When you do actually go to zero, don’t cry.”

Thorbjornsen, it should be noted, has lost at the casino himself.

In September, he says, he got a Telegram message from a friend, inviting him to a Zoom meeting. He accepted and participated in a call with people who had “American voices.”

After the meeting, Thorbjornsen learned that his friend’s Telegram had been hacked. Whoever was

responsible had used the Zoom link to remotely install software on Thorbjornsen’s computer, which “got access to everything”—his email, his crypto wallets, a bitcoin-based retirement fund. It cost him at least \$1.2 million. The blockchain sleuth known as ZachXBT traced the funds and attributed the hack to North Korea.

ZachXBT called it “poetic.”

Ultimately, Thorbjornsen describes himself as a guy who’s had a bad year. He says he had to liquidate his crypto assets because he’s dealing with the fallout of a recent divorce. He also feels he is fending off “threat vectors” left and right. More than once, he asked if I was a private investigator masquerading as a journalist.

Still, his many contradictions don’t inspire confidence. He doesn’t have any more crypto assets, he says. However, the crypto wallet he shared with me so I could pay him back for lunch showed that it contained assets worth more than \$143,000 in US dollars. He now says it wasn’t his wallet. He says he doesn’t control THORChain’s social media, but he’d also told me that he runs the @THORChain X account (later backtracking and saying the account is “delegated” to him for trickier questions).

He insists that he does not care about money. He says that in the robot future, the AI-powered hive mind will become our benevolent overlord, rendering money obsolete, so why give it much thought? Yet as we flew back from the vineyard, he pointed out his new house from the helicopter. It resembles a compound. He says he lives there with his new wife.

Multiple people I spoke with about Thorbjornsen before I met him warned me he wasn’t trustworthy, and he’s undeniably made fishy statements. For instance, the presence of a North Korean flag in a row of decals on the tail of his helicopter suggested an affinity with the country for which THORChain has processed so much crypto. Thorbjornsen insists he had requested the flag of Australia’s Norfolk Island, calling the mix-up “a complete coincidence.” The flags were gone by the time of our flight, apparently removed during a recent repair.

“Money is a meme,” he says. “Money does not exist.” Meme or not, it’s had real repercussions for those who have interacted with THORChain, and those who wound up losing have been looking for someone to blame.

When I spoke with Thorbjornsen again in January, he appeared increasingly concerned that he is that someone. He’s spending more time in Singapore, he told me. Singapore happens to have historically denied extraditions to the US for money-laundering prosecutions. ■

Jessica Klein is a Philadelphia-based freelance journalist covering intimate partner violence, cryptocurrency, and other topics.

Colombian military officials intercepted this 40-foot-long uncrewed fiberglass “narco sub” in the ocean just off Tayrona National Park.

Smooth

Fast, stealthy, and cheap—roboticized, semisubmersible drone boats
The first ones are coming ashore. BY EDUARDO ECHEVERRI LÓPEZ
PHOTOGRAPHS BY CARLOS PARRA RIOS

On a bright morning last April, a surveillance plane operated by the Colombian military spotted a 40-foot-long shark-like silhouette idling in the ocean just off Tayrona National Park. It was, unmistakably, a “narco sub,” a stealthy fiberglass vessel that sails with

its hull almost entirely underwater, used by drug cartels to move cocaine north. The plane’s crew radioed it in, and eventually nearby coast guard boats got the order, routine but urgent: Intercept.

In Cartagena, about 150 miles from the action, Captain Jaime González Zamudio, commander of the regional coast guard

group, sat down at his desk to watch what happened next. On his computer monitor, icons representing his patrol boats raced toward the sub’s coordinates as updates crackled over his radio from the crews at sea. This was all standard; Colombia is the world’s largest producer of cocaine, and its navy has been seizing narco subs for



carrying tons of cocaine could be international law enforcement's nightmare scenario.

criminals

decades. And so the captain was pretty sure what the outcome would be. His crew would catch up to the sub, just a bit of it showing above the water's surface. They'd bring it to heel, board it, and force open the hatch to find two, three, maybe four exhausted men suffocating in a mix of diesel fumes and humidity, and a

cargo compartment holding several tons of cocaine.

The boats caught up to the sub. A crew boarded, forced open the hatch, and confirmed that the vessel was secure. But from that point on, things were different.

First, some unexpected details came over the radio: There was no cocaine on

board. Neither was there a crew, nor a helm, nor even enough room for a person to lie down. Instead, inside the hull the crew found a fuel tank, an autopilot system and control electronics, and a remotely monitored security camera. González Zamudio's crew started sending pictures back to Cartagena: Bolted to the

hull was another camera, as well as two plastic rectangles, each about the size of a cookie sheet—antennas for connecting to Starlink satellite internet.

The authorities towed the boat back to Cartagena, where military techs took a closer look. Weeks later, they came to an unsettling conclusion: This was Colombia's first confirmed *uncrewed* narco sub. It could be operated by remote control, but it was also capable of some degree of autonomous travel. The techs concluded that the sub was likely a prototype built by the Clan del Golfo, a powerful criminal group that operates along the Caribbean coast.

For decades, handmade narco subs have been some of the cocaine trade's most elusive and productive workhorses, ferrying multi-ton loads of illicit drugs from Colombian estuaries toward markets in North America and, increasingly, the rest of the world. Now off-the-shelf technology—Starlink terminals, plug-and-play nautical autopilots, high-resolution video cameras—may be advancing that cat-and-mouse game into a new phase. Uncrewed subs could move more cocaine over longer distances, and they wouldn't put human smugglers at risk of capture. Law enforcement around the world is just beginning to grapple with what the Tayrona sub means for the future—whether it was merely an isolated experiment or the opening move in a new era of autonomous drug smuggling at sea.

Drug traffickers love the ocean. “You can move drug traffic through legal and illegal routes,” says Juan Pablo Serrano, a captain in the Colombian navy and head of the operational coordination center for Orión, a multiagency, multinational counternarcotics effort. The giant container ships at the heart of global commerce offer a favorite approach, Serrano says. Bribe a chain of dockworkers and inspectors, hide a load in one of thousands of cargo boxes, and put it on a totally legal commercial vessel headed to Europe or North America. That route is slow and expensive—involving months of transit and bribes spread across a wide network—but

relatively low risk. “A ship can carry 5,000 containers. Good luck finding the right one,” he says.

Far less legal, but much faster and cheaper, are small, powerful motorboats. Quick to build and cheap to crew, these “go-fasts” top out at just under 50 feet long and can move smaller loads in hours rather than days. But they're also easy for coastal radars and patrols to spot.

Submersibles—or, more accurately, “semisubmersibles”—fit somewhere in the middle. They take more money and engineering to build than an open speedboat, but they buy stealth—even if a bit of the vessel rides at the surface, the bulk stays hidden underwater. That adds another option to a portfolio that smugglers constantly rebalance across three variables: risk, time, and cost. When US and Colombian authorities tightened control over air routes and commercial shipping in the early 1990s, subs became more attractive. The first ones were crude wooden hulls with a fiberglass shell and extra fuel tanks, cobbled together

Brazil. In 2024, police in the Solomon Islands found the first narco sub in the Asia-Pacific region, a semisubmersible probably originating from Colombia on its way to Australia or New Zealand.

If the variables are risk, time, and cost, then the economics of a narco sub are simple. Even if they spend more time on the water than a powerboat, they're less likely to get caught—and a relative bargain to produce. A narco sub might cost between \$1 million and \$2 million to build, but a kilo of cocaine costs just about \$500 to make. “By the time that kilo reaches Europe, it can sell for between \$44,000 and \$55,000,” Serrano says. A typical semisubmersible carries up to three metric tons—cargo worth well over \$160 million at European wholesale prices.

As a result, narco subs are getting more common. Seizures by authorities tripled in the last 20 years, according to Colombia's International Center for Research and Analysis Against Maritime Drug Trafficking (CMCON), and Serrano admits that the Orión alliance has enough

A typical semisubmersible costs under \$2 million to build and can carry three metric tons of cocaine. That's worth over \$160 million in Europe—wholesale.

in mangrove estuaries, hidden from prying eyes. Today's fiberglass semisubmersible designs ride mostly below the surface, relying on diesel engines that can push multi-ton loads for days at a time while presenting little more than a ripple and a hot exhaust pipe to radar and infrared sensors.

Most ferry between South American coasts and handoff points in Central America and Mexico, where allied criminal organizations break up the cargo and slowly funnel it toward the US. But some now go much farther. In 2019, Spanish authorities intercepted a semisubmersible after a 27-day transatlantic voyage from

ships and aircraft to catch only a fraction of what sails.

Until now, though, narco subs have had one major flaw: They depended on people, usually poor fishermen or low-level recruits sealed into stifling compartments for days at a time, steering by GPS and sight, hoping not to be spotted. That made the subs expensive and a risk to drug sellers if captured. Like good capitalists, the Tayrona boat's builders seem to have been trying to obviate labor costs with automation. No crew means more room for drugs or fuel and no sailors to pay—or to get arrested or flip if a mission goes wrong.



Off-the-shelf nautical autopilots, WiFi antennas, Starlink satellite internet connections, and remote cameras are all drug smugglers need to turn semisubmersibles into drone ships.

To find drone subs, international law enforcement will likely have to rely on networks of surveillance systems and, someday, swarms of their own drones.

“If you don’t have a person or people on board, that makes the transoceanic routes much more feasible,” says Henry Shuldiner, a researcher at InSight Crime who has analyzed hundreds of narco-sub cases. It’s one thing, he notes, to persuade someone to spend a day or two going from Colombia to Panama for a big payout; it’s another to ask four people to spend three weeks sealed inside a cramped tube, sleeping, eating, and relieving themselves in the same space. “That’s a hard sell,” Shuldiner says.

An uncrewed sub doesn’t have to race to a rendezvous because its crew can endure only a few days inside. It can move more slowly and stealthily. It can wait out patrols or bad weather, loiter near a meeting point, or take longer and less well-monitored routes. And if something goes wrong—if a military plane appears or navigation fails—its owners can simply scuttle the vessel from afar.

Meanwhile, the basic technology to make all that work is getting more and more affordable, and the potential profit margins are rising. “The rapidly approaching universality of autonomous technology could be a nightmare for the U.S. Coast Guard,” wrote two Coast Guard officers in the US Naval Institute’s journal *Proceedings* in 2021. And as if to prove how good an idea drone narco subs are, the US Marine Corps and the weapons builder Leidos are testing a low-profile uncrewed vessel called the Sea Specter, which they describe as being “inspired” by narco-sub design.

The possibility that drug smugglers are experimenting with autonomous subs isn’t just theoretical. Law enforcement agencies on other smuggling routes have found signs the Tayrona sub isn’t an isolated case. In 2022, Spanish police seized three small submersible drones near Cádiz, on Spain’s southern coast. Two years later, Italian authorities confiscated a remote-controlled minisubmarine they believed was intended for drug runs. “The probability of expansion is high,” says Diego Cánovas, a port and maritime security expert in Spain. Tayrona, the biggest and most technologically advanced uncrewed narco sub found so far, is more likely a preview than an anomaly.



Today, the Tayrona semisubmersible sits on a strip of grass at the ARC Bolívar naval base in Cartagena. It’s exposed to the elements; rain has streaked its paint. To one side lies an older, bulkier narco sub seized a decade ago, a blue cylinder with a clumsy profile. The Tayrona’s hull looks lower, leaner, and more refined.

Up close, it is also unmistakably hand-made. The hull is a dull gray-blue, the fiberglass rough in places, with scrapes and dents from the tow that brought it into port. It has no identifying marks on the exterior—nothing that would tie it to a country, a company, or a port. On the upper surface sit the two Starlink antennas, painted over in the same gray-blue to keep them from standing out against the sea.

I climb up a ladder and drop through the small hatch near the stern. Inside, the air is damp and close, the walls beaded with condensation. Small puddles of fuel have collected in the bilge. The vessel has no seating, no helm or steering wheel, and

not enough space to stand up straight or lie down. It’s clear it was never meant to carry people. A technical report by CMCON found that the sub would have enough fuel for a journey of some 800 nautical miles, and the central cargo bay would hold between 1 and 1.5 tons of cocaine.

At the aft end, the machinery compartment is a tangle of hardware: diesel engine, batteries, pumps, and a chaotic bundle of cables feeding an electronics rack. All the core components are still there. Inside that rack, investigators identified a NAC-3 autopilot processor, a commercial unit designed to steer mid-size boats by tying into standard hydraulic pumps, heading sensors, and rudder-feedback systems. They cost about \$2,200 on Amazon.

“These are plug-and-play technologies,” says Wilmar Martínez, a mechatronics professor at the University of America in Bogotá, when I show him pictures of the inside of the sub. “Midcareer mechatronics students could install them.”

For all its advantages, an autonomous drug-smuggling submarine wouldn't be invincible. Even without a crew on board, there are still people in the chain. Every satellite internet terminal—Starlink or not—comes with a billing address, a payment method, and a log of where and when it pings the constellation. Colombian officers have begun to talk about negotiating formal agreements with providers, asking them to alert authorities when a transceiver's movements match known smuggling patterns. Brazil's government has already cut a deal with Starlink to curb criminal use of its service in the Amazon.

The basic playbook for finding a drone sub will look much like the one for crewed semisubmersibles. Aircraft and ships will use radar to pick out small anomalies and infrared cameras to look for the heat of a diesel engine or the turbulence of a wake. That said, it might not work. "If they wind up being smaller, they're going to be darn near impossible to detect," says Michael

is experimenting with uncrewed platforms in counternarcotics patrols. Across the Atlantic, the European Union's European Maritime Safety Agency operates drones for maritime surveillance.

Today, though, the major screens against oceangoing vessels of all kinds are coastal radar networks. Spain operates SIVE to watch over choke points like the Strait of Gibraltar, and in the Pacific, Australia's over-the-horizon radar network, JORN, can spot objects hundreds of miles away, far beyond the range of conventional radar.

Even so, it's not enough to just spot an uncrewed narco sub. Law enforcement also has to stop it—and that will be tricky.

With a crewed vessel, Colombian doctrine says coast guard units should try to hail the boat first with lights, sirens, radio calls, and warning shots. If that fails, interceptor crews sometimes have to jump aboard and force the hatch. Officers worry that future autonomous craft could be wired to sink or even explode if someone gets too close. "If they get destroyed, we may lose

software that runs the vessel or the networks it uses to talk to satellite constellations. At the cutting edge of these countermeasures are electromagnetic pulses designed to fry electronics outright, turning a million-dollar narco sub into a dead hull drifting at sea.

In reality, the tools that might catch a future Tayrona sub are unevenly distributed, politically sensitive, and often experimental. Powerful cyber or electromagnetic tricks are closely guarded secrets; using them in a drug case risks exposing capabilities that militaries would rather reserve for wars. Systems like Australia's JORN radar are tightly held national security assets, their exact performance specs classified, and sharing raw data with countries on the front lines of the cocaine trade would inevitably mean revealing hints as to how they got it. "Just because a capability exists doesn't mean you employ it," Knickerbocker says.

Analysts don't think uncrewed narco subs will reshape the global drug trade, despite the technological leap. Trafficking organizations will still hedge their bets across those three variables, hiding cocaine in shipping containers, dissolving it into liquids and paints, racing it north in fast boats. "I don't think this is revolutionary," Shuldiner says. "But it's a great example of how resilient cocaine traffickers are, and how they're continuously one step ahead of authorities."

There's still that chance, though, that everything international law enforcement agencies know about drug smuggling is about to change. González Zamudio says he keeps getting requests from foreign navies, coast guards, and security agencies to come see the Tayrona sub. He greets their delegations, takes them out to the strip of grass on the base, and walks them around it, gives them tours. It has become a kind of pilgrimage. Everyone who makes it worries that the next time a narco sub appears near a distant coastline, they'll board it as usual, force the hatch—and find it full of cocaine and gadgets, but without a single human occupant. And no one knows what happens after that. ■

Eduardo Echeverri López is a journalist based in Colombia.

Autonomous drug subs are “a great example of how resilient cocaine traffickers are, and how they’re continuously one step ahead of authorities,” says one researcher.

Knickerbocker, a former US Navy officer who advises defense tech firms.

Even worse, navies already act on only a fraction of their intelligence leads because they don't have enough ships and aircraft. The answer, Knickerbocker argues, is “robot on robot.” Navies and coast guards will need swarms of their own small, relatively cheap uncrewed systems—surface vessels, underwater gliders, and long-endurance aerial vehicles that can loiter, sense, and relay data back to human operators. Those experiments have already begun. The US 4th Fleet, which covers Latin America and the Caribbean,

the evidence,” says Víctor González Badrán, a navy captain and director of CMCON. “That means no seizure and no legal proceedings against that organization.”

That's where electronic warfare enters the picture—radio-frequency jamming, cyber tools, perhaps more exotic options. In the simplest version, jamming means flooding the receiver with noise so that commands from the operator never reach the vessel. Spoofing goes a step further, feeding fake signals so that the sub thinks it's somewhere else or obediently follows a fake set of waypoints. Cyber tools might aim higher up the chain, trying to penetrate the

Hackers tried to destroy Allison Nixon.
They had no idea who they were messing with.

The hunter and the prey

The threats started in spring.

In April 2024, a mysterious someone using the online handles “Waifu” and “Judische” began posting death threats on Telegram and Discord channels aimed at a cybersecurity researcher named Allison Nixon.

“Alison [sic] Nixon is gonna get necklaced with a tire filled with gasoline soon,” wrote Waifu/Judische, both of which are words with offensive connotations. “Decerebration is my fav type of brain death, thats whats gonna happen to alison Nixon.”

It wasn’t long before others piled on. Someone shared AI-generated nudes of Nixon.

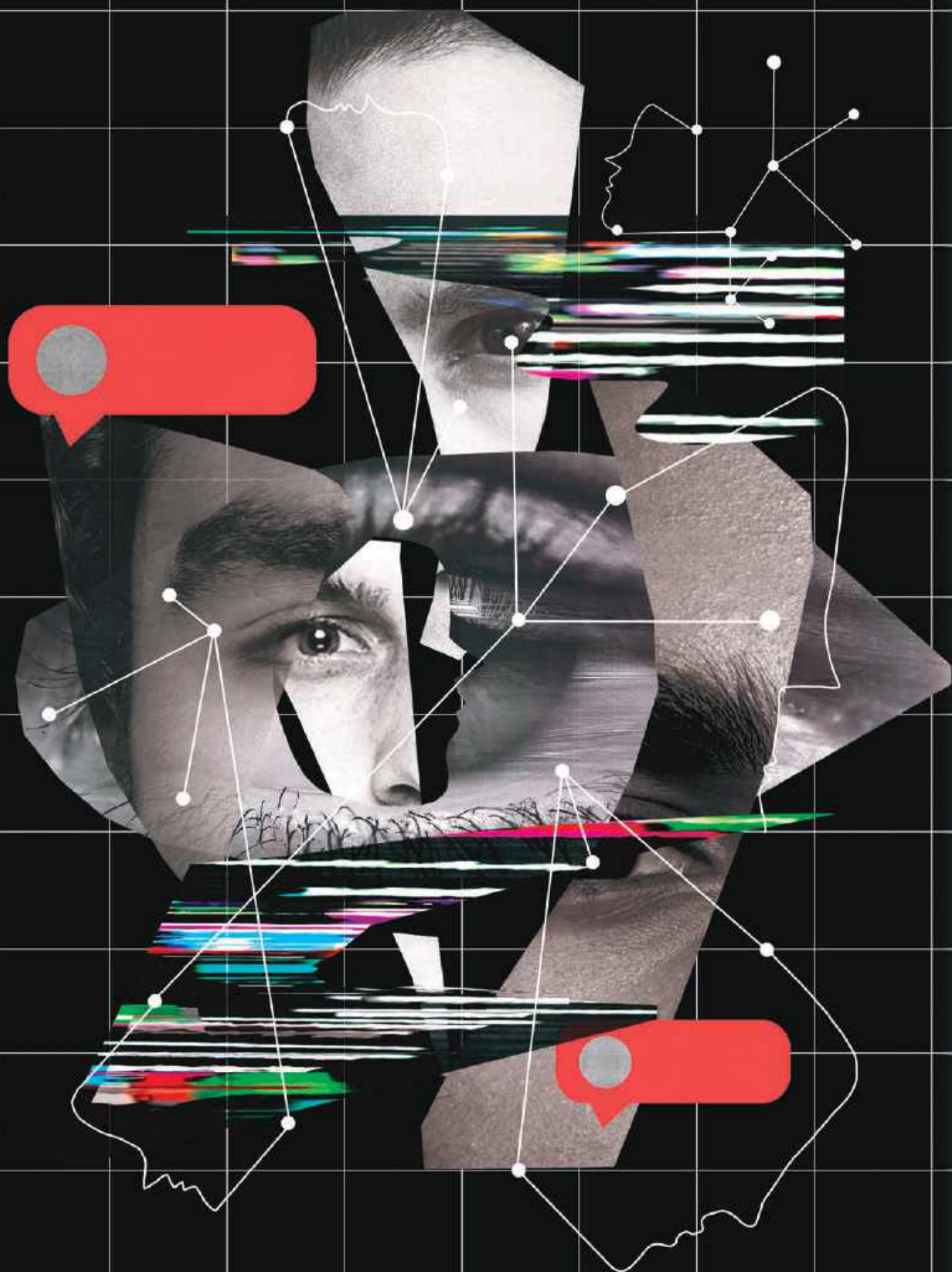
These anonymous personas targeted Nixon because she had become a formidable threat: As chief research officer at the cyber investigations firm Unit 221B, named after Sherlock Holmes’s apartment, she had built a career tracking cybercriminals and helping get them arrested. For years she had lurked quietly in online chat channels or used pseudonyms to engage with perpetrators directly while piecing together clues they’d carelessly drop about themselves and their crimes. This had helped her bring to justice a number of cybercriminals—especially members of a loosely affiliated subculture of anarchic hackers who call themselves the Com.

But members of the Com aren’t just involved in hacking; some of them also engage in offline violence against researchers who track them. This includes bricking (throwing a brick through a victim’s window) and swatting (a dangerous type of hoax that involves reporting a false murder or hostage situation at someone’s home so SWAT teams will swarm it with guns drawn). Members of a Com offshoot known as 764 have been accused of even more violent acts—including animal torture, stabbings, and school shootings—or of inciting others in and outside the Com to commit these crimes.

Nixon started tracking members of the community more than a decade ago, when other researchers and people in law enforcement were largely ignoring them because they were young—many in their teens. Her early attention allowed her to develop strategies for unmasking them.

By
Kim Zetter

Illustrations
Franziska Barczyk



Ryan Brogan, a special agent with the FBI, says Nixon has helped him and colleagues identify and arrest more than two dozen members of the community since 2011, when he first began working with her, and that her skills in exposing them are unparalleled. “If you get on Allison’s and my radar, you’re going [down]. It’s just a matter of time,” he says. “No matter how much digital anonymity and tradecraft you try to apply, you’re done.”

Though she’d done this work for more than a decade, Nixon couldn’t understand why the person behind the Waifu/Judische accounts was suddenly threatening her. She had given media interviews about the Com—most recently on *60 Minutes*—but not about her work unmasking members to get them arrested, so the hostility seemed to come out of the blue. And although she had taken an interest in the Waifu persona in years past for crimes he boasted about committing, he hadn’t been on her radar for a while when the threats began, because she was tracking other targets.

Now Nixon resolved to unmask Waifu/Judische and others responsible for the death threats—and take them down for crimes they admitted to committing. “Prior to them death-threatening me, I had no reason to pay attention to them,” she says.

Com beginnings

Most people have never heard of the Com, but its influence and threat are growing.

It’s an online community comprising loosely affiliated groups of, primarily, teens and twenty-somethings in North America and English-speaking parts of Europe who have become part of what some call a cybercrime youth movement.

Over the last decade, its criminal activities have escalated from simple distributed denial-of-service (DDoS) attacks that disrupt websites to SIM-swapping hacks that hijack a victim’s phone service, as well as crypto theft, ransomware attacks, and corporate data theft. These crimes have affected AT&T, Microsoft, Uber, and others. Com members have also been involved in various forms of sextortion aimed at forcing victims to physically harm themselves or record themselves doing sexually explicit activities. The Com’s impact has also spread beyond the digital realm to kidnapping, beatings, and other violence.

One longtime cybercrime researcher, who asked to remain anonymous because of his work, says the Com is as big a threat in the cyber realm as Russia and China—for one unusual reason.

“There’s only so far that China is willing to go; there’s only so far that Russia or North Korea is willing to go,” he says, referring to international laws and norms, and fears of retaliation, that prevent states from going all out in cyber operations. That doesn’t stop the anarchic Com, he says.

“It is a pretty significant threat, and people tend to ... push it under the rug [because] it’s just a bunch of kids,” he says. “But look at the impact [they have].”

Brogan says the amount of damage they do in terms of monetary losses “can become staggering very quickly.”

There is no single site where Com members congregate; they spread across a number of web forums and Telegram and Discord channels. The group follows a long line of hacking and subculture communities that emerged online over the last two decades, gained notoriety, and then faded or vanished after prominent members were arrested or other factors caused their decline. They differed in motivation and activity, but all emerged from “the same primordial soup,” says Nixon. The Com’s roots can be traced to the Scene, which began as a community of various “warez” groups engaged in pirating computer games, music, and movies.

When Nixon began looking at the Scene, in 2011, its members were hijacking gaming accounts, launching DDoS attacks, and running booter services. (DDoS attacks overwhelm a server or computer with traffic from bot-controlled machines, prevent-

**International laws and norms,
and fears of retaliation,
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That doesn’t stop the anarchic Com.**

ing legitimate traffic from getting through; booters are tools that anyone can rent to launch a DDoS attack against a target of choice.) While they made some money, their primary goal was notoriety.

This changed around 2018. Cryptocurrency values were rising, and the Com—or the Community, as it sometimes called itself—emerged as a subgroup that ultimately took over the Scene. Members began to focus on financial gain—cryptocurrency theft, data theft, and extortion.

The pandemic two years later saw a surge in Com membership that Nixon attributes to social isolation and the forced movement of kids online for schooling. But she believes economic conditions and socialization problems have also driven its growth. Many Com members can’t get jobs because they lack skills or have behavioral issues, she says. A number who have been arrested have had troubled home lives and difficulty adapting to school, and some have shown signs of mental illness. The Com provides camaraderie, support, and an outlet for personal frustrations. Since 2018, it has also offered some a solution to their money problems.

Loose-knit cells have sprouted from the community—Star Fraud, ShinyHunters, Scattered Spider, Lapsus\$—to collaborate on clusters of crime. They usually target high-profile crypto bros



and tech giants and have made millions of dollars from theft and extortion, according to court records.

But dominance, power, and bragging rights are still motivators, even in profit operations, says the cybercrime researcher, which is partly why members target “big whales.”

“There is financial gain,” he says, “but it’s also [sending a message that] I can reach out and touch the people that think they’re untouchable.” In fact, Nixon says, some members of the Com have overwhelming ego-driven motivations that end up conflicting with their financial motives.

“Often their financial schemes fall apart because of their ego, and that phenomenon is also what I’ve made my career on,” she says.

The hacker hunter emerges

Nixon has straight dark hair, wears wire-rimmed glasses, and has a slight build and bookish demeanor that, on first impression, could allow her to pass for a teen herself. She talks about her work in rapid cadences, like someone whose brain is filled with facts that are under pressure to get out, and she exudes a sense of urgency as she tries to make people understand the threat the Com poses. She doesn’t suppress her happiness when someone she’s been tracking gets arrested.

In 2011, when she first began investigating the communities from which the Com emerged, she was working the night shift in the security operations center of the security firm SecureWorks. The center responded to tickets and security alerts emanating from customer networks, but Nixon coveted a position on the company’s counter-threats team, which investigated and published threat-intelligence reports on mostly state-sponsored hacking groups from China and Russia. Without connections or experience, she had no path to investigative work. But Nixon is an intensely curious person, and this created its own path.

Where the threat team focused on the impact hackers had on customer networks—how they broke in, what they stole—Nixon was more interested in their motivations and the personality traits that drove their actions. She assumed there must be online forums where criminal hackers congregated, so she googled “hacking forums” and landed on a site called Hack Forums.

“It was really stupid simple,” she says.

She was surprised to see members openly discussing their crimes there. She reached out to someone on the SecureWorks threat team to see if he was aware of the site, and he dis-

missed it as a place for “script kiddies”—a pejorative term for unskilled hackers.

This was a time when many cybersecurity pros were shifting their focus away from cybercrime to state-sponsored hacking operations, which were more sophisticated and getting a lot of attention. But Nixon likes to zig where others zag, and her colleague’s dismissiveness fueled her interest in the forums. Two other SecureWorks colleagues shared that interest, and the three studied the forums during downtime on their shifts. They focused on trying to identify the people running DDoS booters.

Allison Nixon is chief research officer at the cybersecurity investigations firm Unit 221B, where she tracks cybercriminals and helps bring them to justice.

What Nixon loved about the forums was how accessible they were to a beginner like herself. Threat-intelligence teams require privileged access to a victim's network to investigate breaches. But Nixon could access everything she needed in the public forums, where the hackers seemed to think no one was watching. Because of this, they often made mistakes in operational security, or OPSEC—letting slip little biographical facts such as the city where they lived, a school they attended, or a place they used to work. These details revealed in their chats, combined with other information, could help expose the real identities behind their anonymous masks.

"It was a shock to me that it was relatively easy to figure out who [they were]," she says.

She wasn't bothered by the immature boasting and petty fights that dominated the forums. "A lot of people don't like to do this work of reading chat logs. I realize that this is a very uncommon thing. And maybe my brain is built a little weird that I'm willing to do this," she says. "I have a special talent that I can wade through garbage and it doesn't bother me."

Nixon soon realized that not all the members were script kiddies. Some exhibited real ingenuity and "powerful" skills, she says, but because they were applying these to frivolous purposes—hijacking gamer accounts instead of draining bank accounts—researchers and law enforcement were ignoring them. Nixon began tracking them, suspecting that they would eventually direct their skills at more significant targets—an intuition that proved to be correct. And when they did, she had already amassed a wealth of information about them.

She continued her DDoS research for two years until a turning point in 2013, when the cybersecurity journalist Brian Krebs, who made a career tracking cybercriminals, got swatted.

About a dozen people from the security community worked with Krebs to expose the perpetrator, and Nixon was invited to help. Krebs sent her pieces of the puzzle to investigate, and eventually the group identified the culprit (though it would take two years for him to be arrested). When she was invited to dinner with Krebs and the other investigators, she realized she'd found her people.

"It was an amazing moment for me," she says. "I was like, wow, there's all these like-minded people that just want to help and are doing it just for the love of the game, basically."

Staying one step ahead

It was porn stars who provided Nixon with her next big research focus—one that underscored her skill at spotting Com actors

and criminal trends in their nascent stages, before they emerged as major threats.

In 2018, someone was hijacking the social media accounts of certain adult-film stars and using those accounts to blast out crypto scams to their large follower bases. Nixon couldn't figure out how the hackers had hijacked the social media profiles, but she promised to help the actors regain access to their accounts



if they agreed to show her the private messages the hackers had sent or received during the time they controlled them. These messages led her to a forum where members were talking about how they stole the accounts. The hackers had tricked some of these actors into disclosing the mobile phone numbers of others. Then they used a technique called SIM swapping to reset passwords for social media accounts belonging to those other stars, locking them out.

In SIM swapping, fraudsters get a victim's phone number assigned to a SIM card and phone *they* control, so that calls and messages intended for the victim go to them instead. This includes one-time security codes that sites text to account holders to verify

themselves when accessing their account or changing its password. In some of the cases involving the porn stars, the hackers had manipulated telecom workers into making the SIM swaps for what they thought were legitimate reasons, and in other cases they bribed the workers to make the change. The hackers were then able to alter the password on the actors' social media accounts, lock out the owners, and use the accounts to advertise their crypto scams.

SIM swapping is a powerful technique that can be used to hijack and drain entire cryptocurrency and bank accounts, so Nixon was surprised to see the fraudsters using it for relatively unprofitable schemes. But SIM swapping had rarely been used for financial fraud at that point, and like the earlier hackers Nixon had seen on Hack Forums, the ones hijacking porn star accounts didn't seem to grasp the power of the technique they were using. Nixon suspected that this would change and SIM swapping would soon become a major problem, so she shifted her research focus accordingly. It didn't take long for the fraudsters to pivot as well.

Nixon's work is unique because she engages with the actors in chat spaces to draw out information from them that "would not be otherwise normally available."

Nixon's skill at looking ahead in this way has served her throughout her career. On multiple occasions a hacker or hacking group would catch her attention—for using a novel hacking approach in some minor operation, for example—and she'd begin tracking their online posts and chats in the belief that they'd eventually do something significant with that skill.

They usually did. When they later grabbed headlines with a showy or impactful operation, these hackers would seem to others to have emerged from nowhere, sending researchers and law enforcement scrambling to understand who they were. But Nixon would already have a dossier compiled on them and, in some cases, had unmasked their real identity as well. Lizard Squad was an example of this. The group burst into the headlines in 2014 and 2015 with a series of high-profile DDoS campaigns, but Nixon and colleagues at the job where she worked at the time had already been watching its members as individuals for a while. So the FBI sought their assistance in identifying them.

"The thing about these young hackers is that they ... keep going until they get arrested, but it takes years for them to get arrested," she says. "So a huge aspect of my career is just sitting on this information that has not been actioned [yet]."

It was during the Lizard Squad years that Nixon began developing tools to scrape and record hacker communications online,

though it would be years before she began using these concepts to scrape the Com chatrooms and forums. These channels held a wealth of data that might not seem useful during the nascent stage of a hacker's career but could prove critical later, when law enforcement got around to investigating them; yet the contents were always at risk of being deleted by Com members or getting taken down by law enforcement when it seized websites and chat channels.

Over several years, she scraped and preserved whatever chatrooms she was investigating. But it wasn't until early 2020, when she joined Unit 221B, that she got the chance to scrape the Telegram and Discord channels of the Com. She pulled all of this data together into a searchable platform that other researchers and law enforcement could use. The company hired two former hackers to help build scraping tools and infrastructure for this work; the result is eWitness, a community-driven, invitation-only platform. It was initially seeded only with data Nixon had

collected after she arrived at Unit 221B, but has since been augmented with data that other users of the platform have scraped from Com social spaces as well, some of which doesn't exist in public forums anymore.

Brogan, of the FBI, says it's an incredibly valuable tool, made more so by Nixon's own contributions. Other security firms scrape online criminal spaces as well, but they seldom share the content with outsiders, and Brogan says Nixon's work is unique because she engages with the actors in chat spaces to draw out information from them

that "would not be otherwise normally available."

The preservation project she started when she got to Unit 221B could not have been better timed, because it coincided with the pandemic, the surge in new Com membership, and the emergence of two disturbing Com offshoots, CVLT and 764. She was able to capture their chats as these groups first emerged; after law enforcement arrested leaders of the groups and took control of the servers where their chats were posted, this material went offline.

CVLT—pronounced "cult"—was reportedly founded around 2019 with a focus on sextortion and child sexual abuse material. 764 emerged from CVLT and was spearheaded by a 15-year-old in Texas named Bradley Cadenhead, who named it after the first digits of his zip code. Its focus was extremism and violence.

In 2021, because of what she observed in these groups, Nixon turned her attention to sextortion among Com members.

The type of sextortion they engaged in has its roots in activity that began a decade ago as "fan signing." Hackers would use the threat of doxxing to coerce someone, usually a young female, into writing the hacker's handle on a piece of paper. The hacker would use a photo of it as an avatar on his online accounts—a kind of trophy. Eventually some began blackmailing victims into writing the hacker's handle on their face, breasts, or genitals. With CVLT, this escalated even further; targets were blackmailed into

carving a Com member's name into their skin or engaging in sexually explicit acts while recording or livestreaming themselves.

During the pandemic a surprising number of SIM swappers crossed into child sexual abuse material and sadistic sextortion, according to Nixon. She hates tracking this gruesome activity, but she saw an opportunity to exploit it for good. She had long been frustrated at how leniently judges treated financial fraudsters because of their crimes' seemingly nonviolent nature. But she saw a chance to get harsher sentences for them if she could tie them to their sextortion and began to focus on these crimes.

At this point, Waifu still wasn't on her radar. But that was about to change.

Endgame

Nixon landed in Waifu's crosshairs after he and fellow members of the Com were involved in a large hack involving AT&T customer call records in April 2024.

Waifu's group gained access to dozens of cloud accounts with Snowflake, a company that provides online data storage for customers. One of those customers had more than 50 billion call logs of AT&T wireless subscribers stored in its Snowflake account.

Among the subscriber records were call logs for FBI agents who were AT&T customers. Nixon and other researchers believe the hackers may have been able to identify the phone numbers of agents through other means. Then they may have used a reverse-lookup program to identify the owners of phone numbers that the agents called or that called them and found Nixon's number among them. This is when they began harassing her.

But then they got reckless. They allegedly extorted nearly \$400,000 from AT&T in exchange for promising to delete the call records they'd stolen. Then they tried to re-extort the telecom, threatening on social media to leak the records they claimed to have deleted if it didn't pay more. They tagged the FBI in the post.

"It's like they were begging to be investigated," says Nixon.

The Snowflake breaches and AT&T records theft were grabbing headlines at the time, but Nixon had no idea her number was in the stolen logs or that Waifu/Judische was a prime suspect in the breaches. So she was perplexed when he started taunting and threatening her online.

Over several weeks in May and June, a pattern developed. Waifu or one of his associates would post a threat against her and then post a message online inviting her to talk. She assumes now that they believed she was helping law enforcement investigate the Snowflake breaches and hoped to draw her into a dialogue to extract information from her about what authorities knew. But Nixon wasn't helping the FBI investigate them yet. It was only after she began looking at Waifu for the threats that she became aware of his suspected role in the Snowflake hack.

It wasn't the first time she had studied him, though. Waifu had come to her attention in 2019 when he bragged about framing another Com member for a hoax bomb threat and later talked about his involvement in SIM-swapping operations. He made an impression on her. He clearly had technical skills, but Nixon says he also often appeared immature, impulsive, and emotionally unstable, and he was desperate for attention in his interactions with other members. He bragged about not needing sleep and using Adderall to hack through the night. He was also a bit reckless about protecting personal details. He wrote in private chats to another researcher that he would never get caught because he was good at OPSEC, but he also told the researcher that he lived in Canada—which turned out to be true.

Nixon's process for unmasking Waifu followed a general recipe she used to unmask Com members: She'd draw a large investigative circle around a target and all the personas that communicated with that person online, and then study their

They tried to re-extort the telecom, threatening on social media to leak the records. They tagged the FBI in the post. "It's like they were begging to be investigated," says Nixon.

interactions to narrow the circle to the people with the most significant connections to the target. Some of the best leads came from a target's enemies; she could glean a lot of information about their identity, personality, and activities from what the people they fought with online said about them.

"The enemies and the ex-girlfriends, generally speaking, are the best [for gathering intelligence on a suspect]," she says. "I love them."

While she was doing this, Waifu and his group were reaching out to other security researchers, trying to glean information about Nixon and what she might be investigating. They also attempted to plant false clues with the researchers by dropping the names of other cybercriminals in Canada who could plausibly be Waifu. Nixon had never seen cybercriminals engage in counterintelligence tactics like this.

Amid this subterfuge and confusion, Nixon and another researcher working with her did a lot of consulting and cross-checking with other researchers about the clues they were gathering to ensure they had the right name before they gave it to the FBI.

By July she and the researcher were convinced they had their guy: Connor Riley Moucka, a 25-year-old high school dropout living with his grandfather in Ontario. On October 30, Royal



Canadian Mounted Police converged on Moucka's home and arrested him.

According to an affidavit filed in Canadian court, a plainclothes Canadian police officer visited Moucka's house under some pretense on the afternoon of October 21, nine days before the arrest, to secretly capture a photo of him and compare it with an image US authorities had provided. The officer knocked and rang the bell; Moucka opened the door looking disheveled and told the visitor: "You woke me up, sir." He told the officer his name was Alex; Moucka sometimes used

the alias Alexander Antonin Moucka. Satisfied that the person who answered the door was the person the US was seeking, the officer left. Waifu's online rants against Nixon escalated at this point, as did his attempts at misdirection. She believes the visit to his door spooked him.

Nixon won't say exactly how they unmasked Moucka—only that he made a mistake.

"I don't want to train these people in how to not get caught [by revealing his error]," she says.

The Canadian affidavit against Moucka reveals a number of other violent posts he's alleged to have made online beyond the threats he made against her. Some involve musings about becoming a serial killer or mass-mailing sodium nitrate pills to Black people in Michigan and Ohio; in another, his online persona talks about obtaining firearms to "kill Canadians" and commit "suicide by cop."

Prosecutors, who list Moucka's online aliases as including Waifu, Judische, and two more in the indictment, say he and others extorted at least \$2.5 million from at least three victims whose data they stole from Snowflake accounts. Moucka has been charged with nearly two dozen counts, including conspiracy, unauthorized access to computers, extortion, and wire fraud. He has pleaded not guilty and was extradited to the US last July. His trial is scheduled for October this year, though hacking cases usually end in plea agreements rather than going to trial.

It took months for authorities to arrest Moucka after Nixon and her colleague shared their findings with the authorities, but an alleged associate of his in the Snowflake conspiracy, a US Army soldier named Cameron John Wagenius (Kiberphant0m online), was arrested more quickly.

On November 10, 2024, Nixon and her team found a mistake Wagenius made that helped identify him, and on December 20 he was arrested. Wagenius has already pleaded guilty to two charges around the sale or attempted sale of

confidential phone records and will be sentenced this March.

These days Nixon continues to investigate sextortion among Com members. But she says that remaining members of Waifu's group still taunt and threaten her.

"They are continuing to persist in their nonsense, and they are getting taken out one by one," she says. "And I'm just going to keep doing that until there's no one left on that side." ■

Kim Zetter is a journalist who covers cybersecurity and national security. She is the author of [Countdown to Zero Day](#).

For many people, life in and around Chicago means near-constant surveillance in the name of public safety. Meet some of the people in the city and suburbs engaging with the controversial technologies that make it possible.

By Rod McCullom
Photographs by Akilah Townsend

Inside the panopticon

Early on the morning of September 2, 2024, a Chicago Transit Authority Blue Line train was the scene of a random and horrific mass shooting. Four people were shot and killed on a westbound train as it approached the suburb of Forest Park.

The police swiftly activated a digital dragnet—a surveillance network that connects thousands of cameras in the city.

The process began with a quick review of the transit agency's surveillance cameras, which captured the alleged gunman shooting the victims execution style. Law enforcement followed the suspect, through real-time footage, across the rapid-transit system. Police officials circulated the images to transit staff and to thousands of officers. An officer in the adjacent suburb of Riverdale recognized the suspect from a previous arrest. By the time he was captured at another train station, just 90 minutes after the shooting, authorities already had his name, address, and previous arrest history.

Little of this process would come as much surprise to Chicagoans. The city has tens of thousands of surveillance cameras—up to 45,000, by some estimates. That's among the highest numbers per capita in the US. Chicago boasts one of the largest license plate reader systems in the country, and the ability to access audio and video surveillance from independent agencies such as the Chicago Public Schools, the Chicago Park District, and the public transportation system as well as many residential and commercial security systems such as Ring doorbell cameras.

Law enforcement and security advocates say this vast monitoring system protects public safety and works well. But activists and many residents say it's a surveillance panopticon that creates a chilling effect on behavior and violates guarantees of privacy and free speech.

Black and Latino communities in Chicago have historically been targeted by excessive policing and surveillance, says Lance Williams, a scholar of urban violence at Northeastern Illinois University. That scrutiny has created new problems without delivering the promised safety, he suggests. In order to “solve the problem of crime or violence and make these communities safer,” he says, “you have to deal with structural problems,” such as the shortage of livable-wage jobs, affordable housing, and mental-health services across the city.

Recent years have seen some effective pushback against the surveillance. Until recently, for example, the city was the largest customer of ShotSpotter acoustic sensors, which are designed to detect gunfire and alert police. The system was introduced in a small area on the South Side in 2012. By 2018, an area of about 136 square miles—some 60% of the city—was covered by the acoustic surveillance network.

Critics questioned ShotSpotter's effectiveness and objected that the sensors were installed largely in Black and Latino neighborhoods. Those critiques gained urgency with the fatal shooting in March 2021 of a 13-year-old, Adam Toledo, by police responding to a ShotSpotter alert.

The tragedy became the touchstone of the #StopShotSpotter protest movement and one of the major issues in Brandon Johnson's successful mayoral campaign in 2023. When he reached office, Johnson followed through, ending the city's contract with SoundThinking, the San Francisco Bay Area company behind ShotSpotter. In total, it's estimated the city paid more than \$53 million for the system.

In response to a request for comment, SoundThinking said that ShotSpotter enables law enforcement “to reach the scene faster, render aid to victims, and locate evidence more effectively.” It stated the company “plays no part in the selection of deployment areas” but added: “We believe communities experiencing the highest levels of gun violence deserve the same rapid emergency response as any other neighborhood.”

While there has been successful resistance to police surveillance in the nation's third-largest city, there are also countervailing forces: governments and officials in Chicago and the surrounding suburbs are moving to expand the use of surveillance, also in response to public pressure. Even the victory against acoustic surveillance might be short-lived. Early last year, the city issued a request for proposals for gun violence detection technology.

Many people in and around Chicago—digital privacy and surveillance activists, defense attorneys, law enforcement officials, and ordinary citizens—are part of this push and pull. Here are some of their stories.



"I don't think there is another city in the US that has as an extensive and integrated camera network as Chicago has," former Homeland Security Secretary Michael Chertoff said in 2010.

Alejandro Ruizesparza and Freddy Martinez

Cofounders, Lucy Parsons Labs

Oak Park, a quiet suburb at Chicago's western border, is the birthplace of Ernest Hemingway. It includes the world's largest collection of Frank Lloyd Wright-designed buildings and homes.

Until recently, the village of Oak Park was also the center of a three-year-long campaign against an unwelcome addition to its manicured lawns and Prairie-style architecture: automated license plate readers from a company called Flock Safety. These are high-speed cameras that automatically scan license plates to look for stolen or wanted vehicles, or for drivers with outstanding warrants.

An Oak Park group called Freedom to Thrive—made up of parents, activists, lawyers, data scientists, and many others—suspected that this technology was not a good or equitable addition to their neighborhood. So the group engaged the Chicago-based nonprofit Lucy Parsons Labs to help navigate the often intimidating process of requesting license plate reader data under the Illinois Freedom of Information Act.

Lucy Parsons Labs, which is named for a turn-of-the-century Chicago labor organizer, investigates technologies such as license plate readers, gunshot detection systems, and police bodycams.

LPL provides digital security and public records training to a variety of groups and is frequently called on to help community members audit and analyze surveillance systems that are targeting their neighborhoods. It's led by two first-generation Mexican-Americans from the city's Southwest Side. Alejandro Ruizesparza has a background in community organizing and data science. Freddy Martinez was also a community organizer and has a background in physics.

The group is now approaching its 10th year, but it was an all-volunteer effort until 2022. That's when LPL received its first unrestricted, multi-year operational grant from a large foundation: the



**“We became
I think that sort of speaks to**

Freddy Martinez (left) and Alejandro Ruizesparza (right) direct Lucy Parsons Labs, a charitable organization focused on digital rights.



Chicago-based John D. and Catherine T. MacArthur Foundation, known worldwide for its so-called “genius grants.” A grant from the Ford Foundation followed the next year.

The additional resources—a significant amount compared with the previous all-volunteer budget, acknowledges Ruizesparza—meant the two cofounders and two volunteers became full-time employees. But the group is determined not to become “too comfortable” and lose its edge. There is a tenacity to Lucy Parsons Labs’ work—a “sense of scrappiness,” they say—because “we did so much of this work with no money.”

One of LPL’s primary strategies is filing extensive FOIA requests for raw data sets of police surveillance. The process can take a while, but it often reveals issues.

In the case of Oak Park, the FOIA requests were just one tool that Freedom to Thrive and LPL used to sort out what was going on. The data revealed that in the first 10 months of operation, the eight Flock license plate readers the town had deployed scanned 3,000,000 plates. But only 42 scans led to an alert—an infinitesimal yield of 0.000014%.

At the same time, the impacts of those few flagged license plates were disproportionate. While Oak Park’s population of about 53,000 is only 19% Black, Black drivers made up 85% of those flagged by the Flock cameras, seemingly amplifying what were already concerning racial disparities in the village’s traffic stops. Flock did not respond to a request for comment.

LPL brings a mix of radical politics and critical theory to its mission. Most surveillance technologies are “largely extensions of the plantation systems,” says Ruizesparza.

The comparison makes sense: Many slaveholding communities required enslaved persons to carry signed

almost de facto experts in navigating the process and the law. some of the DIY punk aesthetic.”

Police say Patrol cars raced to the scene.

documents to leave plantations and wear badges with numbers sewn to their clothing. The group says it aims to empower local communities to push back against biased policing technologies through technical assistance, training, and litigation—and to demystify algorithms and surveillance tools in the process.

“When we talk to people, they realize that you don’t need to know how to run a regression to understand that a technology has negative implications on your life,” says Ruizesparza. “You don’t need to understand how circuits work to understand that you probably shouldn’t have all of these cameras embedded in only Black and brown regions of a city.”

The group came by some of its techniques through experimentation. “When LPL was first getting started, we didn’t really feel like FOIA would have been a good way of getting information. We didn’t know anything about it,” says Martinez. “Along the way, we were very successful in uncovering a lot of surveillance practices.”

One of the covert surveillance practices uncovered by those aggressive FOIA requests, for example, was the Chicago Police Department’s use of “Stingray” equipment, portable surveillance devices deployed to track and monitor mobile phones.

The contentious issue of Oak Park’s license plate readers was finally put to a vote in late August. The village trustees voted 5-2 to terminate the contract with Flock Safety.

Since then, community-based groups from across the country—as far away as California—have contacted LPL to say the Chicago collective’s work has inspired their own efforts, says Martinez: “We became almost de facto experts in navigating the process and the law. I think that sort of speaks to some of the DIY punk aesthetic.”

Brian Strockis

Chief, Oak Brook Police Department

If you drive about 20 miles west of Chicago, you’ll find Oakbrook Center, one of the nation’s leading luxury shopping destinations. The open-air mall includes Neiman-Marcus, Louis Vuitton, and Gucci and attracts high-end shoppers from across the region. It’s also become a destination for retail theft crews that coordinate “smash and grabs” and often escape with thousands of dollars’ worth of inventory that can be quickly sold, such as sunglasses or luxury handbags.

In early December, police say, a Chicago man tried to lead officers on what could have been a dangerous high-speed chase from the mall. Patrol cars raced to the scene. So did a “first responder drone,” built by Flock Safety and deployed by the Oak Brook Police Department.

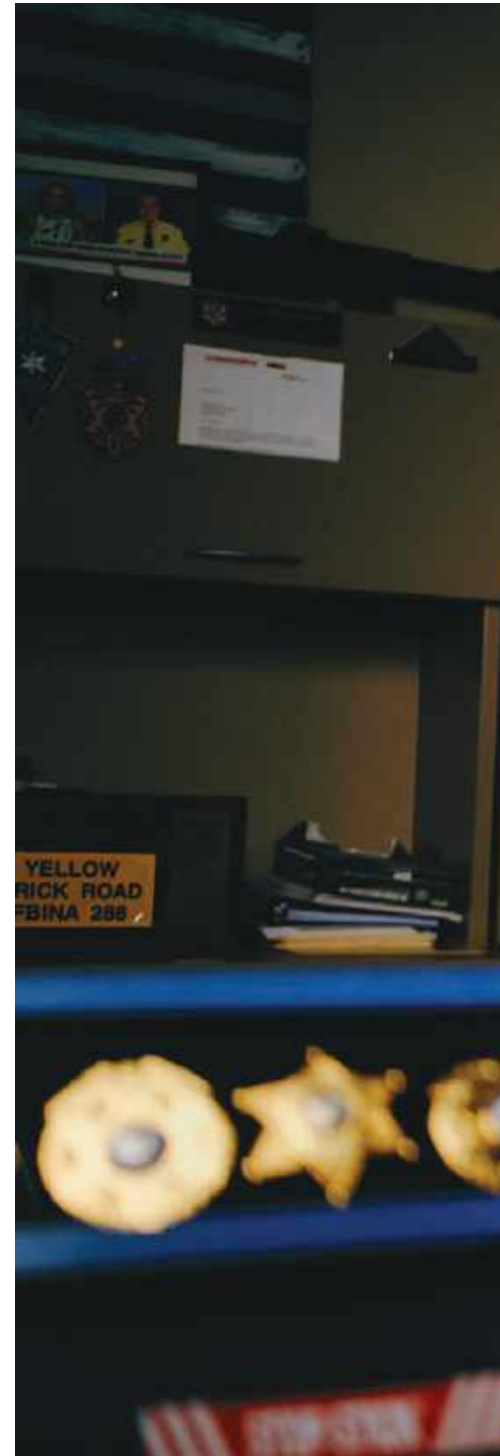
The drone identified the suspect vehicle from the mall parking lot using its license plate reader and snapped high-definition photos that were texted to officers on the ground. The suspect was later tracked to Chicago, where he was arrested.

This was the type of outcome that Brian Strockis, chief of the Oak Brook Police Department, hoped for when he pioneered the “drone as first responder,” or DFR, program in Illinois. A longtime member of the force, he joined the department almost 25 years ago as a patrol officer, worked his way up the brass ladder, and was awarded the top job in 2022.

Oak Brook was the first municipality in Illinois to deploy a drone as a first responder. One of the main reasons, says Strockis, was to reduce the number of high-speed chases, which are potentially dangerous to officers, suspects, and civilians. A drone is also a more effective and cost-efficient way to deal with suspects in fleeing vehicles, says Strockis.

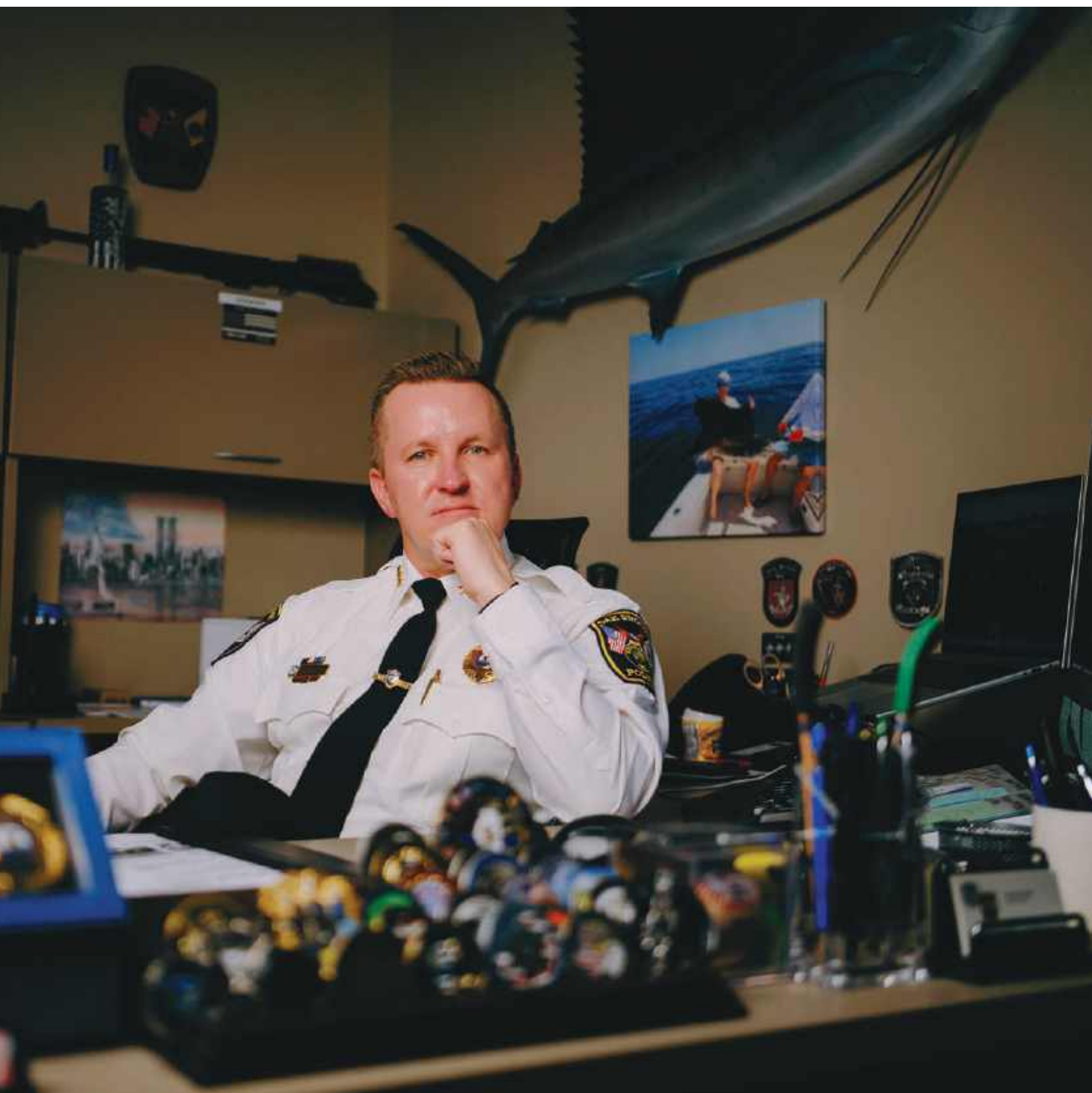
“It’s a force multiplier in that we’re able to do more with less,” says the chief, who spoke with me in his office at Oak Brook’s Village Hall.

The department’s drone autonomously launches from the roof of the building and



Brian Strockis, chief of the Oak Brook Police Department, led the way in introducing drones as first responders in the state of Illinois.

there was the potential for a dangerous high-speed chase.
But the first unit to arrive was a drone.



Urged on by a radio listener, Mark Wallace started organizing against Chicago's red-light and speed cameras, a substantial revenue stream for the city that has been found to disproportionately burden majority Black and Latino areas.

responds to about 10 to 12 service calls per day, at speeds up to 45 miles per hour. It arrives at crime scenes before patrol officers in nine out of every 10 cases.

Next door to Village Hall is the Oak Brook Police Department's real-time crime center, a large room with two video walls that integrates livestreams from the first-responder drone, handheld drones, traffic cameras, license plate readers, and about a thousand private security cameras. When I visited, the two DFR operators demonstrated how the machine can fly itself or be directed to locations from a destination entered on Google Maps. They sent it off to a nearby forest preserve and then directed it to return to the rooftop base, where it docks automatically, changes batteries, and charges. After the demo, one of the drone operators logged the flight, as required by state law.

Strockis says he is aware of the privacy concerns around using this technology but that protections are in place.

For example, the drone cannot be used for random or mass surveillance, he says, because the camera is always pointed straight ahead during flight and does not angle down until it reaches its desired location. The drone's payload does not include facial recognition technology, which is restricted by state law, he says.

The drone video footage is invaluable, he adds, because "you are seeing the events as they're transpiring from an angle that you wouldn't otherwise be privy to."

It's an extra layer of protection for the public as well as for the officers, says the chief: "For every incident that an officer responds to now, you have squad car and bodycam video. You likely have cell-phone video from the public, officers, complainants, from offenders. So adding this element is probably the best video source on a scene that the police are going to anyway."

Mark Wallace

Executive director, Citizens to Abolish Red Light Cameras

Mark Wallace wears several hats. By day he is a real estate investor and mortgage lender. But he is probably best known to many Chicagoans—especially across the city's largely African-American communities on the South and West Sides—as a talk radio host for the station WVON and one of the leading voices against the city's extensive network of red-light and speed cameras.

For the past two decades, city officials have maintained that the cameras—which are officially known as "automated enforcement"—are a crucial safety measure. They are also a substantial revenue stream, generating around \$150 million a year and a total of some \$2.5 billion since they were installed.

"The one thing that the cameras have the ability to do is generate a lot of money," Wallace says. He describes the tickets as a "cash grab" that disproportionately affects Black and Latino communities.

A groundbreaking 2022 analysis by ProPublica found, in fact, that households in majority Black and Latino zip codes were ticketed at much higher rates than others, in part because the cameras in those areas were more likely to be installed near expressway ramps and on wider streets, which encouraged faster speeds. The tickets, which can quickly rack up late fees, were also found to cause more of a financial burden in such communities, the report found.

These were some of the same concerns that many people expressed on the radio and in meetings, Wallace says.

Chicago's automated traffic enforcement began in 2003, and it became the most extensive—and most lucrative—such program in the country. About 300 red-light cameras and 200 speed cameras are set up near schools and parks. The cost of the tickets can quickly double if they are not paid or contested—providing a windfall for the city.

Wallace began his advocacy against the cameras soon after arriving at the radio station in the early 2010s. A younger listener called in and said, he recalls, "that he enjoyed the information that came from WVON but that we didn't do anything." The comment stuck with him, especially in light of WVON's storied history. The station was closely involved in the civil rights movement of the 1960s and broadcast Martin Luther King Jr.'s speeches during his Chicago campaign.

Wallace hoped to change the caller's perception about the station. He had firsthand experience with red-light cameras, having been ticketed himself, and decided to take them on as a cause. He scheduled a meeting at his church for a Friday night, promoting it on his show. "More than 300 people showed up," he remembers, chatting with me in the spacious project studio and office in the basement of his townhouse on the city's South Side. "That said to me there are a lot of people who see this inequity and injustice."

Wallace began using his platform on WVON—*The People's Show*—to mobilize communities around social and economic justice, and many discussions revolved around the automated enforcement program. The cause gained traction after city and state officials were found to have taken thousands of dollars from technology and surveillance companies to make sure their cameras remained on the streets.

Wallace and his group, Citizens to Abolish Red Light Cameras, want to repeal the ordinances authorizing the city's camera programs. That hasn't happened so far, but political pressure from the group paved the way for a Chicago City Council ordinance that required public meetings before any red-light cameras are installed, removed, or relocated. The group hopes for more restrictions for speed cameras, too.

"It was never about me personally. It was about ensuring that we could demonstrate to people that you have power," says Wallace. "If you don't like something, as Barack Obama would say, get a pen and clipboard and go to work to fight to make these changes."





**“There is a need for much broader structural change to how the
use surveillance technology and**

Jonathan Manes litigated cases related to detentions at Guantanamo Bay and the legality of drone strikes before turning his attention to Chicago's implementation of gunshot detection technology.



Jonathan Manes

Senior counsel, MacArthur Justice Center

Derrick Scruggs, a 30-year-old father and licensed armed security guard, was working in the parking lot of an AutoZone on Chicago's Southwest Side on April 19, 2021. That's when he was detained, interrogated, and subjected to a "humiliating body search" by two Chicago police officers, Scruggs later attested. "I was just doing my job when police officers came at me, handcuffed me, and treated me like a criminal—just because I was near a ShotSpotter alert," he says.

The officers found no evidence of a shooting and released Scruggs. But the next day, the police returned and arrested him for an alleged violation related to his security guard paperwork. Prosecutors later dismissed the charges, but he was held in custody overnight and was then fired from his job. "Because of what they did," he says, "I lost my job, couldn't work for months, and got evicted from my apartment."

Scruggs is believed to be among thousands of Chicagoans who've been questioned, detained, or arrested by police because they were near the location of a ShotSpotter alert, according to an analysis by the City of Chicago Office of Inspector General. The case caught the attention of Jonathan Manes, a law professor at Northwestern and senior counsel at the MacArthur Justice Center, a public interest law firm.

Manes previously worked in national security law, but when he joined the justice center about six years ago, he chose to focus squarely on the intersection of civil rights with police surveillance and technology. "My goal was to identify areas that weren't well covered by other civil rights organizations but were a concern for people here in Chicago," he says.

And when he and his colleagues looked into ShotSpotter, they revealed a disturbing problem: The system generated alerts that yielded no evidence of gun-related crimes but were used by police as

a pretext for other actions. There seemed to be "a pattern of people being stopped, detained, questioned, sometimes arrested, in response to a ShotSpotter alert—often resulting in charges that have nothing to do with guns," Manes says.

The system also directed a "massive number of police deployments onto the South and West Sides of the city," Manes says. Those regions are home to most of Chicago's Black and Latino residents. The research showed that 80% of the city's Black population but only 30% of its white population lived in districts covered by the system.

Manes brought Scruggs's case into a lawsuit that he was already developing against the city's use of ShotSpotter. In late 2025, he and his colleagues reached a settlement that prohibits police officers from doing what they did in Scruggs's case—stopping or searching people simply because they are near the location of a gunshot detection alert.

Chicago had already decommissioned ShotSpotter in 2024, but the agreement will cover any future gunshot detection systems. Manes is carefully watching to see what happens next.

Though Manes is pleased with the settlement, he points out that it narrowly focused on how police resources were used *after* the gunshot detection system was operational. "There is a need for much broader structural change to how the city chooses to use surveillance technology and then deploys it," he adds. He supports laws that require disclosure from local officials and law enforcement about what technologies are being proposed and how civil rights could be affected.

More than two dozen jurisdictions nationwide have adopted surveillance transparency laws, including San Francisco, Seattle, Boston, and New York City. But so far Chicago is not on that list. ■

Rod McCullom is a Chicago-based science and technology writer whose focus areas include AI, biometrics, cognition, and the science of crime and violence.

city chooses to
then deploys it."

THE CURIOUS disappearing

A NEW WAVE OF CAR THEFT IS ROCKING THE INDUSTRY—MIXING HIGH TECH WITH OLD-

By CRAIG SILVERMAN

When Sam Zahr first saw the gray Rolls-Royce Dawn convertible with orange interior and orange roof,

he knew he'd found a perfect addition to his fleet. "It was very appealing to our clientele," he told me. As the director of operations at Dream Luxury Rental, he outfits customers in the Detroit area looking to ride in style to a wedding, a graduation, or any other event with high-end vehicles—Rolls-Royces, Lamborghinis, Bentleys, Mercedes G-Wagons, and more.

But before he could rent out the Rolls, Zahr needed to get the car to Detroit from Miami, where he bought it from a used-car dealer.

His team posted the convertible on Central Dispatch, an online marketplace that's popular among car dealers, manufacturers, and owners who want to arrange vehicle shipments. It's not too complicated, at least in theory: A typical listing includes the type of vehicle, zip codes of the origin and destination, dates for pickup and delivery, and the fee. Anyone with a Central Dispatch account can see the job, and an individual carrier or transport broker who wants it can call the number on the listing.

Zahr's team got a call from a transport company that wanted the job. They agreed on the price and scheduled pickup for January 17, 2025. Zahr watched from a few feet away as the car was loaded into an enclosed trailer. He expected the vehicle to arrive in Detroit just a few days later—by January 21.

But it never showed up.



CASE OF THE Lamborghiniis

SCHOOL CHOP SHOP TECHNIQUES TO SNAG LUXURY CARS WHILE THEY'RE IN TRANSPORT.

Illustrations by Richard A. Chance



Zahr called a contact at the transport company to ask what happened.

"He's like, *I don't know what you're talking about.*"

Zahr told me his contact angrily told him they mostly ship Coca-Cola products, not luxury cars. "He was yelling and screaming about it," Zahr said.

Over the years, people have broken into his business to steal cars, or they've rented them out and never come back. But until this day, he'd never had a car simply disappear during shipping. He'd expected no trouble this time around, especially since he'd used Central Dispatch—"a legit platform that everyone uses to transport cars," he said.

"That's the scary part about it, you know?"

Wreaking havoc

Zahr had unwittingly been caught up in a new and growing type of organized criminal enterprise: vehicle transport fraud and theft. Crooks use email phishing, fraudulent paperwork, and other tactics to impersonate legitimate transport companies and get hired to deliver a luxury vehicle. They divert the shipment away from its intended destination and then use a mix of technology, computer skills, and old-school chop shop techniques to erase traces of the vehicle's original ownership and registration.

These vehicles can be retitled and resold in the US or loaded into a shipping container and sent to an overseas buyer. In some cases, the car has been resold or is out of the country by the time the rightful owner even realizes it's missing.

"Criminals have learned that stealing cars via the web portals has become extremely

easy, and when I say easy—it's become seamless," says Steven Yariv, the CEO of Dealers Choice Auto Transport of West Palm Beach, Florida, one of the country's largest luxury-vehicle transport brokers.

Individual cases have received media coverage thanks to the high value of the stolen cars and the fact that some belong to professional athletes and other celebrities. In late 2024, a Lamborghini Huracán belonging to Colorado Rockies third baseman Kris Bryant went missing en route to his home in Las Vegas; R&B singer Ray J told TMZ the same year that two Mercedes Maybachs never arrived in New York as planned; and last fall, NBA Hall of Famer Shaquille O'Neal had a \$180,000 custom Range Rover stolen when the transport company hired to move the vehicle was hacked. "They're saying they think it's probably in Dubai by now, to be honest," an employee of the company that customized the SUV told Shaq in a YouTube video.

But the nationwide epidemic of vehicle transport fraud and theft has remained under the radar, even as it's rocked the industry over the past two years. *MIT Technology Review* identified more than a dozen cases involving high-end vehicles, obtained court records, and spoke to law enforcement, brokers, drivers, and victims in multiple states to reveal how transport fraud is wreaking havoc across the country.

It's challenging to quantify the scale of this type of crime, since there isn't a single entity or association that tracks it. Still, these law enforcement officials and brokers, as well as the country's biggest online car-transport marketplaces, acknowledge that fraud and theft are on the rise.

When I spoke with him in August, Yariv estimated that around 8,000 exotic and high-end cars had been stolen since the spring of 2024, resulting in over \$1 billion in losses. "You're talking 30 cars a day [on] average is gone," he said.

Multiple state and local law enforcement officials told *MIT Technology Review* that the number is plausible. (The FBI did not respond to a request for an interview.)

"It doesn't surprise me," said J.D. Decker, chief of the Nevada Department

of Motor Vehicles' police division and chair of the fraud subcommittee for the American Association of Motor Vehicle Administrators. "It's a huge business."

Data from the National Insurance Crime Bureau (NICB), a nonprofit that works with law enforcement and the insurance industry to investigate insurance fraud and related crimes, provides further evidence of this crime wave. NICB tracks both car theft and cargo theft, a broad category that refers to goods, money, or baggage that is stolen while part of a commercial shipment; the category also covers cases in which a vehicle is stolen via a diverted transport truck or a purloined car is loaded into a shipping container. NICB's statistics about

me that transport thieves often have buyers lined up before they take a car: "When they steal them, they have a plan."

In 2024, Payne spent months trying to locate a Rolls-Royce he'd purchased after it was stolen via transport fraud. It eventually turned up in the Instagram feed of a Mexican pop star, he says. He never got the car back.

The criminals are "gonna keep doing it," he says, "because they make a couple phone calls, make a couple email accounts, and they get a \$400,000 car for free. I mean, it makes them God, you know?"

Out-innovating the industry

The explosion of vehicle transport fraud follows a pattern that has played out across

"Criminals have learned that stealing cars via the web portals has become extremely easy, and when I say easy—it's become seamless."

car theft show that it has declined following an increase during the pandemic—but over the same period cargo theft has dramatically increased, to an estimated \$35 billion annually. The group projected in June that it was expected to rise 22% in 2025.

NICB doesn't break out data for vehicles as opposed to other types of stolen cargo. But Bill Woolf, a regional director for the organization, said an antifraud initiative at the Port of Baltimore experienced a 200% increase from 2023 to 2024 in the number of stolen vehicles recovered. He said the jump could be due to the increased effort to identify stolen cars moving through the port, but he noted that earlier the day we spoke, agents had recovered two high-end stolen vehicles bound for overseas.

"One day, one container—a million dollars," he said.

Many other vehicles are never recovered—perhaps a result of the speed with which they're shipped off or sold. Travis Payne, an exotic-car dealer in Atlanta, told

the economy over the past roughly two decades: A business that once ran on phones, faxes, and personal relationships shifted to online marketplaces that increased efficiency and brought down costs—but the reduction in human-to-human interaction introduced security vulnerabilities that allowed organized and often international fraudsters to enter the industry.

In the case of vehicle transport, the marketplaces are online "load boards" where car owners, dealerships, and manufacturers post about vehicles that need to be shipped from one location to another. Central Dispatch claims to be the largest vehicle load board and says on its website that thousands of vehicles are posted on its platform each day. It's part of Cox Automotive, an industry juggernaut that owns major vehicle auctions, Autotrader, Kelley Blue Book, and other businesses that work with auto dealers, lenders, and buyers.



The system worked pretty well until roughly two years ago, when organized fraud rings began compromising broker and carrier accounts and exploiting loopholes in government licensing to steal loads with surprising ease and alarming frequency.

A theft can start with a phishing email that appears to come from a legitimate load board. The recipient, a broker or carrier, clicks a link in the message, which appears to go to the real site—but logging in sends the victim's username and password to a criminal. The crook logs in as the victim, changes the account's email and phone number to reroute all communications, and begins claiming loads of high-end vehicles. Cox Automotive declined an interview request but said in a statement that the "load board system still works well" and that "fraud impacts a very small portion" of listings.

Criminals also gain access to online marketplaces by exploiting a lax regulatory environment. While a valid US Department of Transportation registration is required

to access online marketplaces, it's not hard for bad actors to register sham transport companies and obtain a USDOT number from the Federal Motor Carrier Safety Administration, the agency that regulates commercial motor vehicles. In other cases, criminals compromise the FMCSA accounts of legitimate companies and change their phone numbers and email addresses in order to impersonate them and steal loads. (USDOT did not respond to a request for comment.)

As Bek Abdullayev, the founder of Super Dispatch, one of Central Dispatch's biggest competitors, explained in an episode of the podcast *Auto Transport Co-Pilot*, "FMCSA [is] authorizing people that are fraudulent companies—people that are not who they say they are." He added that people can "game the system and ... obtain paperwork that makes [them] look like a legitimate company." For example, vehicle carrier insurance can be obtained quickly—if temporarily—by submitting

an online application with fraudulent payment credentials.

The bottom line is that crooks have found myriad ways to present themselves as genuine and permitted vehicle transport brokers and carriers. Once hired to move a vehicle, they often repost the car on a load board using a different fraudulent or compromised account. While this kind of subcontracting, known as "double-brokering," is sometimes used by companies to save money, it can also be used by criminals to hire an unwitting accomplice to deliver the stolen car to their desired location. "They're booking cars and then they're just reposting them and dispatching them out to different routes," says Yariv, the West Palm Beach transport broker.

"A lot of this is cartel operated," says Decker, of the Nevada DMV, who also serves on a vehicle fraud committee for the International Association of Chiefs of Police. "There's so much money in it that it rivals selling drugs."

Even though this problem is becoming increasingly well known, fraudsters continue to steal, largely with impunity. Brokers, auto industry insiders, and law enforcement told *MIT Technology Review* that load boards and the USDOT have been too slow to catch and ban bad actors. (In its statement, Cox Automotive said it has been “dedicated to continually enhancing our processes, technology, and education efforts across the industry to fight fraud.”)

Jake MacDonald, who leads Super Dispatch’s fraud monitoring and investigation efforts, put it bluntly on the podcast with Abdullayev: the reason that fraud is “jumping so much” is that “the industry is slowly moving over to a more technologically advanced position, but it’s so slow that fraud is actually [out-]innovating the industry.”

A Florida sting

As it turns out, the person Zahr’s team hired on Central Dispatch didn’t really work for the transport company.

After securing the job, the fraudster reposted the orange-and-gray Rolls convertible to a load board. And instead of saying that the car needed to go from Miami to the real destination of Detroit, the new job listed an end point of Hallandale Beach, Florida, just 20 or so miles away. It was a classic case of malicious double-brokering: the crooks claimed a load and then reposted it in order to find a new, unsuspecting driver to deliver the car into their possession.

On January 17 of last year, the legitimate driver showed up in a Dodge Ram and loaded the Rolls into an enclosed trailer as Zahr watched.

“The guy came in and looked very professional, and we took a video of him loading the car, taking pictures of everything,” Zahr told me. He never thought to double-check where the driver was headed or which company he worked for.

Not long after a panicked Zahr spoke with his contact at the transport company he thought he was working with, he reported the car as stolen to the Miami police. Detective Ryan Chin was assigned to the case. It fit with a pattern of high-end

auto theft that he and his colleagues had recently been tracking.

“Over the past few weeks, detectives have been made aware of a new method on the rise for vehicles being stolen by utilizing Central Dispatch,” Chin wrote in records obtained by *MIT Technology Review*. “Specific brokers are re-routing the truck drivers upon them picking up vehicles posted for transport and routing them to other locations provided by the broker.”

Chin used Zahr’s photos and video to identify the truck and driver who’d taken the Rolls. By the time police found him, on January 31, the driver had already dropped off Zahr’s Rolls in Hallandale Beach. He’d also picked up and delivered a black Lamborghini Urus and a White Audi R8 for the same client. Each car had been stolen via double-brokering transport fraud, according to court records.

The police department declined to comment or to make Chin available for an interview. But a source with knowledge of the case said the driver was “super cooperative.” (The source asked not to be identified

because they were not authorized to speak to the media, and the driver does not appear to have been identified in court records.)

The driver told police that he had another load to pick up at a dealership in Naples, Florida, later that same day—a second Lamborghini Urus, this one orange. Police later discovered it was supposed to be shipped to California. But the carrier had been hired to bring the car, which retails for about \$250,000, to a mall in nearby Aventura. He told police that he suspected it was going to be delivered to the same person who had booked him for the earlier Rolls, Audi, and Lamborghini deliveries, since “the voice sounds consistent with who [the driver] dealt with prior on the phone.” This drop-off was slated for 4 p.m. at the Waterways Shoppes mall in Aventura.

That was when Chin and a fellow detective, Orlando Rodriguez, decided to set up a sting.

The officers and colleagues across three law enforcement agencies quickly positioned themselves in the Waterways parking

HOT WHEELS

Thieves sometimes catch costly rides belonging to athletes and celebrities.



EXHIBIT A

A **Lamborghini Huracán** belonging to Colorado Rockies third baseman **Kris Bryant** vanished en route to his home in Las Vegas.

The price tag..... **\$340,000**



EXHIBIT B

R&B singer **Ray J** told TMZ in 2024 that **two Mercedes Maybachs** never arrived in New York as planned.

The price tag..... **\$480,000**



EXHIBIT C

NBA Hall of Famer **Shaquille O'Neal's** custom **Range Rover** was stolen in transit between Georgia and Louisiana.

The price tag..... **\$180,000**



lot ahead of the scheduled delivery of the Urus. They watched as, pretty much right on schedule that afternoon, the cooperative driver of the Dodge Ram rolled to a stop in the palm-tree-lined lot, which was surrounded by a kosher supermarket, Japanese and Middle Eastern restaurants, and a physiotherapy clinic.

The driver went inside the trailer and emerged in the orange Lamborghini. He parked it and waited near the vehicle.

Roughly 30 minutes later, a green Rolls-Royce Cullinan (price: \$400,000 and up) arrived with two men and a teenager inside. They got out, opened the trunk, and sat on the tailgate of the vehicle as one man counted cash.

stolen property, grand theft over \$100,000, and organized fraud. (The teenager who arrived in the Rolls was Gevorgyan's son. He was detained and released, according to Richard Cooper, Gevorgyan's attorney.)

As investigators dug into the case, the evidence suggested that this was part of the criminal pattern they'd been following. "I think it's organized," the source told me.

It's something that transport industry insiders have talked about for a while, according to Fred Mills, the owner of Florida-based Advantage Auto Transport, a company that specializes in transporting high-end vehicles. He said there's even a slang term to describe people engaged in transport fraud: the flip-flop mafia.

properly check the paperwork or identity of the person who shows up to transport them.

"'Flip-flop mafia' is an apt nickname because it's surprisingly easy to get a car on a truck and convince somebody that they're a legitimate transport operation when they're not," he says.

Roughly a month after it disappeared, Zahr's Rolls-Royce was recovered by the Miami Beach Police. Video footage obtained by a local TV station showed the gray car with its distinctive orange top being towed into a police garage.

What happens in Vegas

Among the items confiscated from the men in Florida were \$10,796 in cash and a GPS jammer. Law enforcement sources say jammers have become a core piece of technology for modern car thieves—necessary to disable the location tracking provided by GPS navigation systems in most cars. "Once they get the vehicles, they usually park them somewhere [and] put a signal jammer in there or cut out the GPS," the Florida source told me. This buys them time to swap and reprogram the vehicle identification number (VIN), wipe car computers, and reprogram fobs to remove traces of the car's provenance.

No two VINs are the same, and each is assigned to a specific vehicle by the manufacturer. Where they're placed inside a vehicle varies by make and model. The NICB's Woolf says cars also have confidential VINs located in places—including their electronic components—that are supposed to be known only to law enforcement and his organization. But criminals have figured out how to find and change them.

"It's making it more and more difficult for us to identify vehicles as stolen," Woolf says. "Every time we come up with a security measure to prevent the fraudster, they come up with a countermeasure."

All this doesn't even take very much time. "If you know what you're doing, and you steal the car at one o'clock today, you can have it completely done at two o'clock today," says Woolf. A vehicle can be rerouted, reprogrammed, re-VINed, and sometimes even retitled before an owner files a police report.

"Every time we come up with a security measure to prevent the fraudster, they come up with a countermeasure."

"They're doing countersurveillance, looking around," the source told me later. "It's a little out of the ordinary, you know. They kept being fixated [on] where the truck was parked."

The transport driver and the three males who arrived in the Rolls-Royce did not interact. But soon enough, another luxury vehicle, a Bentley Continental GT, which last year retailed for about \$250,000 and up, pulled in. The Bentley driver got out, took the cash from one of the men sitting on the back of the Rolls, and walked over to the transport driver. He handed him \$700 and took the keys to the Lamborghini.

That's when more than a dozen officers swooped in.

"They had nowhere to go," the source told me. "We surrounded them."

The two men in the Rolls were later identified as Arman Gevorgyan and Hrant Nazarian, and the man in the Bentley as Yuriy Korotovsky. The three were arrested and charged with dealing in

It has multiple meanings. One is that the people who show up to transport or accept a vehicle "are out there wearing, you know, flip-flops and slides," Mills says.

The second refers to how fraudsters "flip" from one carrier registration to another as they try to stay ahead of regulators and complaints.

In addition to needing a USDOT number, carriers working across states need an interstate operating authority (commonly known as an MC number) from the USDOT. Both IDs are typically printed on the driver and passenger doors. But the rise of double-brokering—and of fly-by-night and fraudulent carriers—means that drivers increasingly just tape IDs to their door.

Mills says fraudsters will use a USDOT number for 10 or 11 months, racking up violations, and then tape up a new one. "They just wash, rinse, and repeat," he says.

Decker from the Nevada DMV says a lot of high-end vehicles are stolen because dealerships and individual customers don't



That appears to have been the plan in the case of the stolen light-gray 2023 Lamborghini Huracán owned by the Rockies' Kris Bryant.

On September 29, 2024, a carrier hired via a load board arrived at Bryant's home in Cherry Hills, Colorado, to pick up the car. It was supposed to be transported to Bryant's Las Vegas residence within a few days. It never showed up there—but it was in fact in Vegas.

Using Flock traffic cameras, which capture license plate information in areas across the country, Detective Justin Smith of the Cherry Hills Village Police Department tracked the truck and trailer that had picked up the Lambo to Nevada, and he alerted local police.

On October 7, a Las Vegas officer spotted a car matching the Lamborghini's description and pulled it over. The driver said the Huracán had been brought to his auto shop by a man whom the police were able to identify as Dat Viet Tieu. They arrested Tieu later

that same day. In an interview with police, he identified himself as a car broker. He said he was going to resell the Lamborghini and that he had no idea that the car was stolen, according to the arrest report.

Police searched a Jeep Wrangler that Tieu had parked nearby and discovered it had been stolen—and had been re-VINed, retitled, and registered to his wife. Inside the car, police discovered "multiple fraudulent VIN stickers, key fobs to other high-end stolen vehicles, and fictitious placards," their report said.

One of the fake VINs matched the make and model of Bryant's Lamborghini. (Representatives for Bryant and the Rockies did not respond to a request for comment.)

Tieu was released on bail. But after he returned to LVPD headquarters two days later, on October 9, to reclaim his personal property, officers secretly placed him under surveillance with the hope that he'd lead them to one of the other stolen cars matching the key fobs they'd found in the Jeep.

It didn't take long for them to get lucky. A few hours after leaving the police station, Tieu drove to Harry Reid International Airport, where he picked up an unidentified man. They drove to the Caesars Palace parking garage and pulled in near a GMC Sierra. Over the next three hours, the man worked on a laptop inside and outside the vehicle, according to a police report. At one point, he and Tieu connected jumper cables from Tieu's rented Toyota Camry to the Sierra.

"At 2323 hours, the white male adult enters the GMC Sierra, and the vehicle's ignition starts. It was readily apparent the [two men] had successfully re-programmed a key fob to the GMC Sierra," the report said.

An officer watched as the man gave two key fobs to Tieu, who handed the man an unknown amount of cash. Still, the police let the men leave the garage.

The police kept Tieu and his wife under surveillance for more than a week. Then, on October 18, fearing the couple was about to leave town, officers entered Nora's

Italian Restaurant just off the Vegas Strip and took them into custody.

“Obviously, we meet again,” a detective told Tieu.

“I’m not surprised,” Tieu replied.

Police later searched the VIN on the Sierra from the Caesars lot and found that it had been reported stolen in Tremonton, Utah, roughly two weeks earlier. They eventually returned both the Sierra and Kris Bryant’s Lamborghini to their owners.

Tieu pleaded guilty to two felony counts of possession of a stolen vehicle and one count of defacing, altering, substituting, or removing a VIN. In October, he was sentenced to up to one year of probation; if it’s completed successfully, the plea agree-

ment says, the counts of possession of a stolen vehicle will be dismissed. His attorneys, David Z. Chesnoff and Richard A. Schonfeld, said in a statement that they were “pleased” with the court’s decision, “in light of [Tieu’s] acceptance of responsibility.”

when you sign people up?” says Payne. (The spokesperson for Cox Automotive said that it has “a robust verification process for all clients ... who sign up.”)

“If the crap hits the fan, it’s on us as a broker, or it’s on the trucking company, or the clients’ insurance, [which means] that they have no liability in the whole transaction process,” says Mills. “So it definitely frosted a lot of people’s feathers.”

“If the crap hits the fan, it’s on us as a broker, or it’s on the trucking company ... they have no liability in the whole transaction process. So it definitely frosted a lot of people’s feathers.”

Over the last year, Central Dispatch has made changes to further secure its platform. It introduced two-factor authentication for user accounts and started enabling shippers to use its app to track loads in real time, among other measures. It also kicked off an awareness campaign that includes online

educational content and media appearances to communicate that the company takes its responsibilities seriously.

“We’ve removed over 500 accounts already in 2025, and we’ll continue to take any of that aggressive action where it’s needed,” said Lainey Sibble, Central Dispatch’s head of business, in a sponsored episode of the *Auto Remarketing Podcast*. “We also recognize this is not going to happen in a silo. Everyone has a role to play here, and it’s really going to take us all working together in partnership to combat this issue.”

Mills says Central Dispatch got faster at shutting down fraudulent accounts toward the end of last year. But it’s going to take time to fix the industry, he adds: “I compare it to a 15-year opioid addiction. It’s going to take a while to detox the system.”

Yariv, the broker in West Palm Beach, says he has stopped using Central Dispatch and other load boards altogether. “One person has access here, and that’s me. I don’t even log in,” he told me. His team

has gone back to working the phones, as evidenced by the din of voices in the background as we spoke.

“[The fraud is] everywhere. It’s constant,” he said. “The only way it goes away is the dispatch boards have to be shut down—and that’ll never happen.”

It also remains to be seen what kind of accountability there will be for the alleged thieves in Florida. Korotovskyy and Nazarian pleaded not guilty; as of press time, their trials were scheduled to begin in May. (Korotovskyy’s lawyer, Bruce Prober, said in a statement that the case “is an ongoing matter” and his client is “presumed innocent,” while Nazarian’s attorney, Yale Sanford, said in a statement, “As the investigation continues, Mr. Nazarian firmly asserts his innocence.”) A spokesperson with Florida’s Office of the State Attorney emailed a statement: “The circumstances related to these arrests are still a matter of investigation and prosecution. It would be inappropriate to be commenting further.”)

In contrast, Gevorgyan, the third man arrested in the Florida sting, pleaded guilty to four charges.

Yet he maintains his innocence, according to Cooper, his lawyer: “He was pleading [guilty] to get out and go home.” Cooper describes his client as a wealthy Armenian national who runs a jewelry business back home, adding that he was deported to Armenia in September.

Cooper says his client’s “sweetheart” plea deal doesn’t require him to testify or otherwise supply information against his alleged co-conspirators—or to reveal details about how all these luxury cars were mysteriously disappearing across South Florida. Cooper also says prosecutors may have a difficult time convicting the other two men, arguing that police acted prematurely by arresting the trio without first seeing what, if anything, they intended to do with the Lamborghini.

“All they ever had,” Cooper says, “was three schmucks sitting outside of the Lamborghini.” ■

Taking the heat

Many vehicles stolen via transport fraud are never recovered. Experts say the best way to stop this criminal cycle would be to disrupt it before it starts.

That would require significant changes to the way that load boards operate. Bryant’s Lamborghini, Zahr’s and Payne’s Rolls-Royces, and the orange Lamborghini Urus in Florida were all posted for transport on Central Dispatch. Both brokers and shippers argue that the company hasn’t taken enough responsibility for what they characterize as weak oversight.

“You’re Cox Automotive—you’re the biggest car company in the world for dealers—and you’re not doing better screenings

Craig Silverman is an award-winning journalist and the cofounder of *Indicator*, a publication that reports on digital deception.

Three books unpack our infatuation with prediction, and what outsourcing this task to machines might mean for our future.

By Bryan Gardiner
Illustration by Eva Redamonti

Prophet motives

To be human is, fundamentally, to be a forecaster. Occasionally a pretty good one. Trying to see the future, whether through the lens of past experience or the logic of cause and effect, has helped us hunt, avoid *being* hunted, plant crops, forge social bonds, and in general survive in a world that does not prioritize our survival. Indeed, as the tools of divination have changed over the centuries, from tea leaves to data sets, our conviction that the future can be known (and therefore controlled) has only grown stronger.

Today, we are awash in a sea of predictions so vast and unrelenting that most of us barely even register them. As I write this sentence, algorithms on some remote server are busy trying to guess my next

word based on those I have already typed. If you're reading this online, a separate set of algorithms has likely already served you an ad deemed to be one you are most likely to click. (To the die-hards reading this story on paper, congratulations! You have escaped the algorithms ... for now.)

So how did all this happen? People's desire for reliable forecasting is understandable. Still, nobody signed up for an omnipresent, algorithmic oracle mediating every aspect of their life. A trio of new books tries to make sense of our future-focused world—how we got here, and what this change means. Each has its own prescriptions for navigating this new reality, but they all agree on one thing: Predictions are ultimately about power and control.



**The Means of Prediction:
How AI Really Works
(and Who Benefits)**

Maximilian Kasy
UNIVERSITY OF CHICAGO PRESS,
2025



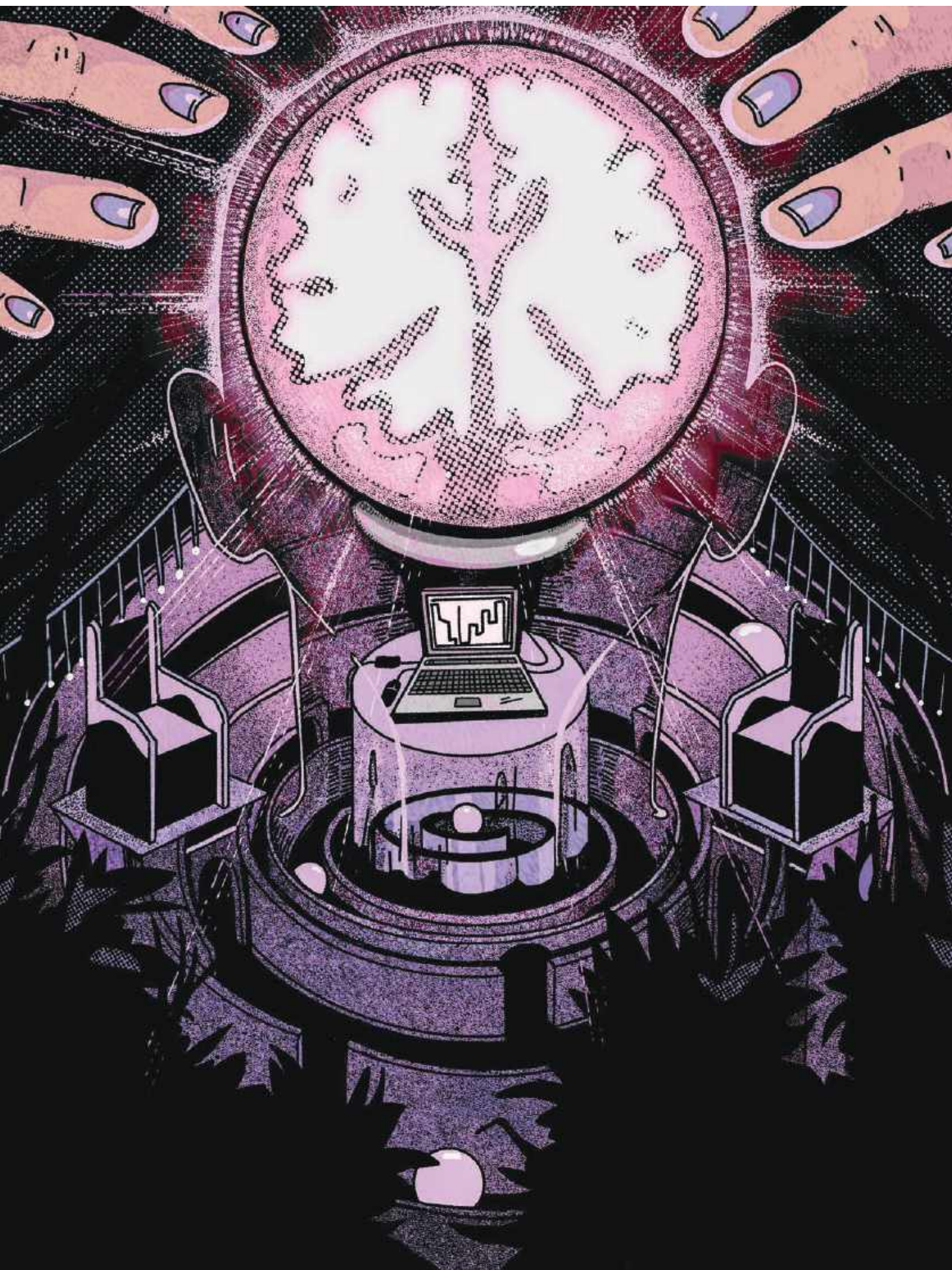
**The Irrational Decision:
How We Gave Computers
the Power to Choose for Us**

Benjamin Recht
PRINCETON UNIVERSITY PRESS,
2026



**Prophecy: Prediction, Power,
and the Fight for the Future,
from Ancient Oracles to AI**

Carissa Véliz
DOUBLEDAY, 2026



In *The Means of Prediction: How AI Really Works (and Who Benefits)*, the Oxford economist Maximilian Kasy explains how most predictions in our lives are based on the statistical analysis of patterns in large, labeled data sets—what’s known in AI circles as supervised learning. Once “trained” on such data sets, algorithms for supervised learning can be presented with all kinds of new information and then deliver their best guess as to some specific future outcome. Will you violate your parole, pay off your mortgage, get promoted if hired, perform well on your college exams, be in your home when it gets bombed? More and more, our lives are shaped (and, yes, occasionally shortened) by a machine’s answer to these questions.

If the thought of a ubiquitous, mostly invisible predictive layer secretly grafted onto your life by a bunch of profit-hungry corporations makes you uneasy ... well, same here. This arrangement is leading to a crueler, blander, more instrumentalized world, one where life’s possibilities are foreclosed, age-old prejudices are entrenched, and everyone’s brain seems to be actively turning into goo. It’s an outcome, according to Kasy, that was entirely predictable.

AI adherents might frame those consequences as “unintended,” or mere problems of optimization and alignment. Kasy, on the other hand, argues that they represent the system working as intended. “If an algorithm selecting what you see on social media promotes outrage, thereby maximizing engagement and ad clicks,” he writes, “that’s because promoting outrage is good for profits from ad sales.” The same holds true for an algorithm that nixes job candidates “who are likely to have family-care responsibilities outside the workplace,” and the ones that “screen out people who are likely to develop chronic health problems or disabilities.” What’s good for a company’s bottom line may not be good for your job-hunting prospects or life expectancy.

Where Kasy differs from other critics is that he doesn’t think working to create less biased, more equitable algorithms will fix any of this. Trying to rebalance the scales

can’t change the fact that predictive algorithms rely on past data that’s often racist, sexist, and flawed in countless other ways. And, he says, the incentives for profit will always trump attempts to eliminate harm. The only way to counter this is with broad democratic control over what Kasy calls “the means of prediction”: data, computational infrastructure, technical expertise, and energy.

A little more than half of *The Means of Prediction* is devoted to explaining how this might be accomplished—through mechanisms including “data trusts” (collective public bodies that make decisions about how to process and use data on behalf of their contributors) and corporate taxing schemes that try to account for the social harm AI inflicts. There’s a lot of economist talk along the way, about how “agents of change” might help achieve “value alignment” in order to “maximize social welfare.” Reasonable, I guess, though a skeptic might point out that Kasy’s rigorous, systematic approach to building new public-serving institutions comes at a time when public trust in institutions has never been lower. Also, there’s the brain goo problem.

To his credit, Kasy is a realist here. He doesn’t presume that any of these proposals will be easy to implement. Or that it will happen overnight, or even in the near future. The troubling question at the end of his book is: Do we have that kind of time?

Reading Kasy’s blueprint for seizing control of the means of prediction raises another pressing question. How on earth did we reach a point where machine-mediated prediction is more or

If the thought of a ubiquitous, mostly invisible predictive layer secretly grafted onto your life by a bunch of profit-hungry corporations makes you uneasy ... well, same here.

less inescapable? *Capitalism*, might be Marx’s pithy response. Fine, as far as it goes, but that doesn’t explain why the same kinds of algorithms that currently model climate change are for some reason also deciding whether you get a new kidney or I get a car loan.

If you ask Benjamin Recht, author of *The Irrational Decision: How We Gave Computers the Power to Choose for Us*, he’d likely tell you our current predicament has a lot to do with the idea and ideology of decision theory—or what economists call rational choice theory. Recht, a polymathic professor in UC Berkeley’s Department of Electrical Engineering and Computer Science, prefers the term “mathematical rationality” to describe the narrow, statistical conception that stoked the desire to build computers, informed how they would eventually work, and influenced the kinds of problems they would be good at solving.

This belief system goes all the way back to the Enlightenment, but in Recht’s telling, it truly took hold at the tail end of World War II. Nothing focuses the mind on risk and quick decision-making like war, and the mathematical models that proved especially useful in the fight against the Axis powers convinced a select group of scientists and statisticians that they might also be a logical basis for designing the first computers. Thus was born the idea of a computer as an ideal rational agent, a machine capable of making optimal decisions by quantifying uncertainty and maximizing utility.

Intuition, experience, and judgment gave way, says Recht, to optimization, game

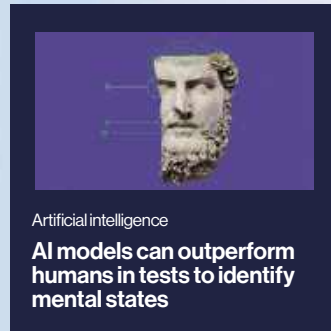
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theory, and statistical prediction. “The core algorithms developed in this period drive the automated decisions of our modern world, whether it be in managing supply chains, scheduling flight times, or placing advertisements on your social media feeds,” he writes. In this optimization-driven reality, “every life decision is posed as if it were a round at an imaginary casino, and every argument can be reduced to costs and benefits, means and ends.”

Today, mathematical rationality (wearing its human skin) is best represented by the likes of the pollster Nate Silver, the Harvard psychologist Steven Pinker, and an assortment of Silicon Valley oligarchs, says Recht. These are people who fundamentally believe the world would be a better place if more of us adopted their analytic mindset and learned to weigh costs and benefits, estimate risks, and plan optimally. In other words, these are people who believe we should all make decisions like computers.

It’s a ridiculous idea for multiple reasons, he says. To name just one, it’s not as if humans couldn’t make evidence-based decisions before automation. “Advances in clean water, antibiotics, and public health brought life expectancy from under 40 in the 1850s to 70 by 1950,” Recht writes. “From the late 1800s to the early 1900s, we had world-changing scientific breakthroughs in physics, including new theories of thermodynamics, quantum mechanics, and relativity.” We also managed to build cars and airplanes without a formal system of rationality and somehow came up with societal innovations like modern democracy without optimal decision theory.

So how might we convince the Pinkers and Silvers of the world that most decisions we face in life are not in fact grist for the unrelenting mill of mathematical rationality? Moreover, how might we demonstrate that (unquantifiable) human intuition, morality, and judgment might be better ways of addressing some of the world’s most important and vexing problems?

One might start by reminding the rationalists that any prediction, computational or otherwise, is really just a *wish*—but one

How might we demonstrate that (unquantifiable) human intuition, morality, and judgment are better ways of addressing some of the world’s most important and vexing problems?

with a powerful tendency to self-fulfill. This idea animates Carissa Véliz’s wonderfully wide-ranging polemic *Prophecy: Prediction, Power, and the Fight for the Future, from Ancient Oracles to AI*.

A philosopher at the University of Oxford, Véliz sees a prediction as “a magnet that bends reality toward itself.” She writes, “When the force of the magnet is strong enough, the prediction becomes the cause of its becoming true.”

Take Gordon Moore. While he doesn’t come up in *Prophecy*, he does figure somewhat prominently in Recht’s history of mathematical rationality. A cofounder of the tech giant Intel, Moore is famous for his 1965 prediction that the density of transistors in integrated circuits would double every two years. “Moore’s Law” turned out to be true, and remains true today, although it does seem to be running out of steam thanks to the physical size limits of the silicon atom.

One story you can tell yourself about Moore’s Law is that Gordon was just a prescient guy. His now-classic 1965 opinion piece “Cramming More Components onto Integrated Circuits,” for *Electronics* magazine, simply extrapolated what computing trends might mean for the future of the semiconductor industry.

Another story—the one I’m guessing Véliz might tell—is that Moore put an informed prediction out into the world, and an entire industry had a collective interest in making it come true. As Recht makes clear, there were and remain obvious financial incentives for companies to make faster and smaller computer chips.

And while the industry has likely spent billions of dollars trying to keep Moore’s Law alive, it’s undoubtedly profited even more from it. Moore’s Law was a helluva strong magnet.

Predictions don’t just have a habit of making themselves come true, says Véliz. They can also distract us from the challenges of the here and now. When an AI boomer promises that artificial general intelligence will be the last problem humanity needs to solve, it not only shapes how we think about AI’s role in our lives; it also shifts our attention away from the very real and very pressing problems of the present day—problems that in many cases AI is causing.

In this sense, the questions around predictions (Who’s making them? Who has the right to make them?) are also fundamentally about power. It’s no accident, Véliz says, that the societies that rely most heavily on prediction are also the ones that tend toward oppression and authoritarianism. Predictions are “veiled prescriptive assertions—they tell us how to act,” she writes. “They are what philosophers call speech *acts*. When we believe a prediction and act in accordance with it, it’s akin to obeying an order.”

As much as tech companies would like us to believe otherwise, technology is not destiny. Humans make it and choose how to use it ... or not use it. Maybe the most appropriate (and human) thing we can do in the face of all the uninvited daily predictions in our lives is to simply defy them. ■

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Radioactive rhinos and rapid eel DNA tests

A range of technologies are being deployed in the fight against wildlife trafficking. By Matthew Ponsford

Every year, poachers shoot hundreds of rhinos, fishing crews haul millions of sharks out of protected seas, and smugglers carry countless animals and plants across borders. This illegal activity is incredibly hard to disrupt, since it's backed by sophisticated criminal networks and the perpetrators know that their chances of being caught are slim. With an annual value of \$20 billion, according to Interpol, it's the world's fourth-most-lucrative criminal enterprise after trafficking in drugs, weapons, and people.

The United Nations seeks to end trafficking in protected species by 2030. But the environmental guardians facing up to these nefarious networks—dispersed alliances of rangers, community groups, and law enforcement officers—have long been ill equipped and underfunded. A recent report by the UN Office on Drugs and Crime found “no reason for confidence” that the 2030 target would be reached.

Still, there is genuine hope that tech could help turn the tide. Tools initially

developed for cities and research facilities are increasingly moving into the planet's wild places, allowing environmental agencies and self-motivated communities in both richer and poorer countries to step up their efforts to detect illegal goods, trace smuggling networks, and prevent poaching at the source.

In December, Interpol announced it had seized record numbers of live animals, thanks in part to a set of sophisticated tools that had helped to expose hidden networks behind trafficking. Its Operation Thunder 2025 coordinated law enforcement agencies from 134 countries and seized 30,000 live animals, from apes to butterflies, using a suite of technologies including digital forensics and AI-driven detection. “The success of Thunder 2025 shows that modern threats demand modern tools,” says José Adrián Sanchez Romero, an operations coordinator at Interpol's environmental security subdirectorate.

Here are five examples of technologies that are arming conservationists and others in the battle to end wildlife crime.



Tagging rhinos

In July, a group of South African researchers announced they had won government approval for one of the most eyebrow-raising attempts to prevent wildlife crime: drilling radioactive substances into the horns of rhinoceroses.

In an effort dubbed the Rhisotope Project, the group worked in 2024 and 2025 to fit 33 rhinos from Limpopo Rhino Orphanage in South Africa with pellets containing low-level radioactive isotopes. The project is supported by the International Atomic Energy Agency.

Blood samples and veterinary exams have shown that the pellets don't affect the health of the rhinos, the rangers, or the surrounding environment. But the isotope emits enough radiation for the horns to be detected by radiation portal monitors, devices that can scan cargo containers and vehicles to detect illicit



sources of radiation. Eleven thousand such monitors are already in operation at airports and shipping terminals worldwide, in addition to thousands of personal monitors worn by border security. In November 2024, Rhisotope tested the system at New York airports and harbors in collaboration with the US Customs and Border Patrol. The group found that border guards could detect an individual horn the team had planted inside a full 40-foot shipping container.

The project was pioneered by James Larkin, director of the radiation and health physics unit at the University of the Witwatersrand in South Africa. Though the country is currently home to 15,000 rhinos, the majority of Africa's total population, poachers have killed 10,000 rhinos there since 2007. In the past, the common approach to deterring poachers was to eliminate the part they're seeking, preemptively cutting off the animal's entire horn. But dehorning requires rhinos to be

sedated for long periods, and it's a stressful and costly process that must be repeated every 18 to 24 months, as rhino horns grow back. The act also renders rhinos less able to protect themselves, and they tend to withdraw from social interactions and competition for mates.

The new approach is far less painful and time-consuming. Each dose costs 21,500 South African rand (about \$1,300) per animal and remains active for five years. Warning signs along perimeter fences make it clear the animals have been tagged, helping to deter poachers.

Larkin, who spent his career as a nuclear safety expert, says he was initially wary when conservationists suggested to him that radioactive substances could help prevent rhino poaching, joking that he didn't want to end up in jail if anyone got hurt. But he changed his mind when he realized there was a dose that would be harmless to bystanders while making the horns both worthless to smugglers and readily

detectable. Poachers will kill a rhino for even a small amount of horn, which can fetch \$60,000 per kilogram as an ingredient for traditional medicines. Adding isotopes, though, renders the horns potentially unsafe to consume, and it's hard for smugglers to reverse: "It's almost impossible to remove isotopes unless you are a skilled radiation protection officer who knows what they are looking for," Larkin says. Even so, he's tight-lipped about the compound the pellets are made from and what they look like: "I don't want to help criminals," he explains.

The South African health agency has now approved Rhisotope to roll out the program across the country. "We have a goal ultimately to treat up to 500 rhinos a year," says Jessica Babich, chief executive of the project. At the same time, the group is working to adapt its approach to other popular poaching targets—elephant tusks and pangolin scales—as well as trafficked plants like cycads.



Scanning signatures

For many exotic pets, from birds to pythons, there are two parallel trades: a legal one in farmed or captive-bred animals and an illicit one in creatures taken from the wild. But faced with a lizard or a parrot, how can law enforcement know its origin story?

In Australia, some conservationists have been trying to follow the numbers. It's very hard to breed the egg-laying mammals known as short-beaked echidnas. US zoos have yielded only 19 echidna babies, or “puggles,” in a century of efforts. So Indonesia's yearly export of dozens of “captive-bred” echidnas has long raised suspicions.

To address the issue, a team at Australia's Taronga Conservation Society, led by Kate Brandis, has developed an x-ray fluorescence (XRF) gun that can analyze elemental signatures in keratin—the stuff of quills, feathers, and hair. Wild echidnas, for instance, forage for a diverse diet of beetle larvae, ants, and grubs, while captive animals tend to be raised on a low-diversity diet of

commercial feed. Each of these dietary histories leaves a record in the mammals' porcupine-like spines, which can be read with high accuracy using a handheld XRF gun. Similar evidence can be found in other species, like cockatoos, pangolins, and turtles, which the team has used to test the device.

There is certainly plenty more to be done: Australia, home to many unique species that live nowhere else on the planet, is a target for collectors from Asia, Europe, and the US. Brandis is targeting some of the species most often trafficked out of the country, including shingleback and blue-tongue lizards. Not long ago, Australian environmental authorities led a trial study at post offices across the country, using the XRF gun alongside AI-equipped parcel scanners, which Brandis's team had trained to recognize concealed species in real time. The trial uncovered more than 100 legally protected lizards that were being shipped out of Australia; a distributor was sentenced to more than three years in jail.

AI in the sky

Commercial fishing, scuba diving, and oil exploration are all prohibited in the Papahānaumokuākea Marine National Monument near Hawaii, an expanse of the Pacific larger than all US national parks combined. It is just one of a number of vast marine protected areas that have emerged in recent years, along with global pacts to conserve 30% of Earth's land and sea.

But establishing these reserves is just one step. Enforcing their protection is another matter. And for many marine reserves—especially those in the Global South—there is no real way to do that, says Ted Schmitt, senior director of conservation at the nonprofit Allen Institute for AI (AI2). Thousands of square kilometers of open ocean is a lot to monitor. Even with satellites scanning the marine areas, the reality until recently was that you had to know what you were looking for: “When you have the vastness of the ocean, you can have analysts who are very well trained, looking for vessels,” he says. Even then, there is little chance of finding wrongdoing without intelligence from the ground.

In 2017, Microsoft cofounder Paul Allen began developing a tool called Skylight to provide analysts with more of that intelligence, using AI to help analyze satellite and ship-tracking data to detect suspicious behavior. The project moved to AI2 after



Allen's death in 2018, and the technology has since been adopted by more than 200 organizations in more than 70 countries. "We're basically monitoring the entire ocean 24-7-365, and surfacing all these vessels," says Schmitt.

To see how coast guards use the system, Schmitt points to a series of arrests in Panama in early 2025. That January, satellites found 16 boats about 200 kilometers off the coast inside the Coiba Ridge marine reserve, which serves as a migratory highway for sharks, rays, and large fish like yellowfin tuna. Skylight's AI algorithms, trained to recognize the signature movements of various types of fishing, detected long-line fishing and requested higher-resolution images of the site from a commercial satellite flying overhead. The images and Skylight's analysis were used by Panama's environmental agency and military, which deployed ships and aircraft to the area, ultimately seizing six vessels and thousands of kilograms of illegally taken fish.

Skylight AI detects around 300,000 vessels per week, according to the company's platform analytics. Stories like Coiba Ridge make clear that AI can benefit partners who are working tirelessly on the ground, says Schmitt: "The Panama case really was one of those 'wow' moments, not because the technology finally proved itself, but because the agencies that needed to operationalize it, and actually take it to a legal finish, did it."

Rapid DNA tests



When the conservation scientist Natalie Schmitt was researching snow leopards in remote areas of Nepal, she worked with people who could point out signs of these elusive big cats—often a pile of droppings. But the results weren't reliable: Leopard scat can easily be confused with the poop of wolves and foxes, which share the same habitat and prey, she explains. What Schmitt wanted was a tool that could identify the animal involved, right on the spot—ideally by finding a way to sequence the DNA in the scat.

While some laboratories can take DNA samples of such material and identify species of interest, they are few and far between in rich countries and usually nonexistent in poorer countries, meaning that this process can be weeks long and involve shipping samples cross-country or across borders. This is a problem not just for field research but for wildlife trafficking enforcement. Imagine a border agent who has just opened a box of shark-like fins or a shipment of live parrots and needs to know whether the particular species is one that can legally be captured and transported. People in this situation don't have weeks to spare.

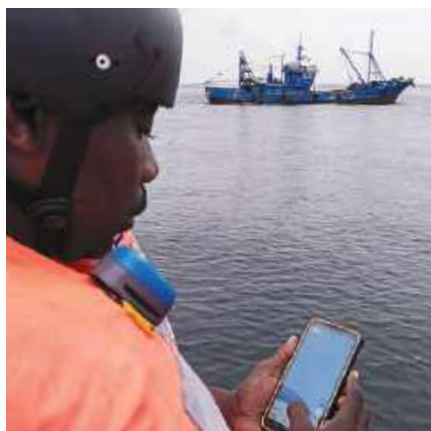
In 2020, Schmitt founded WildTechDNA, which has developed a DNA test that aims to do that work on the fly. The test, which is about as easy and fast as a home pregnancy test, employs a simple

two-step process. First, a new extraction method—"Literally, put the sample in the extraction tube and squeeze 10 times," she says—can cut the time it takes to pull DNA out of a sample from a day to about three minutes. Then, to actually test that DNA, the company took inspiration from the covid pandemic. The researchers found they could use technology similar to rapid at-home tests to identify whether the DNA in question belongs to a specific species: "Our tests use very simple lateral-flow strips to tell you whether a sample belongs to your target species of interest, yes or no."

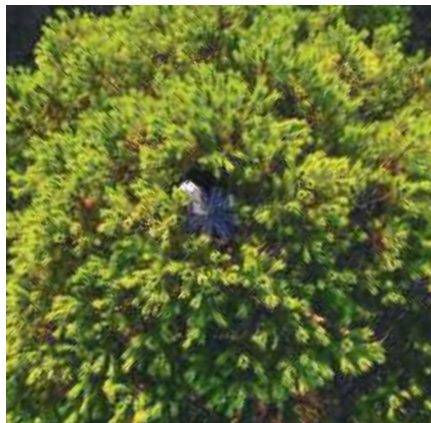
The strips can be tailored to test for a wide range of targets, from big cats to microbes, opening up diverse applications in the wild. They can tell if samples of hair belong to a snow leopard, or if a frog has been infected with the fungi that cause chytridiomycosis, a disease that has devastated amphibians worldwide and wiped out at least 90 species.

WildTechDNA's earliest adopter was the Canadian government, which wanted to detect European eels—a critically endangered species that is effectively impossible to identify by appearance. This confusion has allowed €3 billion of European eels to be smuggled each year, disguised as other eel species. Some of that passes into Canada on its way to suppliers in Japan and China, and in some cases on to Canadian restaurants and consumers. "When a shipment is suspected to contain European eel, they'll randomly sample it and they'll send those samples off to a lab across the country, which will take three weeks," says Schmitt of traditional tracking methods. WildTechDNA developed tests specific to European eels and taught Canadian enforcement officers how to use them, so that they could launch a "nationwide European eel blitz," she says.

In a 2025 campaign, European eels turned up in fewer than 1% of shipments. Schmitt says Canadian authorities have not disclosed details about investigations but are encouraged by the results—significantly below the rates detected using older technologies in 2016, an improvement they attribute to better surveillance.



Listening in



The world's forests are increasingly filled with snooping devices. In addition to affordable camera traps and animal-mounted GPS tags, low-cost solar-powered microphones have proved to be strikingly effective at revealing what's living in some of the planet's most densely inhabited and biodiverse environments.

Rainforest Connection, a non-profit founded by the physicist turned conservation-tech entrepreneur Topher White in 2014, was a pioneer in bioacoustic monitoring for conservation. The group initially repurposed old phones into low-cost monitoring devices but has since developed a standardized device called the Guardian that has now been deployed in more than 600 locations.

Guardians are designed to capture a broad soundscape of the rainforest: "They sit out in the rainforest for long periods of time, up in treetops. They're solar-powered, they can last for years, and we listen to all the sounds continuously and transmit that up to the cloud, where we are then able to analyze it for all sorts of things," says White. From the outset, the aim was to use these devices to pick up immediate threats—"chainsaws, logging trucks, gunshots, things like that," White says—and relay real-time alerts to local partners, including police, Indigenous groups, and local communities that protect the land.

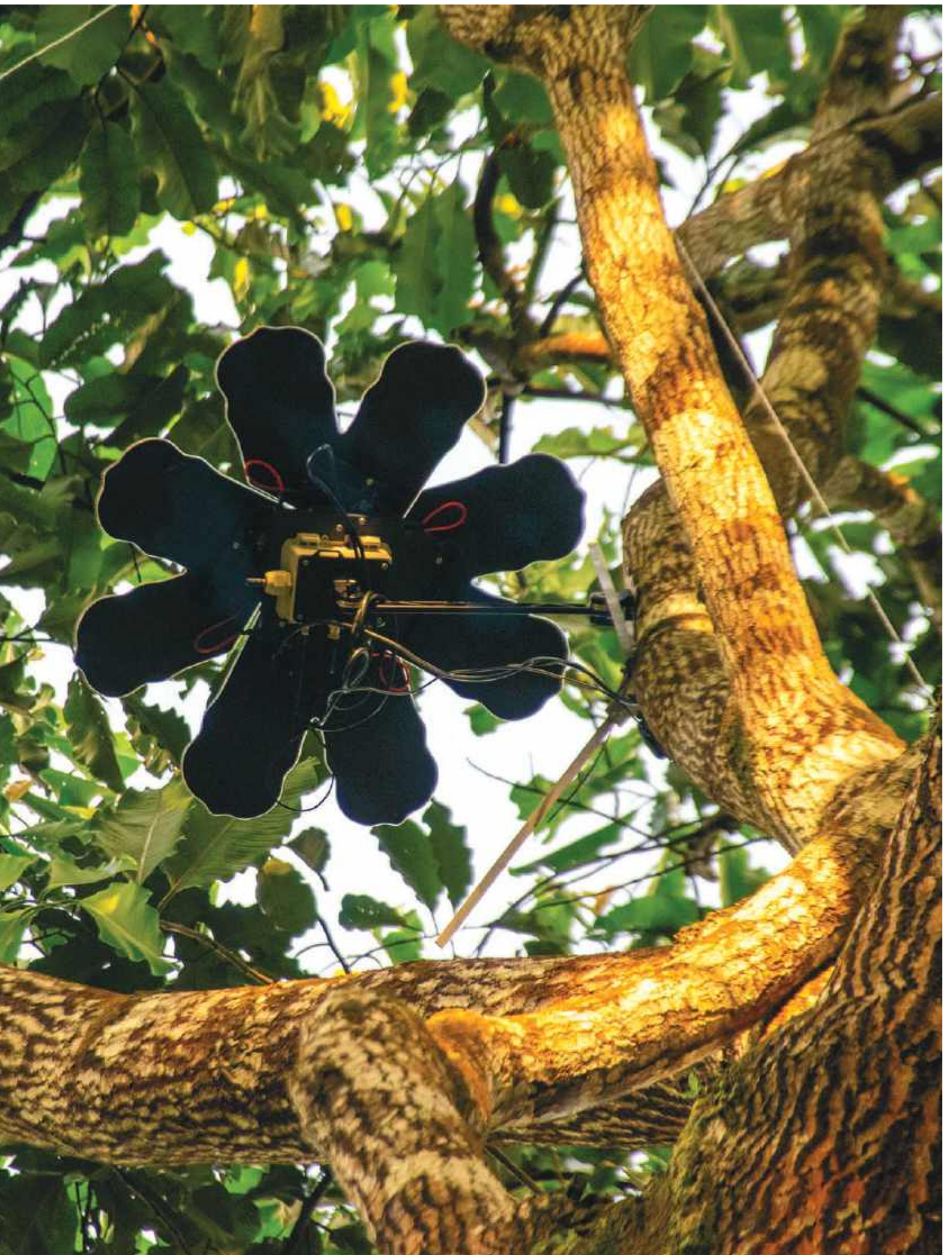
Bioacoustic monitoring devices have rapidly advanced in recent years. Many can now analyze data before transmitting it, and they've become cheaper to make as batteries have gotten smaller. By today's standards, Rainforest Connection's sensors are "over-engineered," says White.

But having a large number of detectors already deployed means there is ample data that can be mined for signals beyond well-known red flags, like gunshots. "An area for a lot more innovation going forward is to use the soundscape itself as a detector," White says. Rainforest Connection and the German software firm SAP tested this approach on the island of Sumatra and found they could identify human intruders by using machine learning to hunt for "uncharacteristic sudden changes to the soundscape." For example, tracking animal calls—and noting when those animals go silent—could reveal the arrival of poachers. In 2026, Rainforest Connection will roll out this approach to reserves in Thailand, Jamaica, and Romania by building a unique model for each environment, trained on thousands of hours of audio and verified using camera traps. "We have a lot of eyes and ears in the forest already, all of which are aware and reacting to each other and to new stimuli," White says.

For the rest of us, Rainforest Connection's unfiltered stream has another use: an app where you can listen to the livestream from the Ecuadorian rainforest, taking in the complete soundscape of birdsong, frog chatter, and cicada chirps. ■



COURTESY OF RAINFOREST CONNECTION





Fiction

Community service

by

Micaiah Johnson

photograph

Henry Horenstein

THE BIRD IS A BEAUTIFUL SILVER-GRAY, AND AS SHE DIES twitching in the lasernet I'm grateful for two things: First, that she didn't make a sound. Second, that this will be the very last time.

They're called corpse doves—because the darkest part of their gray plumage surrounds the lighter part, giving the impression that skeleton faces are peeking out from behind trash cans and bushes—and their crime is having the ability to carry diseases that would be compatible with humans. I open my hand, triggering the display from my imprinted handheld, and record an image to verify the elimination. A ding from my palm lets me know I've reached my quota for the day and, with that, the year.

I'm tempted to give this one a send-off, a real burial with holy words and some flowers, but then I hear a pack of street-rats hooting beside me. My city-issued vest is reflective and nanopainted so it projects a slight glow. I don't know if it's to keep us safe like they say, or if it's just that so many of us are

ex-cons working court-ordered labor, and civilians want to be able to keep an eye on us. Either way, everyone treats us like we're invisible—everyone except children.

I switch the lasernet on the bird from *electrocute* to *incinerate* and watch as what already looked like a corpse becomes ashes.

"Hey, executioner!" says a girl.

"Executioner" is not my official title. The branch of city government we work for is called the Department of Mercy, and we're only ever called technicians. But that doesn't matter to the child, who can't be more than eight but has the authority of a judge as she holds up a finger to point me out to her friends.

"Guys, look!" she says, then turns her attention to me. "You hunting something big?"

I shake my head, slowly packing up my things.

"Something small?" she asks. Then her eyes darken. "You're not a cat killer, are you?"

"No," I say quickly. "I do horseflies."

I don't know why I lied, but as the suspicion leaves her face and a smile returns, I'm glad I did.

"You should come down by the docks. We've got flies! Make your quota in a day."

The girl tosses her hair, making the tinfoil charms she's wrapped around her braids tinkle like wind chimes.

"It's my last day. But if I get flies again for next year, I'll swing by."

Another lie, because we both know the city would never send anyone to the docks for flies. Flies are killed because they are a nuisance, which means people only care about clearing them out of suburbs and financial districts. They'd only send a tech down to the docks to kill something that put the city proper at risk through disease, or by using up more resources than they wanted to spare.

LeeLee is expecting me home to sit through the reassignments with her and it's already late, so I hand out a couple of the combination warming and light sticks I get for winter to the pack of children with nowhere to go. As I walk away, the children are laughing so loud it sounds like screaming. They toss the sticks in the air like signal flares, small bright cries for help that no one will see.

LEELEE'S ANXIETY TAKES THE FORM OF CARETAKING, AND AS soon as I've stepped through the door I can smell bread warming and soup on the stove. I take off my muffling boots. Another day, I'd leave them on and sneak up on her just to be irritating, and she'd turn and threaten me with whatever kitchen utensil was at hand. But she'll be extra nervous today, so I remove the shoes that let me catch nervous birds, and step hard on my way in.

Sometimes it seems impossible that I can spend a year killing every fragile and defenseless thing I've encountered but still take such care with Lee. But I tell myself that the killing isn't me—it's just my sentence, and what I do when I have a choice is the only thing that really says anything about me. For the first six months and 400 birds, I believed it.

LeeLee flicks on a smile that lasts a whole three seconds when she sees me, then clouds over again.

"Soup's too thin. There wasn't enough powder for a real broth."

"I like thin soup," I say.

"Not like this. It doesn't even cover up the taste of the water."

"I like the taste of the water," I say, which breaks her out of her spiraling enough to roll her eyes.

I put my hands on her shoulder to stop her fussing.

"The soup is going to be fine," I say. "So will the reassignment."

I'm not much taller than she is, but when we met in juvie she hadn't hit her last growth spurt yet, so she still tilts her head back to look me in the eyes. "What if it's not?"

"It will—"

"What if you get whatever assignment Jordan got?"

There it is. Because two of us didn't leave juvie together to start community service—three of us did. But Jordan didn't last three weeks into his assignment before he turned his implements inward.

I notice she doesn't say *What if I get what Jordan got?* Because LeeLee is more afraid of being left alone than of having to kill something innocent.

"We don't know what his assignment was," I say.

It's true, but we do know it was bad. Two weeks into our first stretch, a drug meant to sterilize the city's feral cat population accidentally had the opposite effect. Everyone was pulled off their assigned duty for three days to murder litters of new kittens instead. It nearly broke me and Lee, but Jordan seemed almost grateful.

"Besides, we don't know if his assignment had anything to do with ... what he did. You're borrowing trouble. Worry in"—I check my palm—"an hour, when you actually know there's something to worry about."

You'd think it would hover over us too insistently to be ignored, but after we sit down and talk about our day I'm at ease, basking in the warmth of her storytelling and the bread that's more beige than gray today. When the notification comes in, I am well and truly happy, and I can only hope it isn't for the last time.

We both stiffen when we hear the alert. She looks at me, and I give her a smile and a nod, and then we look down. In the time between hearing the notification and checking it, I imagine all kinds of horrors that could be in my assignment slot. I imagine a picture of kittens, reason enough for the girl I met earlier to condemn me. For a moment, just a flash, I imagine looking down and seeing my own face as my target, or LeeLee's.

But when I finally see the file, the relief that comes over me softens my spine. It's a plant. Faceless, and bloodless.

I look up, and LeeLee's eyes are dark as she leans forward, studying my face, looking for whatever crack she failed to see in Jordan. I force myself to smile wide for her.

"It's a plant. I got a plant, Lee."

She reaches forward and squeezes my hands. Hers are shaking.

"What did you get?" I ask.

She waves away my question. "I got rats. I can handle it. I was just worried about you."

I spend the rest of the night unbelievably happy. For the next year, I get to kill a thing that does not scream.

"YOU GET ALL THAT?" THE MAN BEHIND THE DESK ASKS, AND I nod even though I didn't.

I've traded in my boots and lasernet for a hazmat suit and a handheld mister with two different solutions. The man had been talking to me about how to use the solutions, but I can't process verbal information very well. The whole reason I was sent to the correctional facility as a teen was that too many teachers mistook my processing delays for behavioral infractions. I'm planning to

"THEY WERE SUPPOSED TO ADD SULFUR TO THIS BATCH, BUT THEY DIDN'T. SO YOU WON'T SMELL IT. IT WON'T MAKE YOU COUGH OR YOUR EYES WATER. IT'LL JUST BE LIGHTS OUT. GOOD NIGHT. YOU GOT ME?"

read the manual on my own time before I start in a few hours, but when I pick up the mister and look down the barrel, the equipment guy freaks out.

"Did you not hear me? Don't even look at that thing without your mask on." He takes a breath, calmer now that I've lowered my hands. "Look, the first solution—it's fine. It's keyed to the plant itself and just opens its cells up for whatever solution we put on it. You could drink the stuff. But that second? The orange vial? Don't even put it in the mister without your mask on. It dissipates quickly, so you're good once you're done spraying, but not a second before."

He looks around, then leans in. "They were supposed to add sulfur to this batch, but they didn't. So you won't smell it. It won't make you cough or your eyes water. It'll just be lights out. Good night. You got me?"

I nod again as I grab the mask I hadn't noticed before. This time when I thank him, I mean it.

IT TAKES ME AN HOUR TO FIND THE FIRST PLANT, AND WHEN I DO it's beautiful. Lush pink on the inside and dark green on the outside, it looks hearty and primitive. Almost Jurassic. I can see why it's only in the sewers now: it would be too easy to spot and destroy aboveground in the sea of concrete.

After putting on my mask, I activate the mister and then stand back as it sprays the plant with poison. Nothing happens. I remember the prepping solution and switch the cartridges to coat it in that first. The next time I try the poison, the plant wilts instantly, browning and shrinking like a tire deflating. I was wrong. Plants this size don't die silently. It makes a wheezing sound, a deep sigh. By the third time I've heard it, I swear I can make out the word *Please*.

When I get home, LeeLee's locked herself in the bathroom, which doesn't surprise me. I heard that they moved to acid for rats, and the smell of a corpse dissolving is impossible to get used to and even harder to get out of your hair. I eat dinner,

read, change for bed, and she's still in the bathroom. I brush my teeth in the kitchen.

THE NEXT MORNING, I HAVE TO TAKE A TRANSPORT TO THE plant's habitat on the other end of the city, so I spend the time looking through the file that came with the assignment. Under "Characteristics," some city government scientist has written, "Large, dark. Resource-intensive. Stubborn."

I stare at the last word. Its own sentence, tacked on like an afterthought. *Stubborn*. The same word that was written in my file when I got sent from school to the facility where I met LeeLee and Jordan. Large, dark, stubborn, and condemned. I've never been called *resource-intensive*. But I have been called a waste.

And maybe that's why I do it.

When I get to my last plant of the day, I don't reach for the mister. This one is small, young, the green still neon-bright and the teeth at the edges still soft. I pick it up, careful with its roots, and carry it home. I find a discarded water container along the way and place it inside. When I get home I knock on LeeLee's door. She doesn't answer, so I leave the plant on the floor as an offering. They aren't proper flowers, but they smell nice and earthy. It might keep the residual odor from melted organs, fur, and bones from taking over her room.

"KILLING THINGS IS A DUMB JOB," SAYS THE GIRL.

After a week of hearing the death cries of its cousins, I was moved to use some of my allowance to buy cheap fertilizer and growth serum for my plant. The girl and her friends, fewer than before, were panhandling at the megastore across the way. She ran over, braids jingling, as soon as she saw me. I thought she'd leave once I gave her more glowsticks for her friends, but she stayed in step and kept following me.

"It's not a dumb job," I say, even though it is.

"What's the point?"

I shift my bag to point at the bottom of my vest. Beneath "Mercy Dept." the department's slogan is written in cursive: *Killing to Save!*

"See?"

She sees the text but doesn't register it, and I have to remind myself that even getting kicked out of school is a privilege. The city had decided to stop wasting educational resources on me. They'd never even tried with her or the other streetrats.

"It just means we kill to help."

"That doesn't make sense."

Suddenly, all I can think about is Jordan. "Maybe they don't mind."

"What?"

I think of the plants. Maybe they hadn't been pleading. Maybe they'd been sighing with relief. I think of the birds that eventually stopped running away.

"Maybe they're tired. The city's right, and their existence isn't compatible with the world we made. And that's our fault for being stupid and cruel, but it makes their lives so hard. We've made it so they can only live half a life. Maybe the least we can do is finish the job."

It's a terrible thing to say—even worse to a kid.

Her eyes go hard. "What are you killing now, executioner?"

The question surprises me. "Sewer plants. Why?"

"I don't believe you."

I'd wanted her to leave me alone, but when she runs away I feel suddenly empty.

I HAVE AN ISSUE AT WORK WHEN I CAN'T FIND MY POISON VIAL.

I tell them it rolled away in the sewer and I couldn't catch it in time, because I don't want to tell them I was unobservant enough to let a street kid steal from me. After a stern warning and a mountain of forms, they issue a new vial and don't add to my service time.

Pulling overtime to make up for the day I didn't have my poison means it's days before I get to fertilize my houseplant. LeeLee's door is open, so I bring in the fertilizer and serum. She's put the plant on her windowsill, but it prefers indirect sunlight, so I move it to the shelf next to her boxes of knickknacks and trinkets. I add the fertilizer to its soil and am about to spray it with the growth serum when I get an idea. I get the mister from my kit and set it up to spray the prepping solution on the little plant to prime it. I open the window and put on my mask, just in case, but I'm sure the man was telling the truth when he called the first liquid harmless. After its cells are open, I spray it with my store-bought growth serum.

I'm halfway through making dinner when I hear the crash and run into LeeLee's room.

"Shit!"

The plant has grown huge, turning adult instantly, and its new weight has taken down LeeLee's shelf. Dainty keepsake boxes are shattered on our concrete floor.

I bend to my knees quickly, so focused on fixing my mistake that I don't register the oddness of the items I'm picking up—jacks, kids' toys, a bow—until my fingers touch something small and shimmering. It's a scrap of silver, still rounded in the shape of the braids it was taken from.

I got rats. I can handle it.

I'd forgotten the city has more than one kind.

I'M WAITING UP WHEN LEE GETS HOME. I DON'T MAKE HER

tell me. I just grab her kit and rummage through it. Where my kit has a hazmat suit, hers has a stealth mesh to render her invisible. Where I keep my mister, she has a gun loaded with vials too large for rats. I have a mini-vac to suck up excess plant matter to prevent seeds from sprouting. She has zip ties.

**I TELL HER HOW WE'LL MAKE
PLAYGROUNDS OF DEAD DATA
CENTERS AND USE HOSES
TO FILL THE HOLES WHERE
SKYSCRAPERS WERE, AND
KIDS WILL PLAY MARCO POLO
SWIMMING OVER A CEO'S
SUNKEN OFFICE.**

By the time I'm done, she's already cracking under the weight of everything she tried to protect me from. Within moments she's sobbing on the floor. I carry her to her bed and get in beside her. I try not to listen too closely as she recounts every horrible moment, but I'm listening at the end, when she tells me she can't do it anymore. When she confesses that she's the one who stole my poison, and has only been waiting to take it because she didn't have the stomach to do to me what Jordan did to us.

I leave her for just a moment, but by the time I lie back in bed beside her I've figured it out.

I tell her that she won't have to take her shift tomorrow. I tell her I'm going to go around the city with my mister and my growth serum. That I'll move plants from sewers to the yards around City Hall and every public space and the support pylons of important people's companies, and then spray them so they become huge. The city will freak. I tell her it will be like the kittens, but this time we'll all be pulled off our assignments to kill plants. And maybe the serum will work too well. Maybe the city was right to fear these plants, and they will grow and grow and eat our concrete while the roots crack our foundations and cut our electricity and everything will crumble. And the people with something to lose might suffer, but the rest of us will just laugh at the perfection of rubble. I tell her how we'll make playgrounds of dead data centers and use hoses to fill the holes where skyscrapers were, and kids will play Marco Polo swimming over a CEO's sunken office.

She asks if I'll put any at our old detention center.

I tell her, *Hundreds*.

I talk long enough that her eyes close, and loud enough that neither of us can hear the sound of my mister blowing. The man who gave it to me was right. Even without the mask, it doesn't smell like sulfur. It doesn't smell like anything. ■

Micaiah Johnson's debut novel, *The Space Between Worlds*, a *Sunday Times* bestseller and *New York Times* Editors' Choice pick, was named one of the best books of 2020 and one of the best science fiction books of the last decade by NPR. Her first horror novel, *The Unhaunting*, is due out in fall 2026.

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Cue the confession! Meta's outgoing chief AI scientist, Yann LeCun, says **the company fudged Llama 4's benchmarks** to puff up its performance.



Online detectives used AI to unmask the ICE agent who shot Renee Good. **They got the wrong guy.**

Grok and Gemini: **Uncle Sam wants you!** Just don't be all woke about it, okay?



New research says AI is about to **hit the labor market like a tsunami**. Gen Z workers are spiraling.



Got a mysterious rash? **Dr. GPT will see you now.**



Workers across China are donning **VR headsets and exoskeleton suits** to teach robots how to fold laundry. Who works for whom, again?



Step aside, ChatGPT. Claude Code isn't just a pretty coder. It can **analyze DNA, crank out research papers, and deal with your email backlog.**



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Disney hands **Mickey Mouse and his friends** to OpenAI's video platform, Sora. Is the House of Mouse getting sloppy?

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*Dear Diary: I want to "flip to a for-profit" and become a billionaire. xoxo, OpenAI president Greg Brockman. The **Musk-OpenAI lawsuit** keeps coughing up secrets.*



AI toys are teaching kids how to sharpen a knife and parroting Chinese propaganda. We take back everything, AI is awesome.



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